

《计算机组成原理实验》

(预备实验)

厦门大学信息学院软件工程系

2026年3月

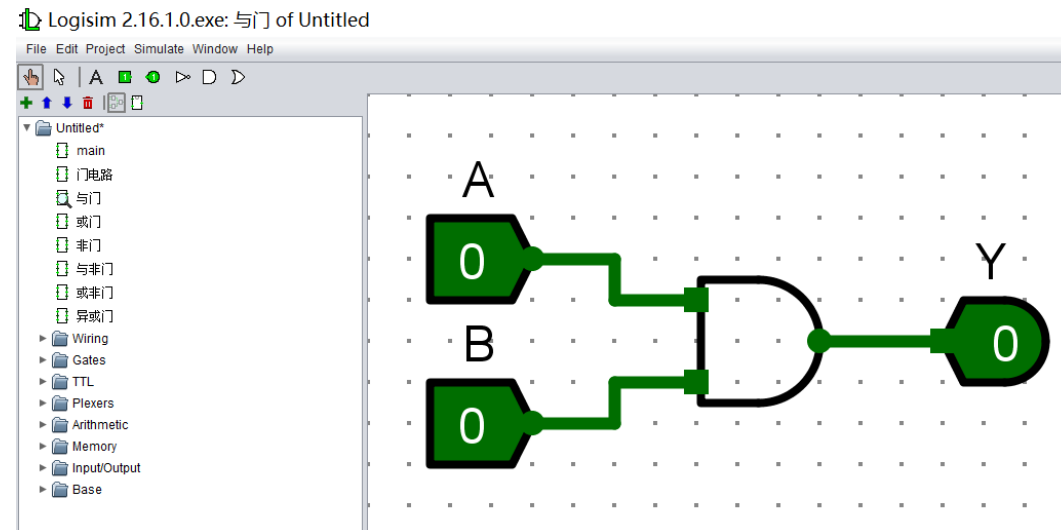
请同学们在第1周利用课余时间完成预备实验

预备实验 Logisim的使用

参考设计文件：[Logisim的使用.circ](#)

Logisim简介

- **Logisim**是一种用于设计和模拟**数字逻辑电路**的教育工具。凭借其简单的工具栏界面和构建它们时的电路仿真，它非常简单，有助于学习与逻辑电路相关的最基本概念。由于能够从较小的子电路构建更大的电路，并通过鼠标拖动来绘制电线束，因此可以使用Logisim来设计和仿真用于相关课程的教学目的。



- 查看电脑上是否已有JDK?
 - 打开“命令提示符”，执行：
 - `java -version`



```
C:\> 命令提示符
Microsoft Windows [版本 10.0.19044.1645]
(c) Microsoft Corporation。保留所有权利。

C:\Users\lily2002>java -version
java version "1.8.0_121"
Java(TM) SE Runtime Environment (build 1.8.0_121-b13)
Java HotSpot(TM) 64-Bit Server VM (build 25.121-b13, mixed mode)

C:\Users\lily2002>
```

- 如果没有，点击“**jdk-8u121-windows-x64.exe**”，安装JDK。

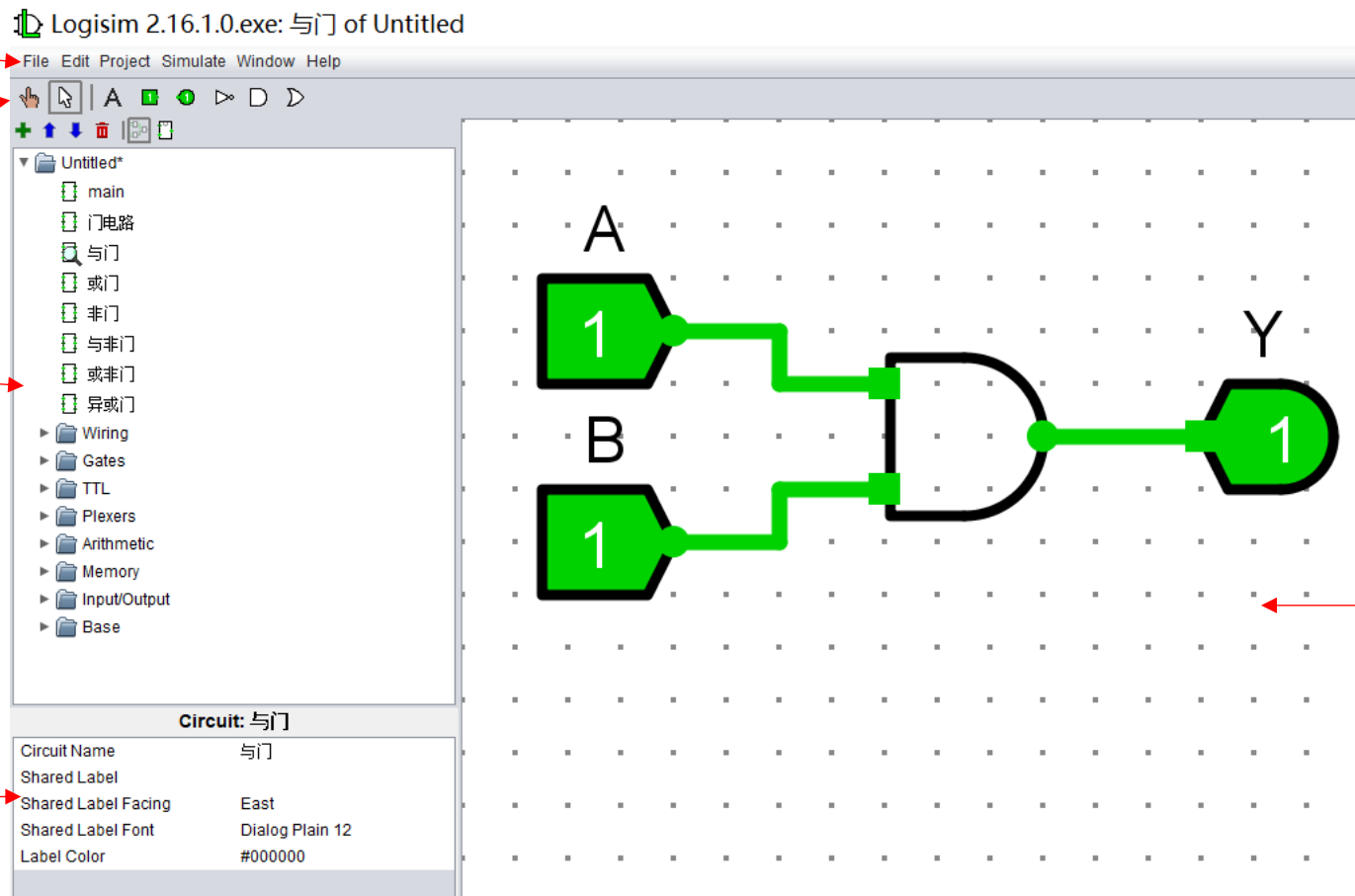
- 点击“**logisim-hust-20200118.exe**”，运行Logisim。

菜单栏

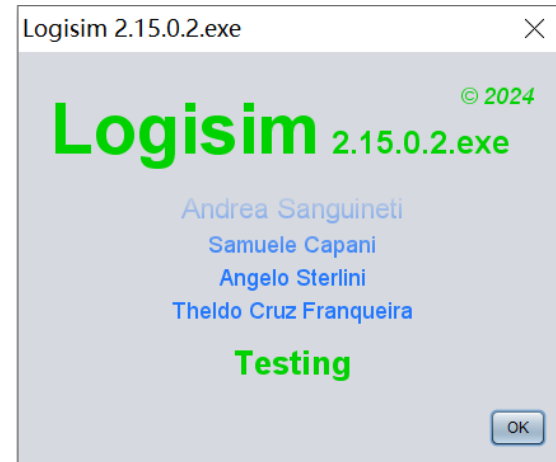
工具栏

管理窗

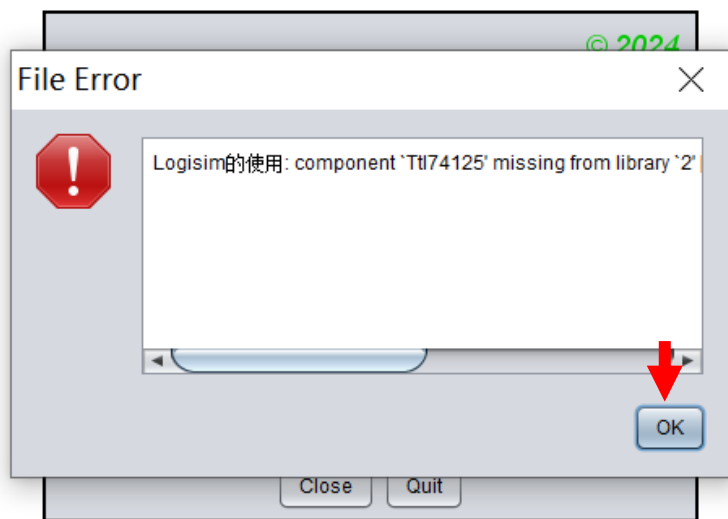
属性表



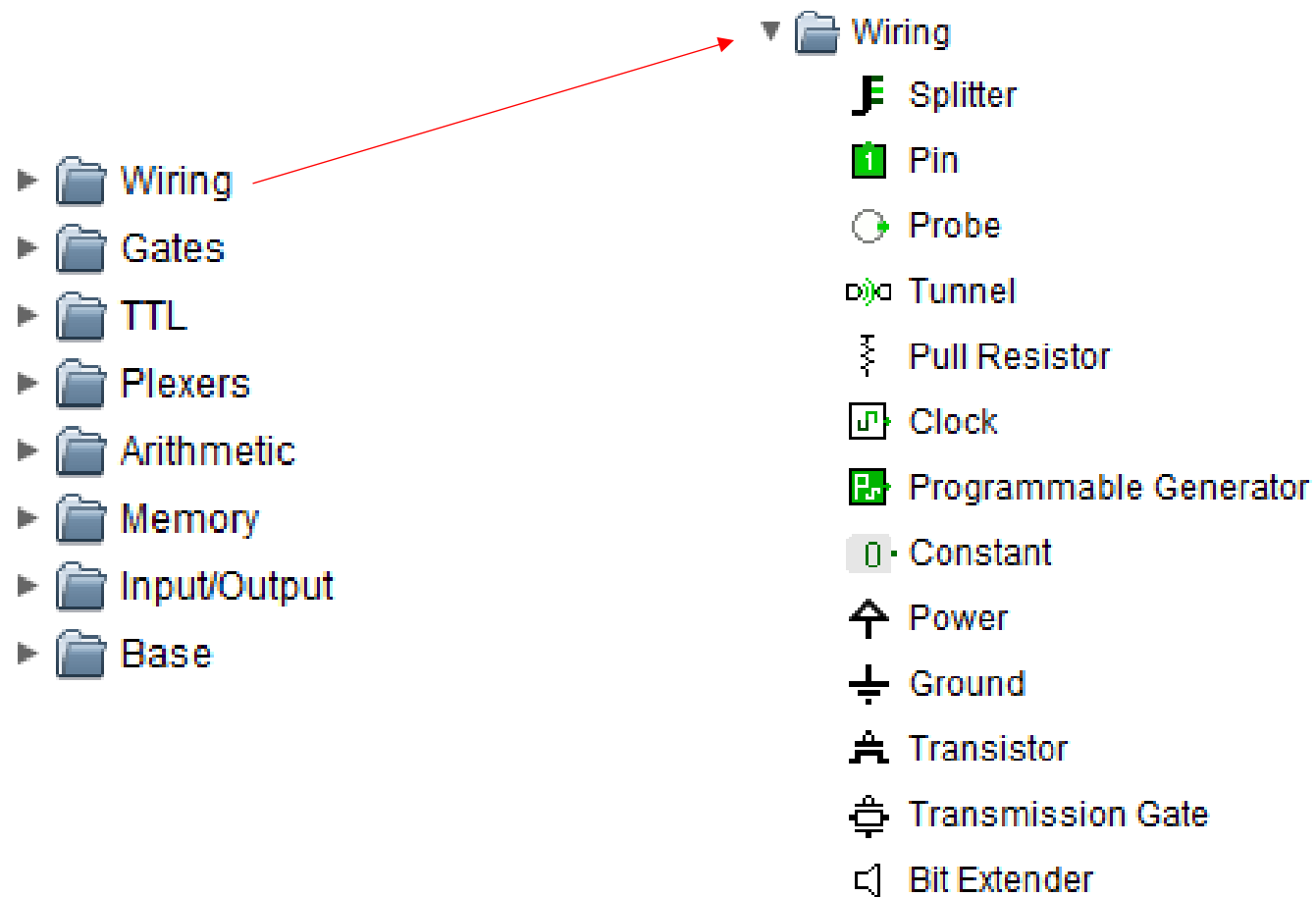
画布区域



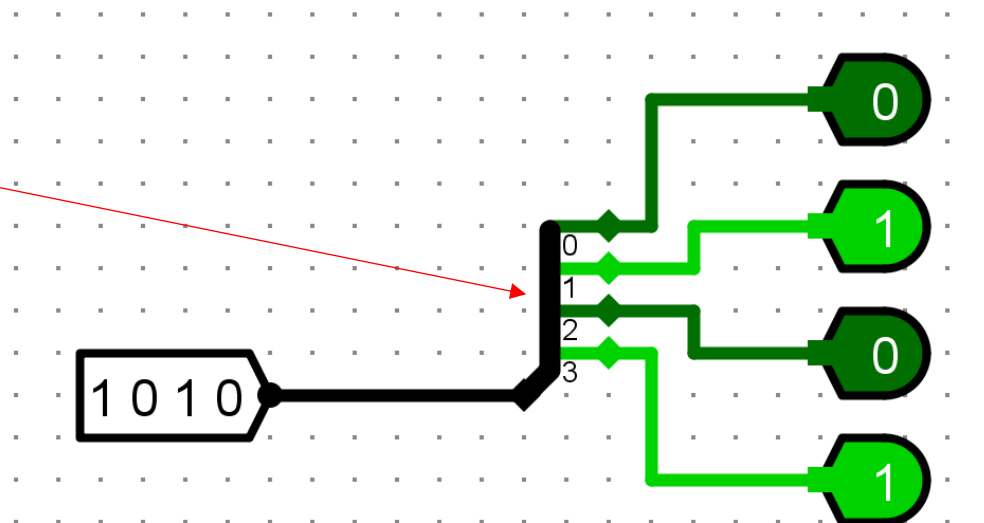
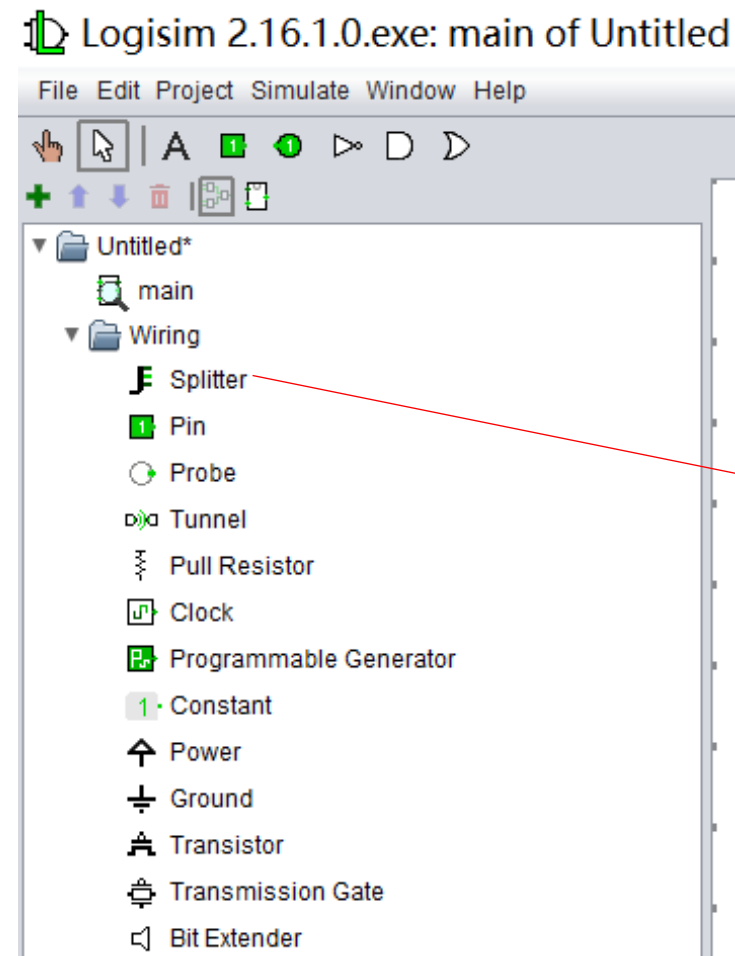
- 在Logisim中打开“Logisim的使用.circ”。



(一) 线路 (Wiring)



分线器 (Splitter) ——用于输出

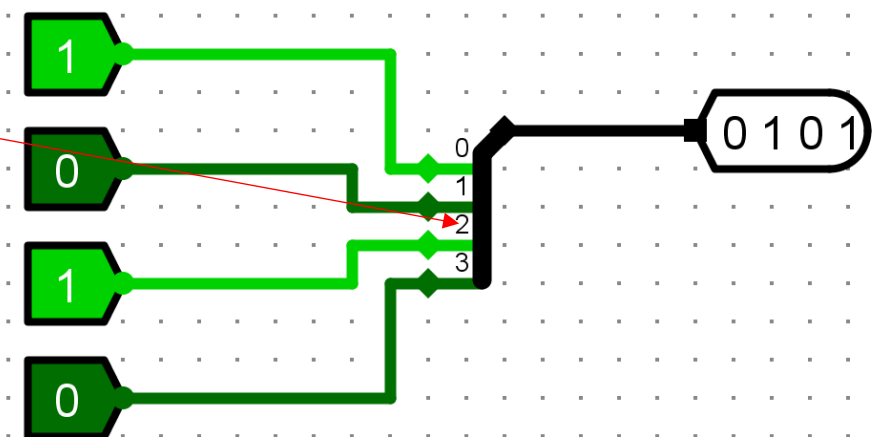
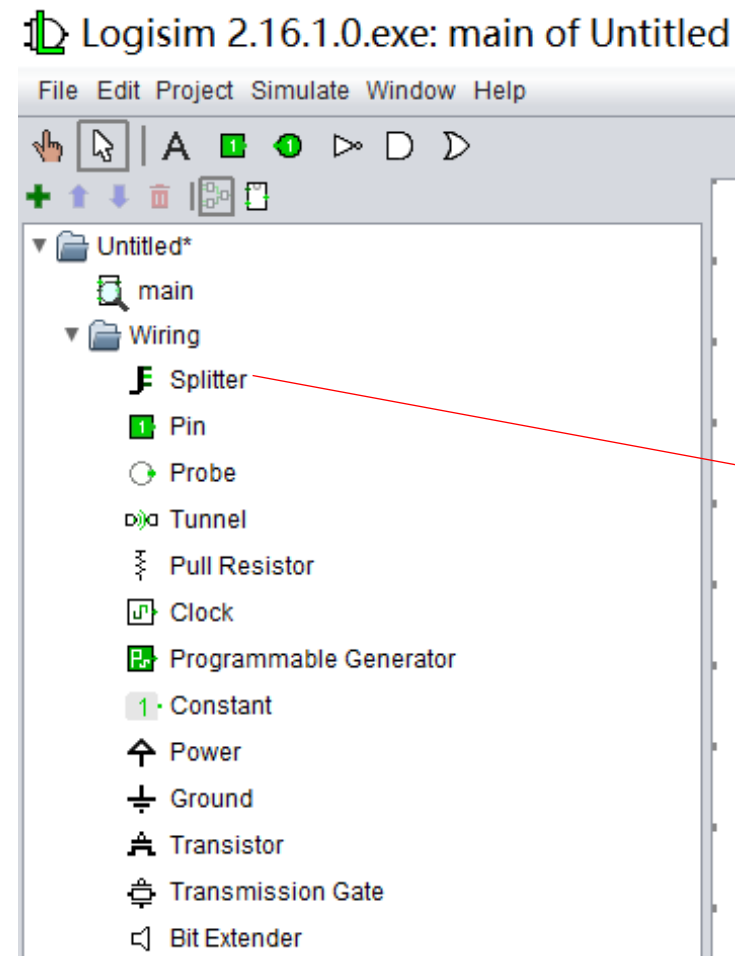


一变多

Splitter的属性 (可以改变)

Selection: Splitter	
Facing	East
Fan Out	4
Bit Width In	4
Appearance	Left-handed
Bit 0	0 (Top)
Bit 1	1
Bit 2	2
Bit 3	3 (Bottom)

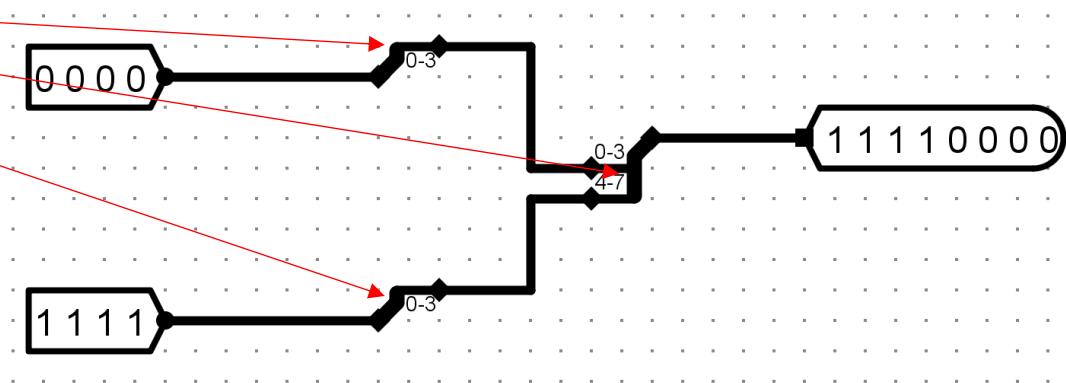
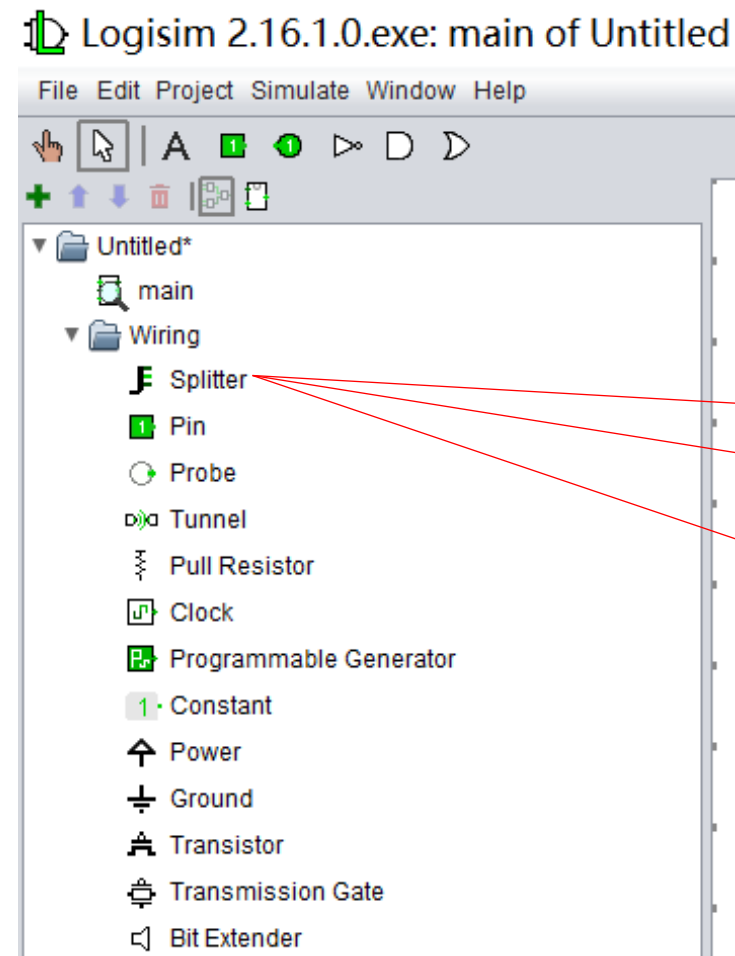
分线器 (Splitter) ——用于输入



Splitter的属性 (可以改变)

Selection: Splitter	
Facing	East
Fan Out	4
Bit Width In	4
Appearance	Left-handed
Bit 0	0 (Top)
Bit 1	1
Bit 2	2
Bit 3	3 (Bottom)

分线 (Splitter) —— 拼接

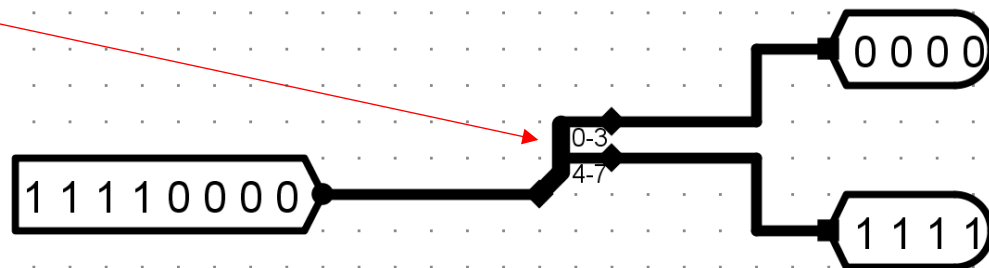
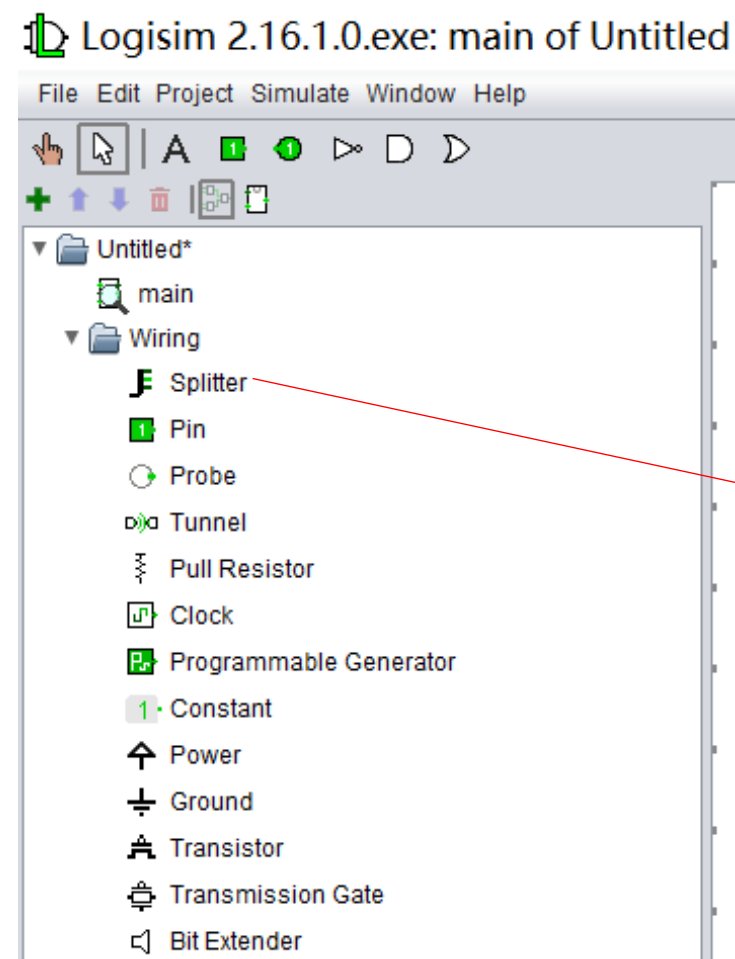


2个4位输入拼接成1个8位输出

Splitter的属性 (可以改变)

Selection: Splitter	
Facing	East
Fan Out	4
Bit Width In	4
Appearance	Left-handed
Bit 0	0 (Top)
Bit 1	1
Bit 2	2
Bit 3	3 (Bottom)

分线器 (Splitter) —— 分线



1个8位输入分成2个4位输出

Splitter的属性（可以改变）

Selection: Splitter	
Facing	East
Fan Out	4
Bit Width In	4
Appearance	Left-handed
Bit 0	0 (Top)
Bit 1	1
Bit 2	2
Bit 3	3 (Bottom)

输入、输出 (Pin)

Pin的属性 (可以改变)

Selection: Pin	
Facing	East
Output?	No
Data Bits	1
Three-state?	No
Pull Behavior	Unchanged
Label	
Label Location	North
Label Font	Dialog Plain 12
Label Color	#000000

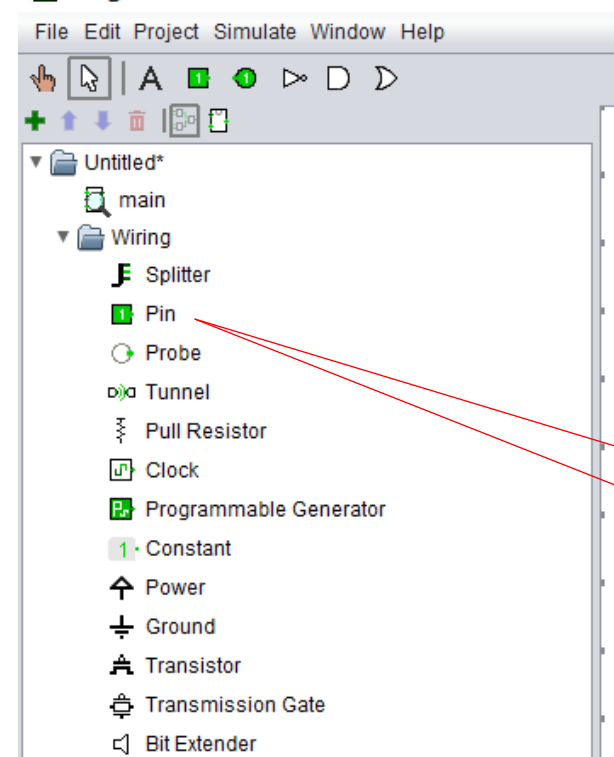
输入

Pin的属性 (可以改变)

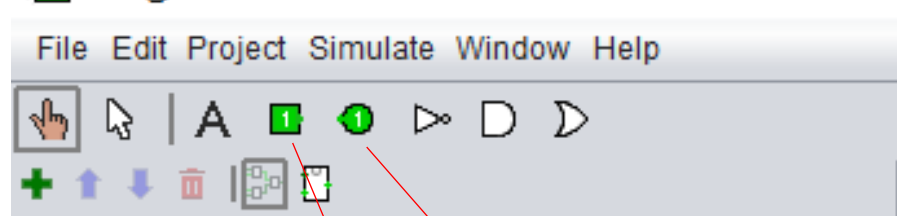
Pin	
Facing	West
Output?	Yes
Data Bits	1
Three-state?	Yes
Pull Behavior	Unchanged
Label	
Label Location	North
Label Font	Dialog Plain 12
Label Color	#000000

输出

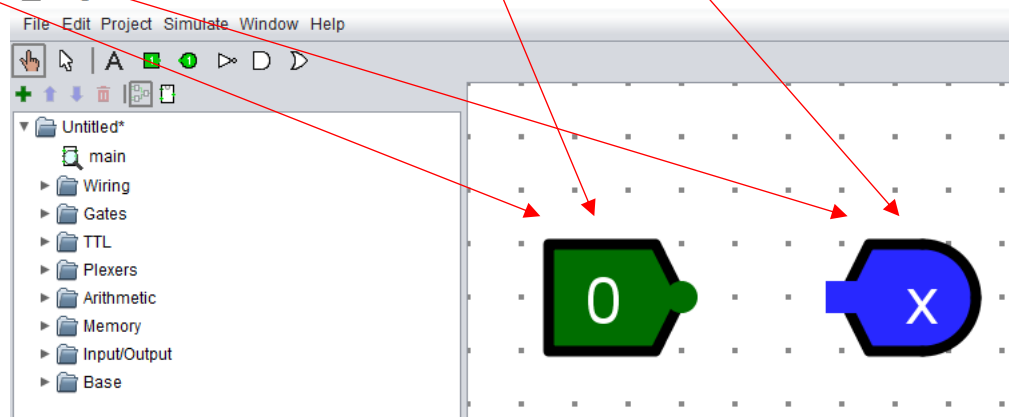
Logisim 2.16.1.0.exe: main of Untitled



Logisim 2.16.1.0.exe: main of Untitled



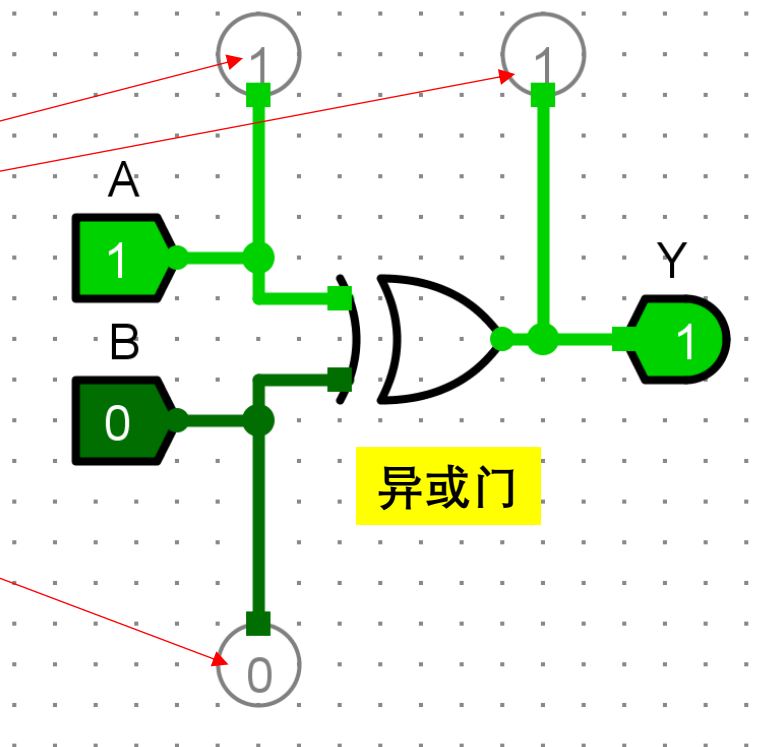
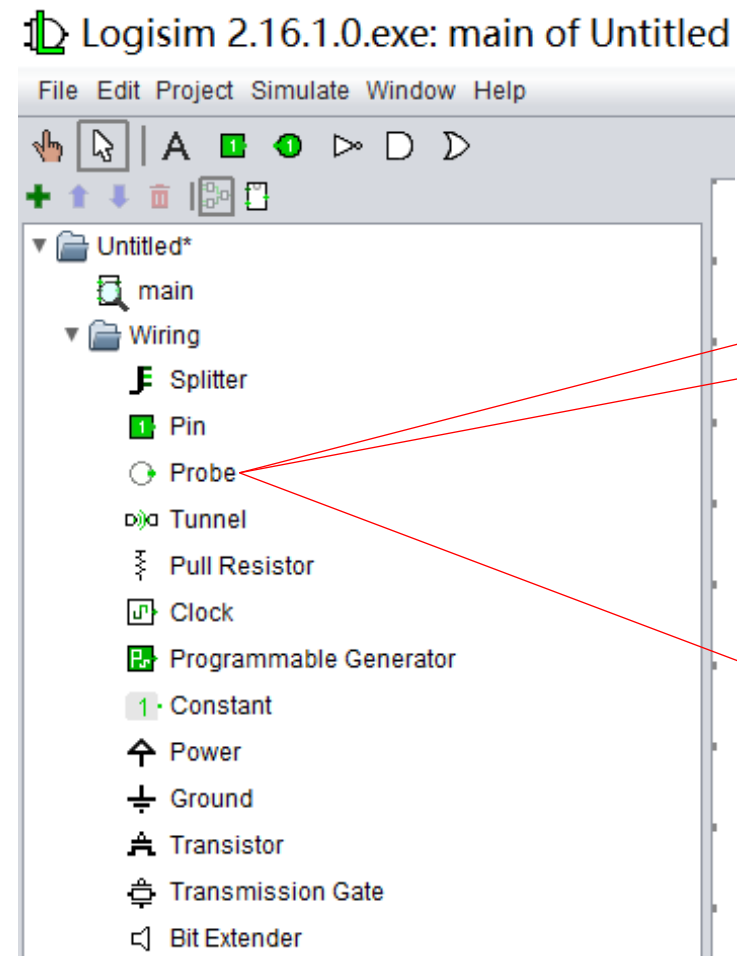
Logisim 2.16.1.0.exe: main of Untitled



输入

输出

探针 (Probe)

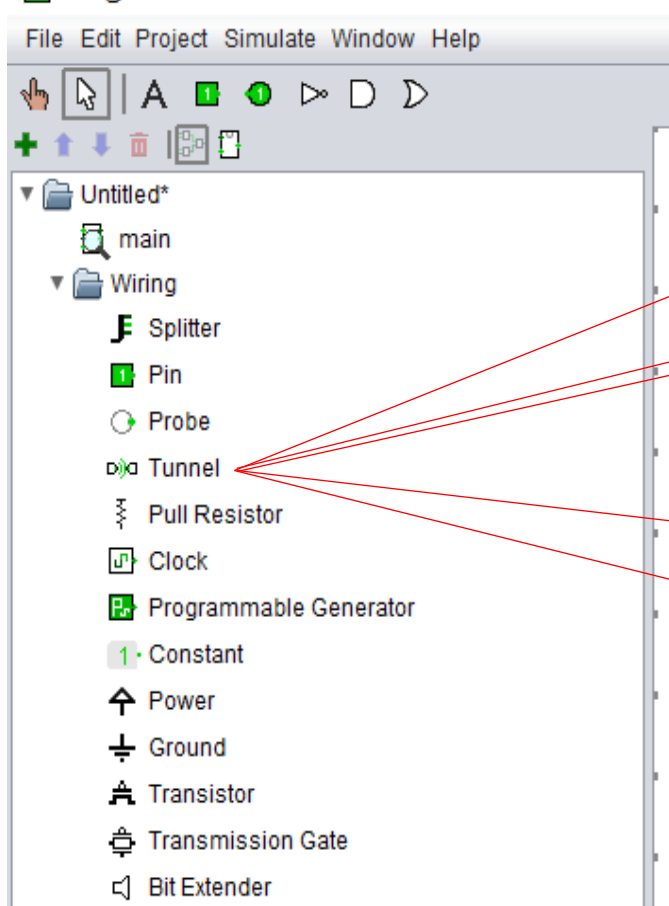


Probe的属性 (可以改变)

Selection: Probe	
Facing	South
Radix	Unsigned Decimal
Label	
Label Location	North
Label Font	Dialog Plain 12
Label Color	#000000

隧道 (Tunnel)

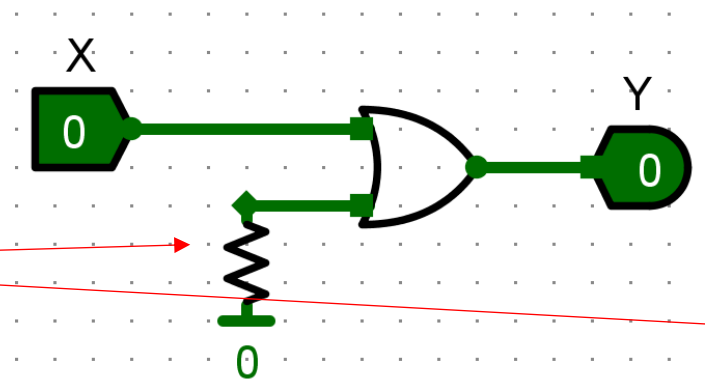
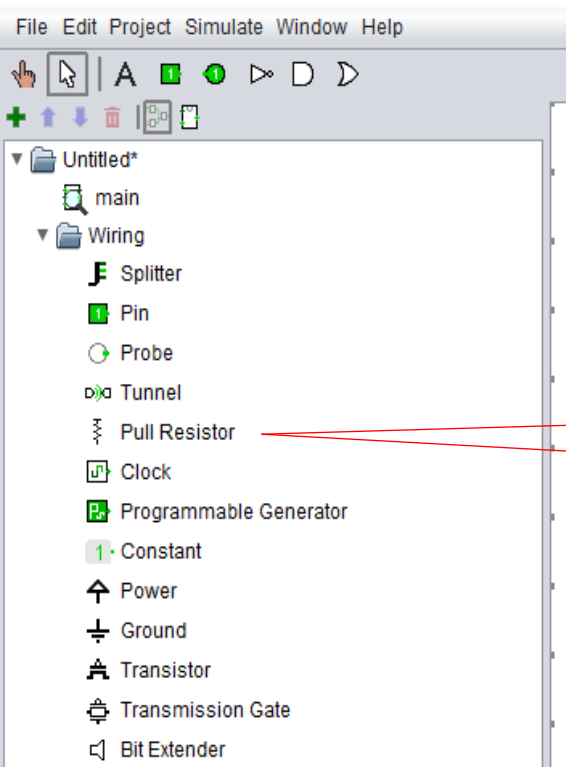
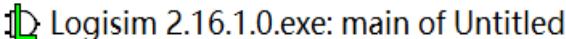
Logisim 2.16.1.0.exe: main of Untitled



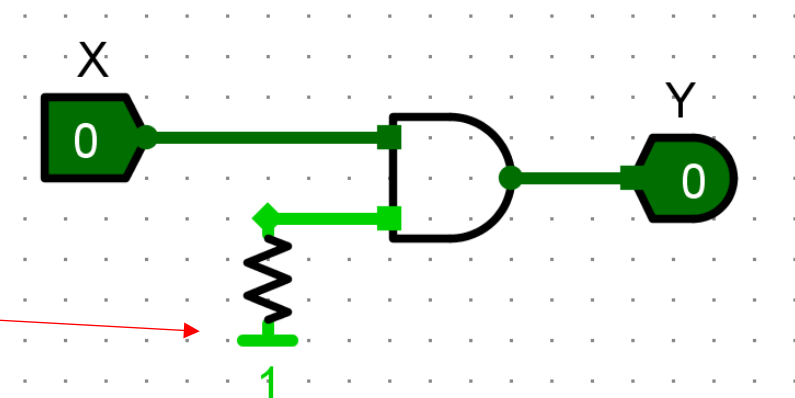
Tunnel的属性 (可以改变)

Selection: Tunnel	
Facing	West
Data Bits	1
Label	B
Label Font	Dialog Plain 12

上拉电阻 (Pull Resistor)



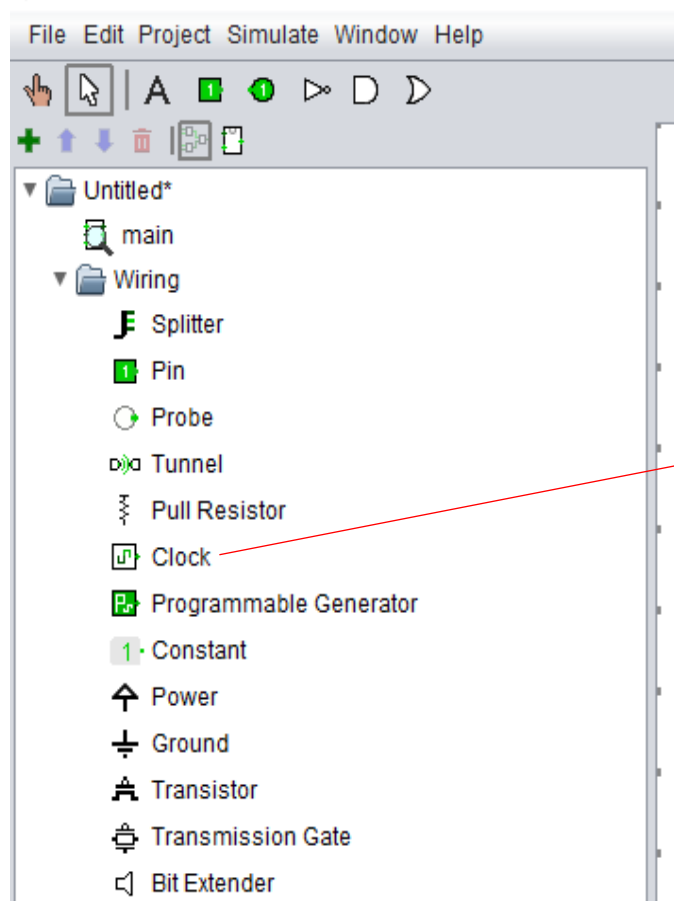
$$Y = X + 0 = X$$



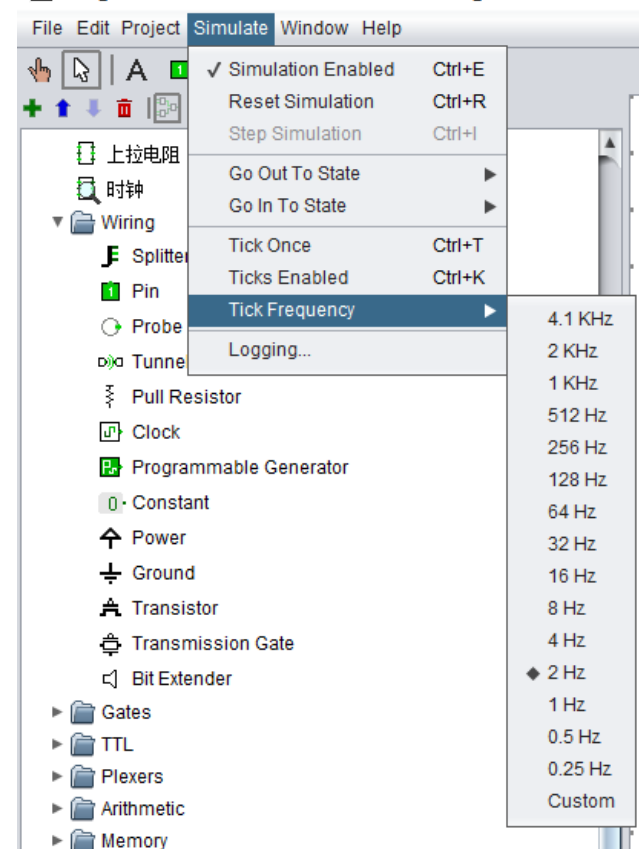
$$Y = X \cdot 1 = X$$

时钟 (Clock)

Logisim 2.16.1.0.exe: main of Untitled



Logisim 2.15.0.2.exe: 时钟 of Logisim使用

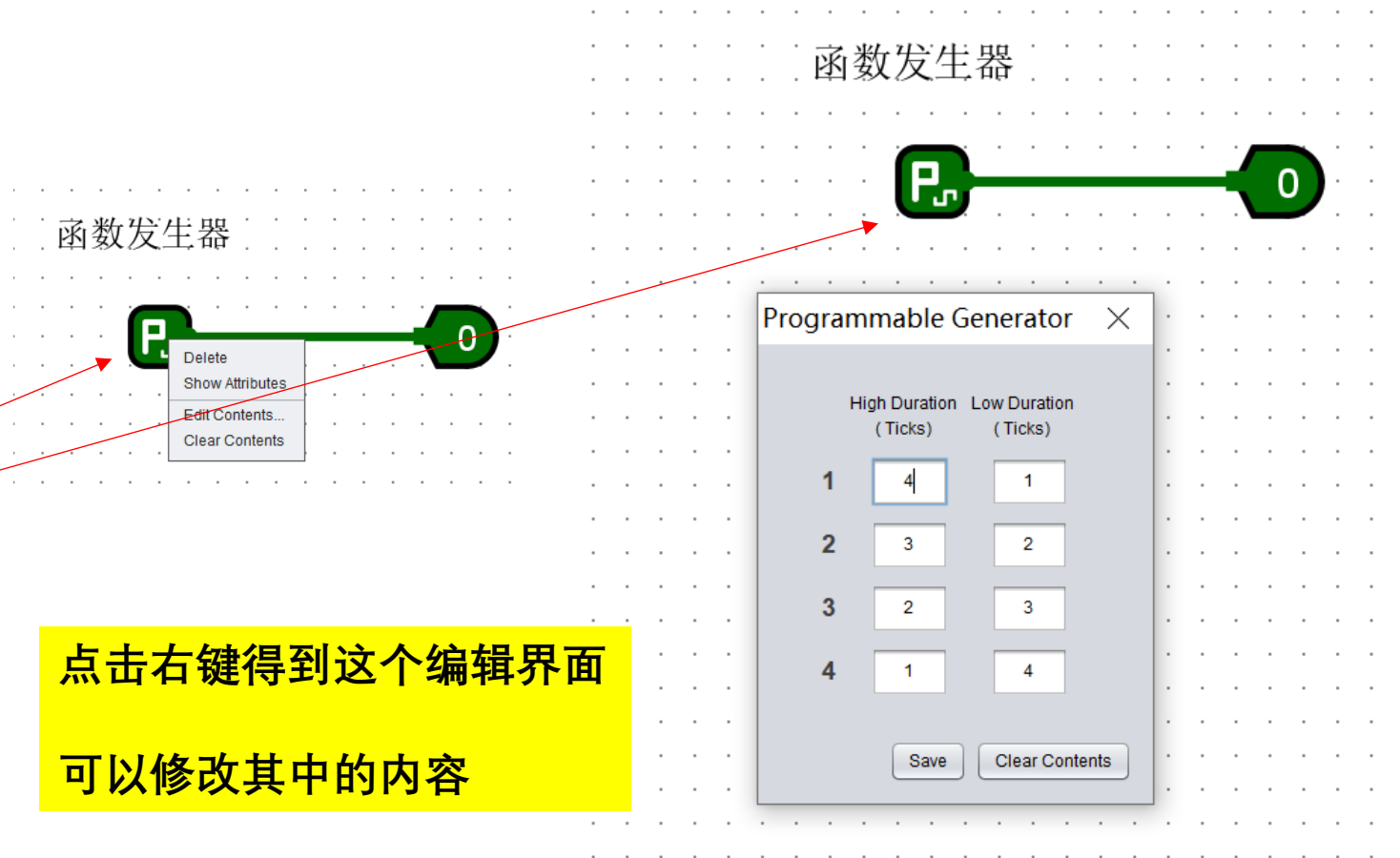
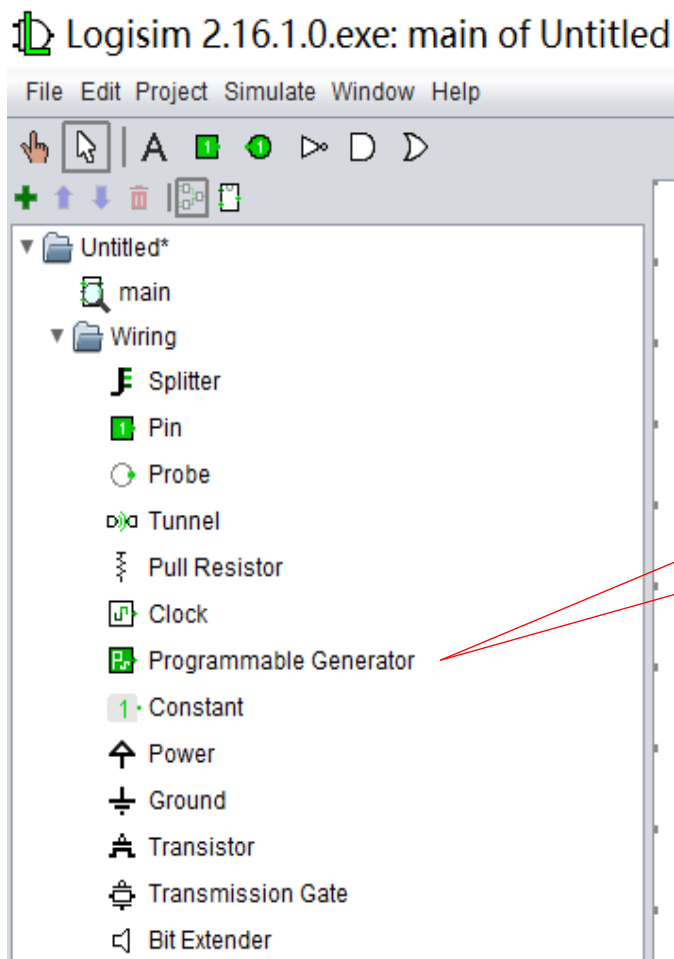


按Ctrl+T: 产生1次时钟

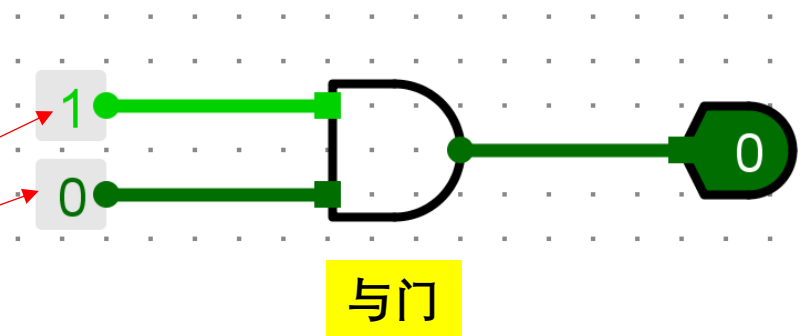
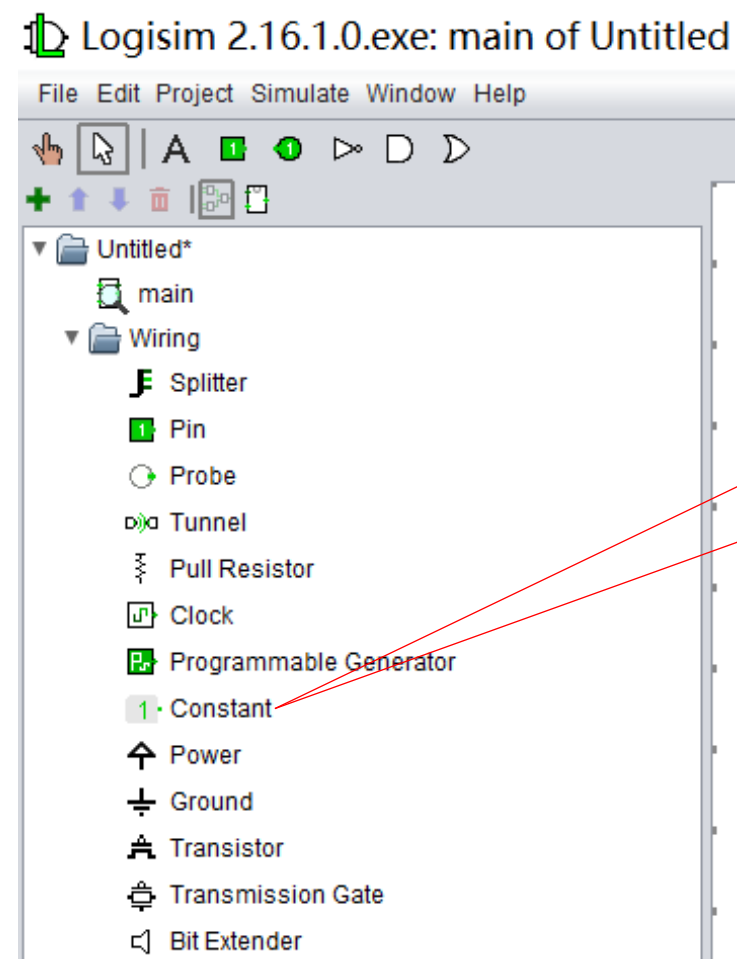
按Ctrl+K: 产生连续的时钟

时钟频率可以设置

函数发生器 (Programmable Generator)



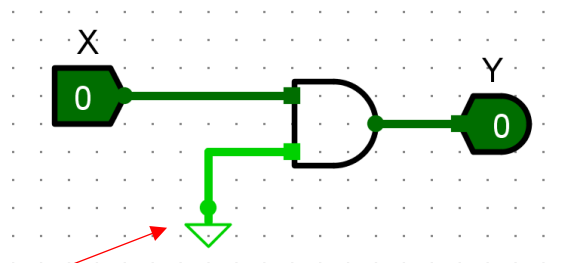
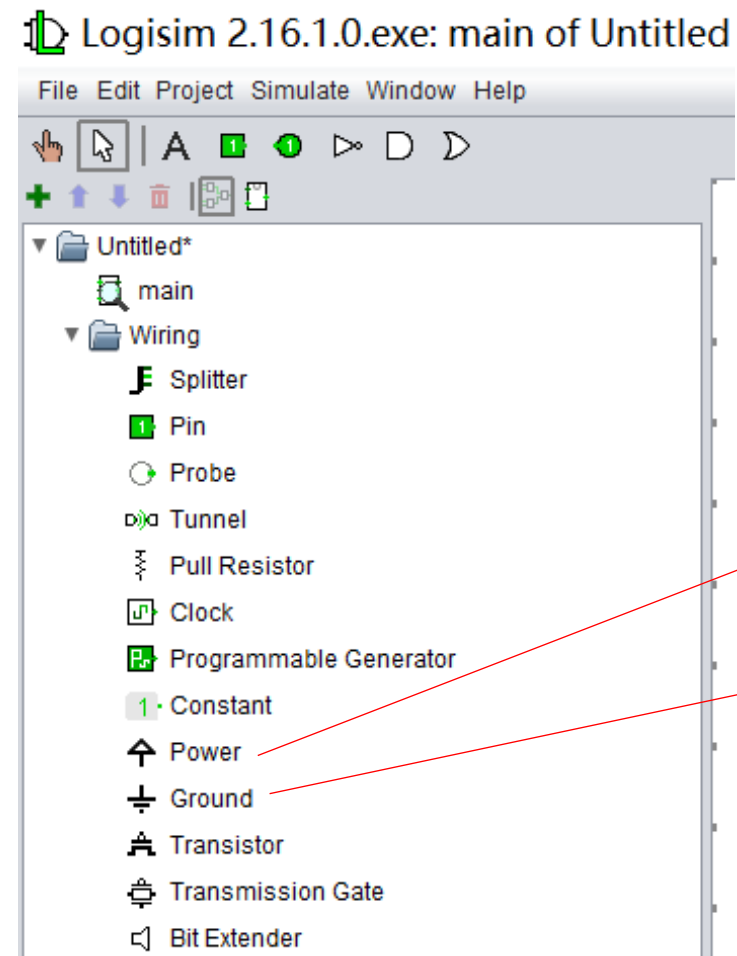
常量 (Constant)



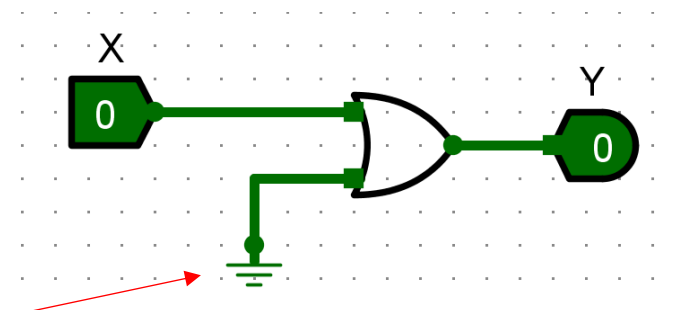
Constant的属性 (可以改变)

Selection: Constant	
Facing	East
Data Bits	1
Value	0x1

电源 (Power) 、接地 (Ground)

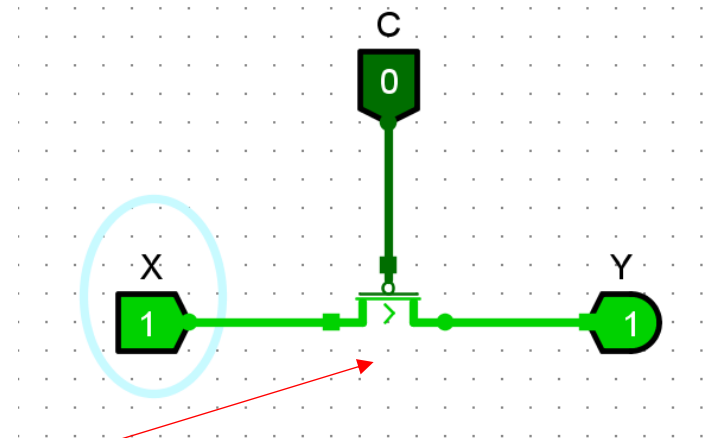
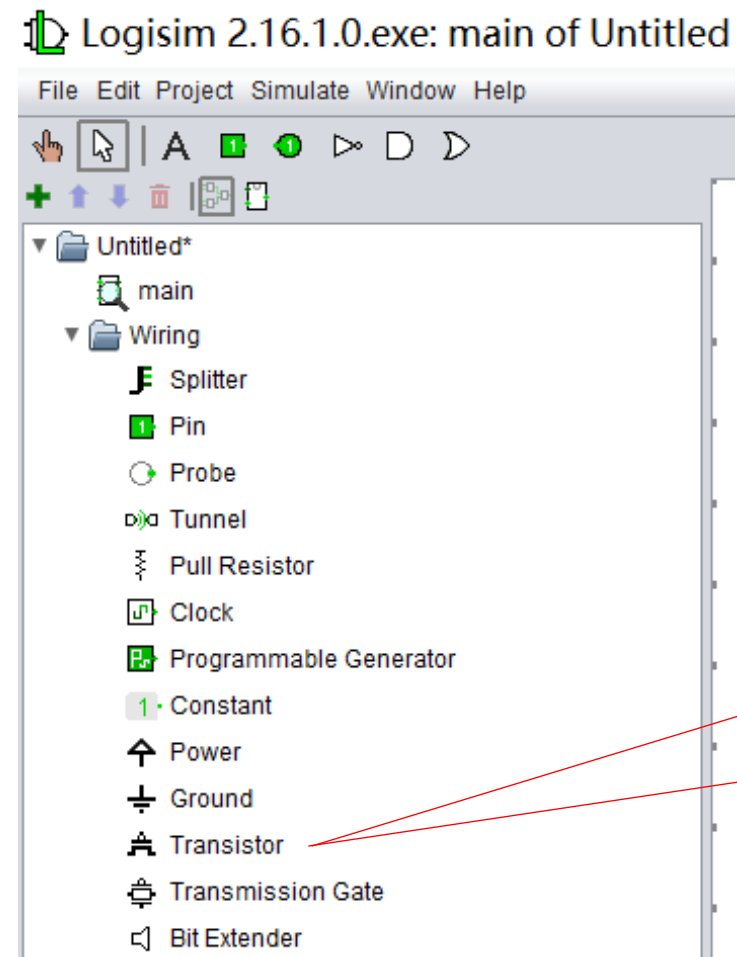


$$Y = X \cdot 1 = X$$

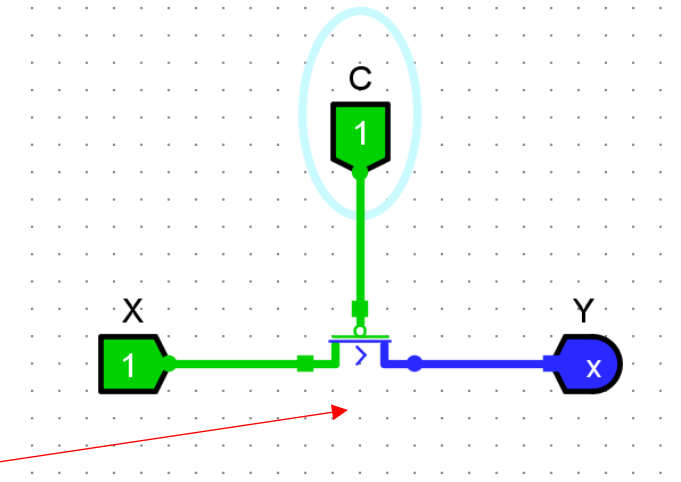


$$Y = X + 0 = X$$

晶体管 (Transistor)

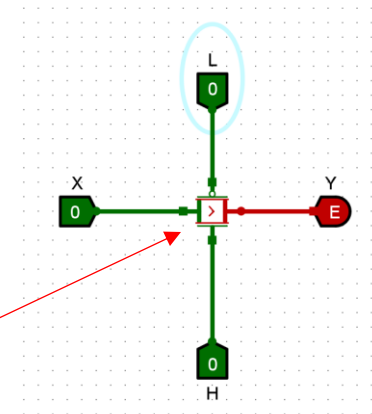


C=0 Y=X

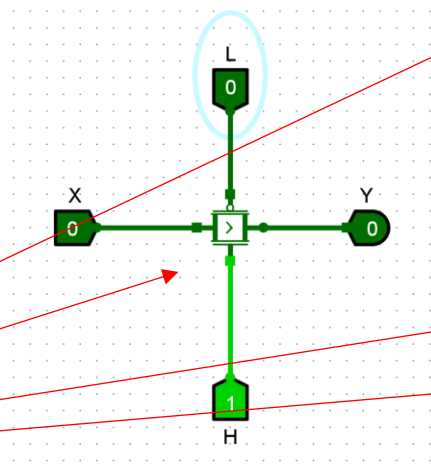
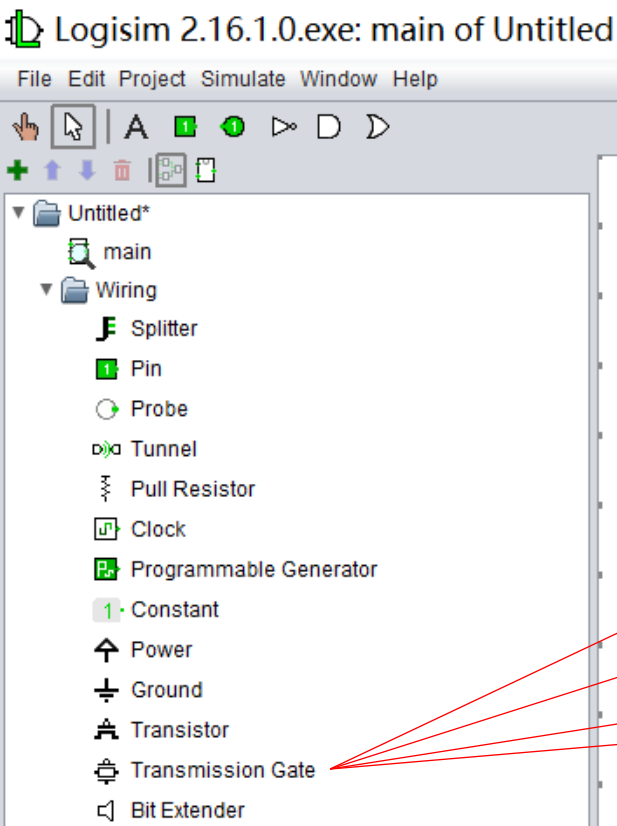


C=1 Y=高阻

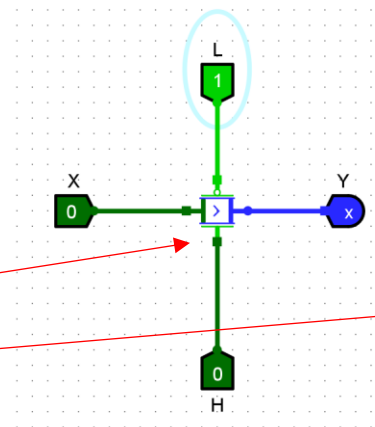
传输门 (Transmission Gate)



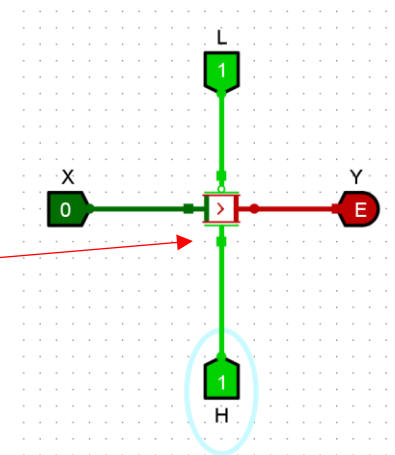
H=0、L=0 Y出错



H=1、 L=0 Y=X

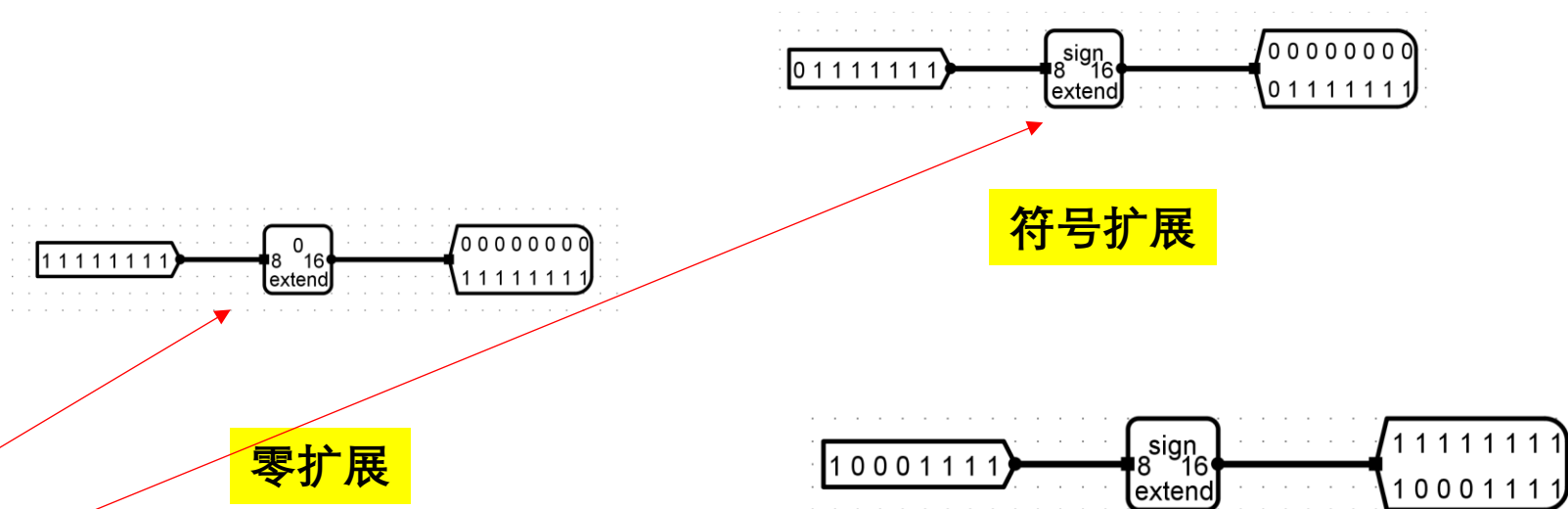
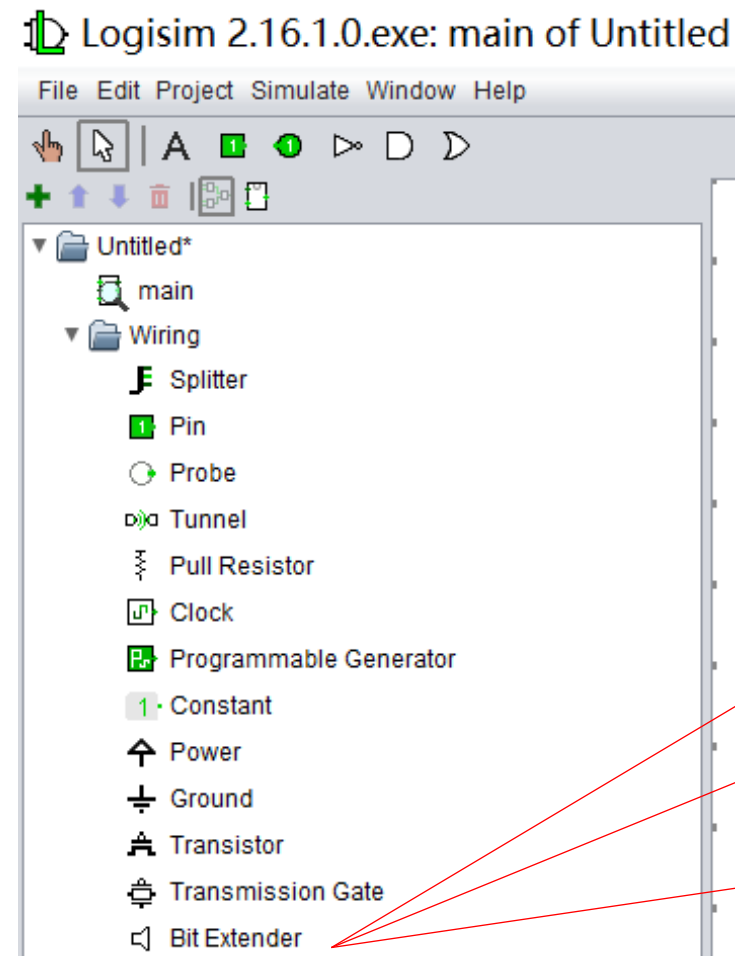


H=0、L=1 Y=高阻

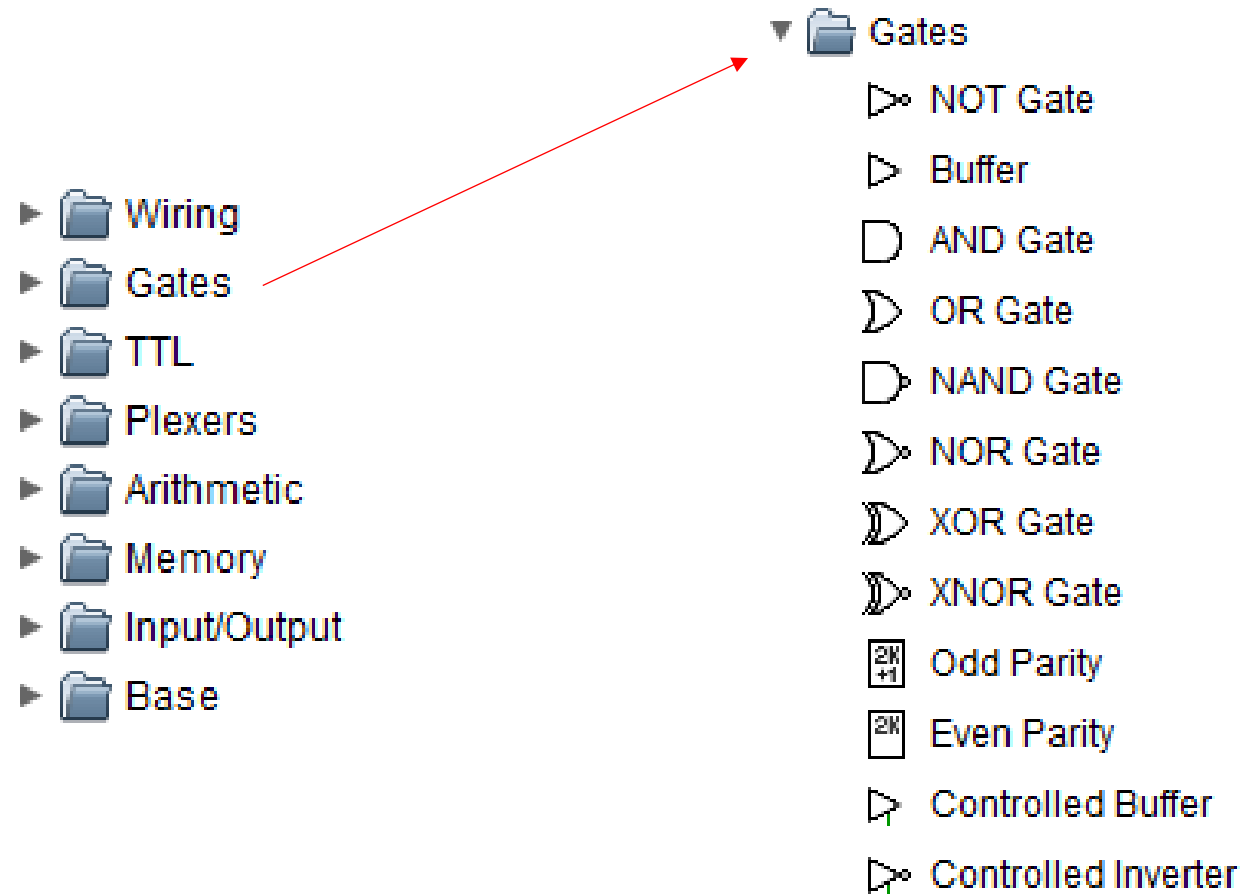


H=1、L=1 Y出错

位扩展器 (Bit Extender)



(二) 门电路 (Gates)



非门 (NOT Gate)

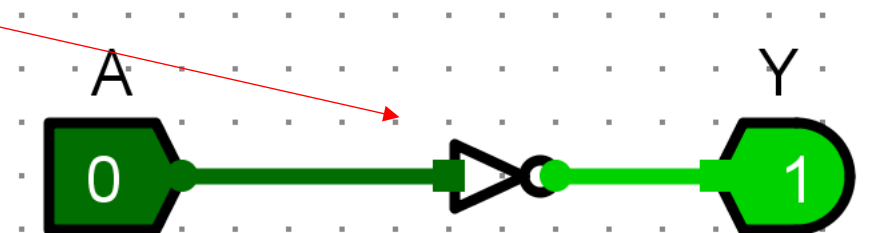
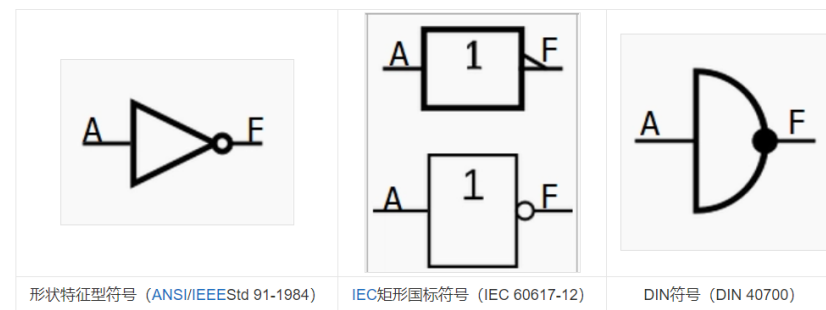
• 非门: $Y = \overline{A}$

▼ Gates

- ▷ NOT Gate
- ▷ Buffer
- ▷ AND Gate
- ▷ OR Gate
- ▷ NAND Gate
- ▷ NOR Gate
- ▷ XOR Gate
- ▷ XNOR Gate
- 2K+1 Odd Parity
- 2K Even Parity
- ▷ Controlled Buffer
- ▷ Controlled Inverter

真值表

输入A	输出Y
0	1
1	0

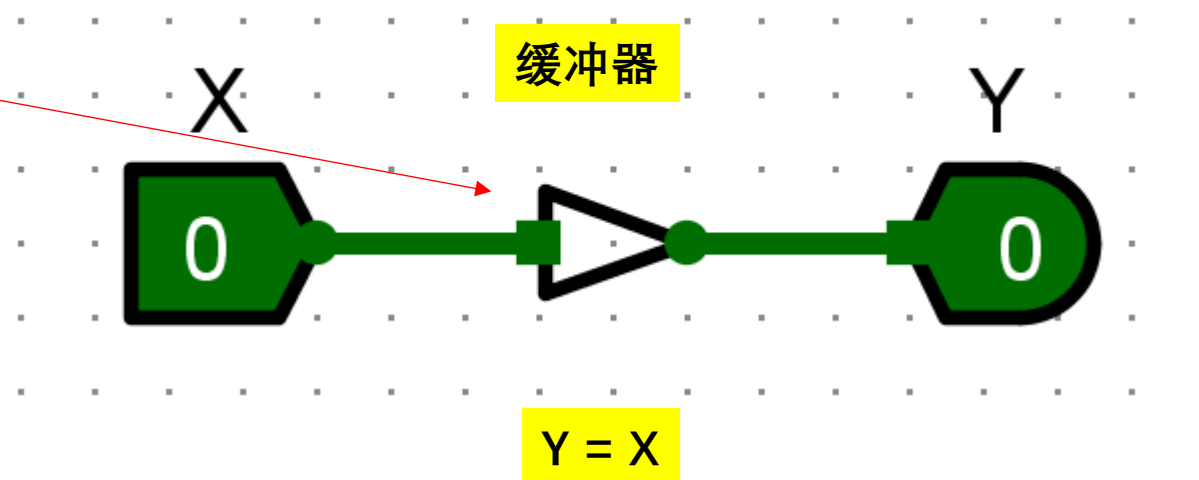


非门

缓冲器 (Buffer)

▼ Gates

- ▷ NOT Gate
- ▷ Buffer
- ▷ AND Gate
- ▷ OR Gate
- ▷ NAND Gate
- ▷ NOR Gate
- ▷ XOR Gate
- ▷ XNOR Gate
- 2K+1 Odd Parity
- 2K Even Parity
- ▷ Controlled Buffer
- ▷ Controlled Inverter

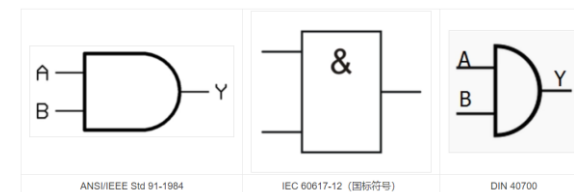


与门 (AND Gate)

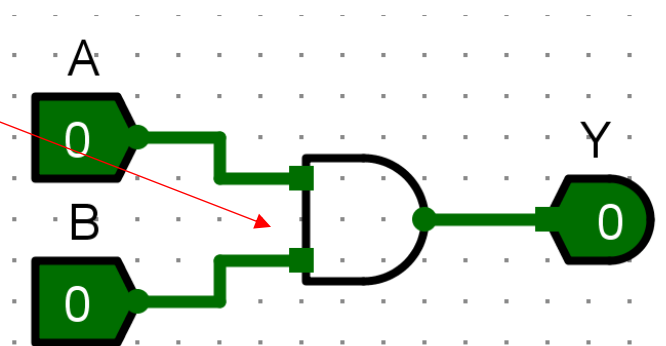
• 与门: $Y=A \cdot B$

真值表

输入A	输入B	输出Y
0	0	0
0	1	0
1	0	0
1	1	1



- ▼ Gates
 - ▷ NOT Gate
 - ▷ Buffer
 - ▷ AND Gate
 - ▷ OR Gate
 - ▷ NAND Gate
 - ▷ NOR Gate
 - ▷ XOR Gate
 - ▷ XNOR Gate
 - [2K +1] Odd Parity
 - [2K] Even Parity
 - ▷ Controlled Buffer
 - ▷ Controlled Inverter



与门

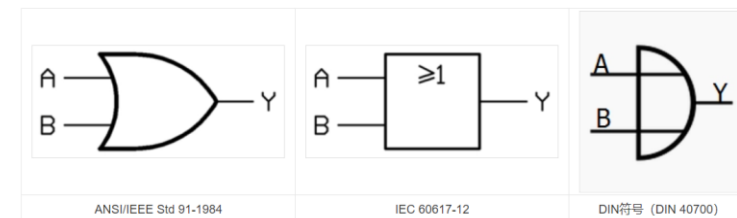
AND Gate

Facing	East
Data Bits	1
Gate Size	Narrow
Number Of Inputs	2
Output Value	0/1
Label	
Label Font	Dialog Plain 12
Label Color	#000000
Negate 1 (Top)	No
Negate 2 (Bottom)	No

与门的属性

或门 (OR Gate)

• 或门: $Y=A+B$

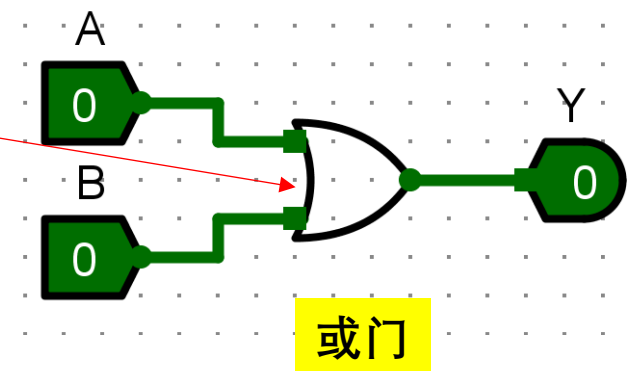


▼ Gates

- ▷ NOT Gate
- ▷ Buffer
- ▷ AND Gate
- ▷ OR Gate
- ▷ NAND Gate
- ▷ NOR Gate
- ▷ XOR Gate
- ▷ XNOR Gate
- 2N+1 Odd Parity
- 2N Even Parity
- ▷ Controlled Buffer
- ▷ Controlled Inverter

真值表

输入A	输入B	输出F
0	0	0
0	1	1
1	0	1
1	1	1

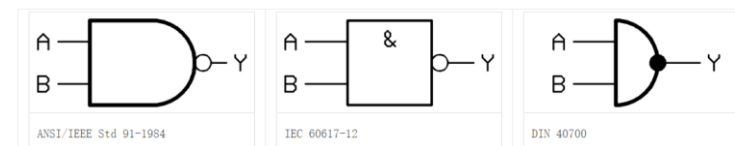


与非门 (NAND Gate)

• 与非门: $Y = \overline{A \cdot B} = \overline{A} + \overline{B}$

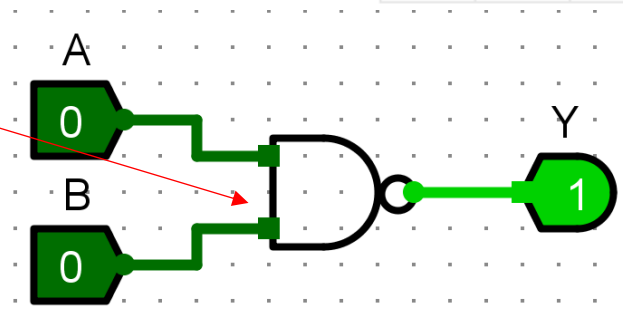
真值表

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

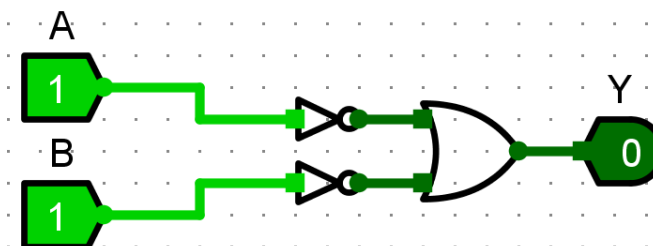


▼ Gates

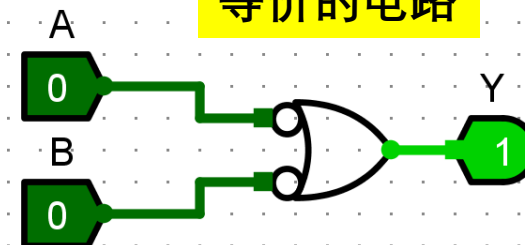
- ▷ NOT Gate
- ▷ Buffer
- ▷ AND Gate
- ▷ OR Gate
- ▷ NAND Gate
- ▷ NOR Gate
- ▷ XOR Gate
- ▷ XNOR Gate
- 2K+1 Odd Parity
- 2K Even Parity
- ▷ Controlled Buffer
- ▷ Controlled Inverter



与非门



等价的电路

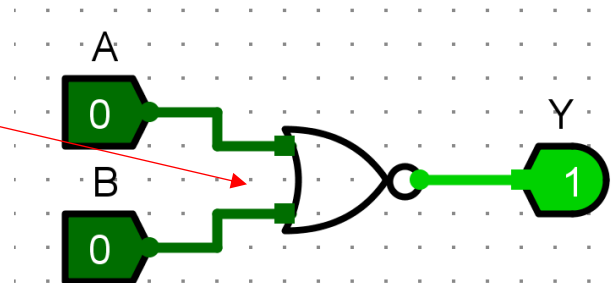
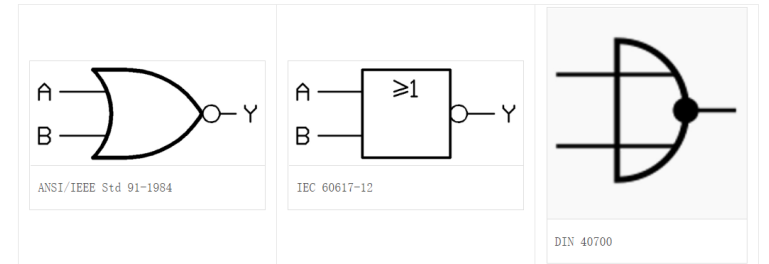


或非门 (NOR Gate)

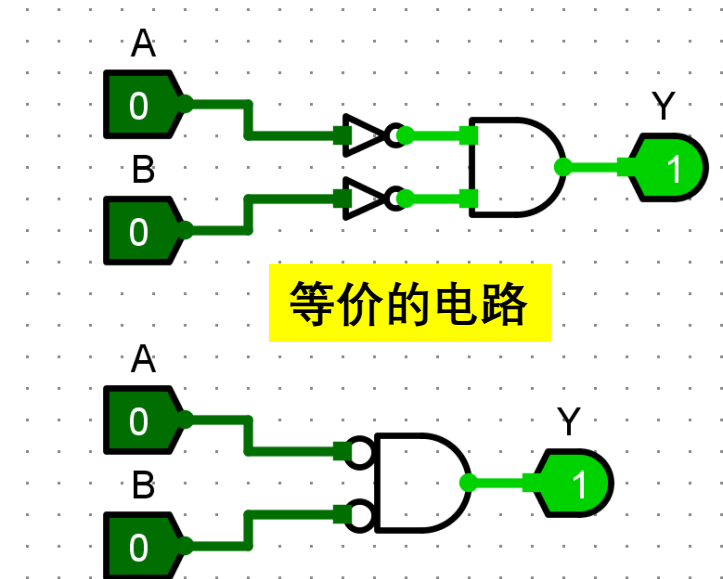
• 或非门: $Y = \overline{A+B} = \overline{A} \cdot \overline{B}$

真值表

输入A	输入B	输出Y
0	0	1
0	1	0
1	0	0
1	1	0



或非门



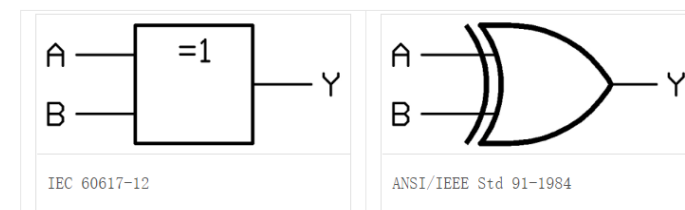
异或门 (XOR Gate)

• 异或门: $Y = A \oplus B = \overline{A} \cdot B + A \cdot \overline{B}$

真值表

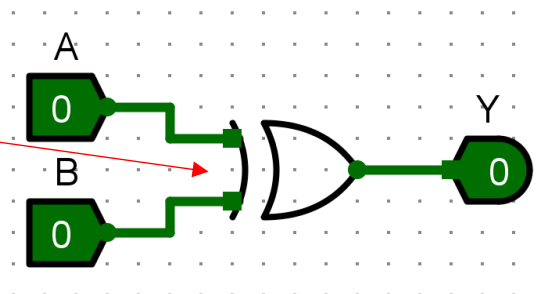
A	B	输出Y
0	0	0
0	1	1
1	0	1
1	1	0

图1

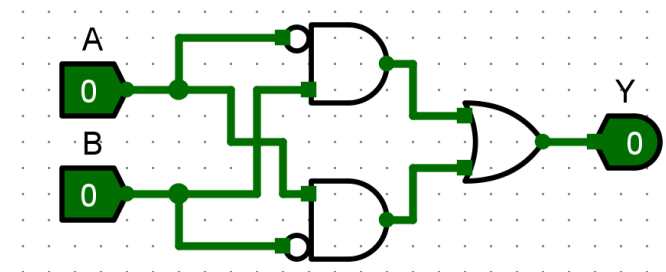


▼ Gates

- ▷ NOT Gate
- ▷ Buffer
- ▷ AND Gate
- ▷ OR Gate
- ▷ NAND Gate
- ▷ NOR Gate
- ▷ XOR Gate
- ▷ XNOR Gate
- 2W+1 Odd Parity
- 2W Even Parity
- ▷ Controlled Buffer
- ▷ Controlled Inverter



异或门



等价的电路

异或非门 (XNOR Gate)

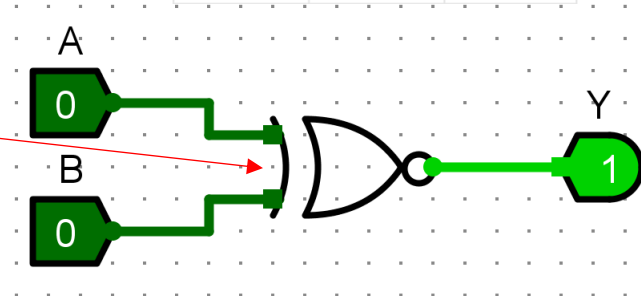
- 异或非门（同或门）： $Y = \overline{A \oplus B} = A \cdot B + \overline{A} \cdot \overline{B}$

▼ Gates

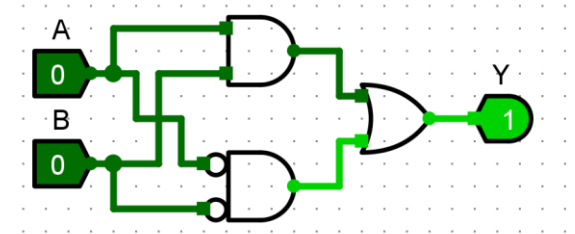
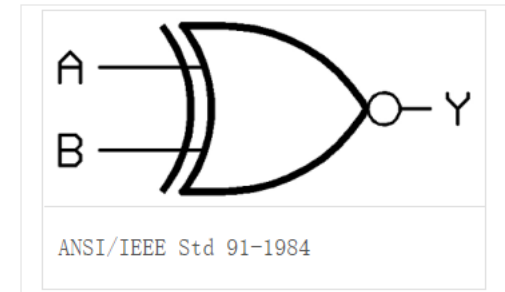
- ▷ NOT Gate
- ▷ Buffer
- ▷ AND Gate
- ▷ OR Gate
- ▷ NAND Gate
- ▷ NOR Gate
- ▷ XOR Gate
- ▷ XNOR Gate
- ▷ Odd Parity
- ▷ Even Parity
- ▷ Controlled Buffer
- ▷ Controlled Inverter

真值表

输入A	输入B	输出F
0	0	1
0	1	0
1	0	0
1	1	1



异或非门（同或门）

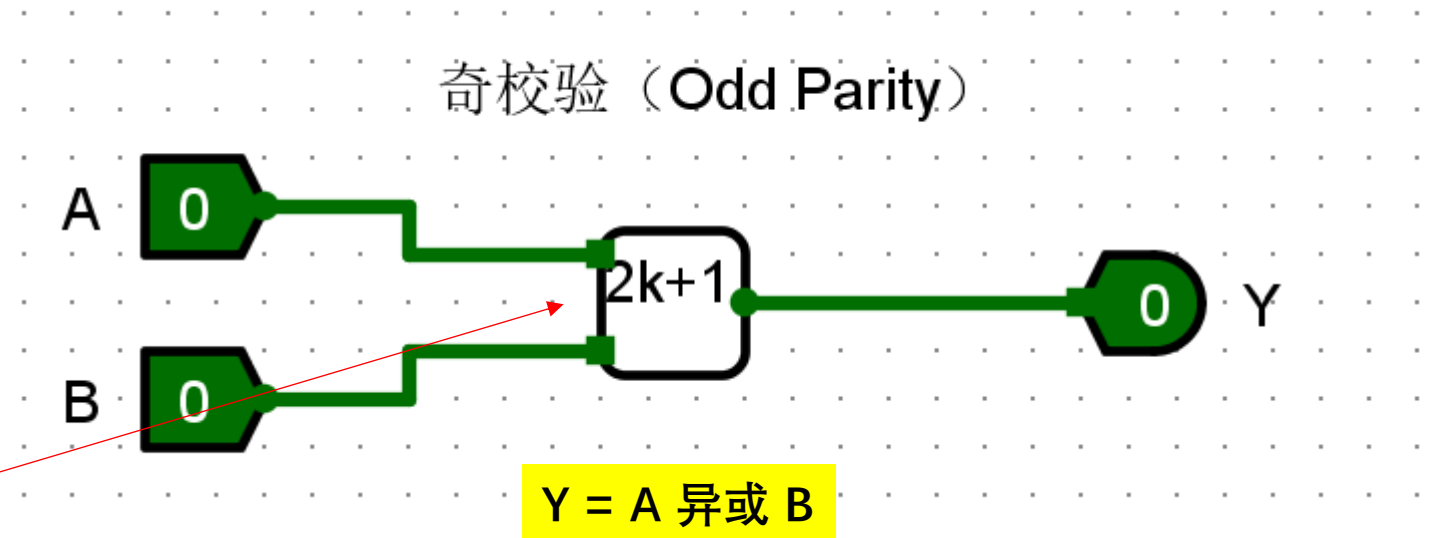


等价的电路

奇校验 (Odd Parity)

▼ Gates

- ▶ NOT Gate
- ▶ Buffer
- ◻ AND Gate
- ◻ OR Gate
- ◻ NAND Gate
- ◻ NOR Gate
- ◻ XOR Gate
- ◻ XNOR Gate
-  Odd Parity
-  Even Parity
- ▶ Controlled Buffer
- ▶ Controlled Inverter



偶校验 (Even Parity)

▼ Gates

▷ NOT Gate

▷ Buffer

▷ AND Gate

▷ OR Gate

▷ NAND Gate

▷ NOR Gate

▷ XOR Gate

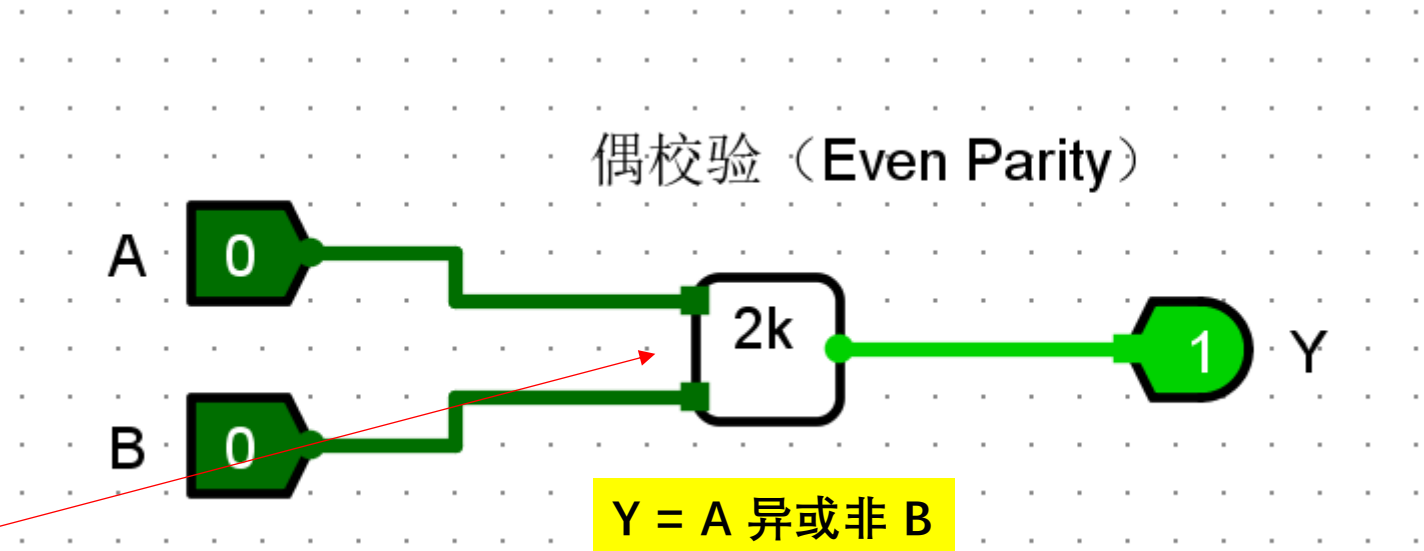
▷ XNOR Gate

 Odd Parity

 Even Parity

▷ Controlled Buffer

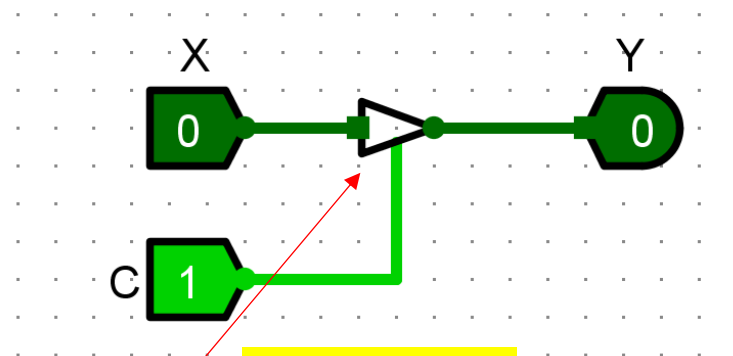
▷ Controlled Inverter



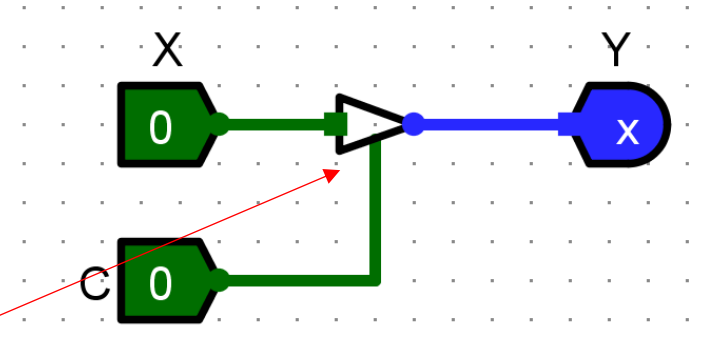
可控缓冲器 (Controlled Buffer)

▼ Gates

- ▶ NOT Gate
- ▶ Buffer
- ▶ AND Gate
- ▶ OR Gate
- ▶ NAND Gate
- ▶ NOR Gate
- ▶ XOR Gate
- ▶ XNOR Gate
- ▶ Odd Parity
- ▶ Even Parity
- ▶ Controlled Buffer
- ▶ Controlled Inverter



C=1 Y=X



C=0 Y=高阻

可控反相器 (Controlled Inverter)

▼ Gates

▷ NOT Gate

▷ Buffer

▷ AND Gate

▷ OR Gate

▷ NAND Gate

▷ NOR Gate

▷ XOR Gate

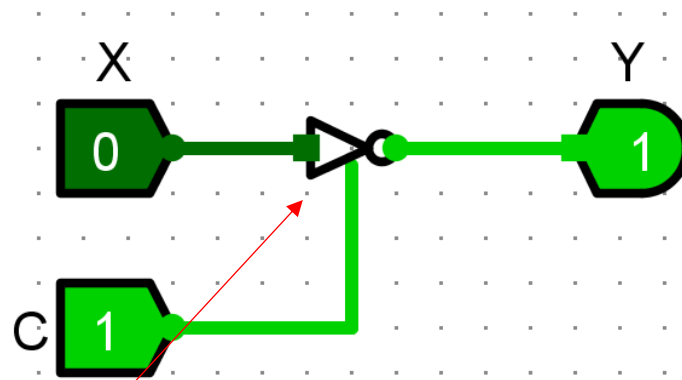
▷ XNOR Gate

▷ Odd Parity

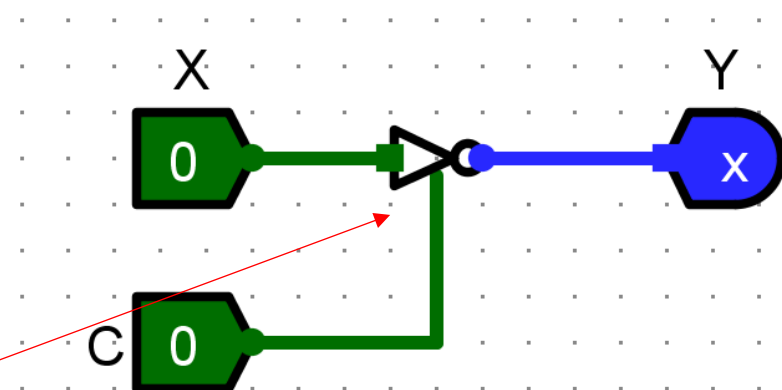
▷ Even Parity

▷ Controlled Buffer

▷ Controlled Inverter

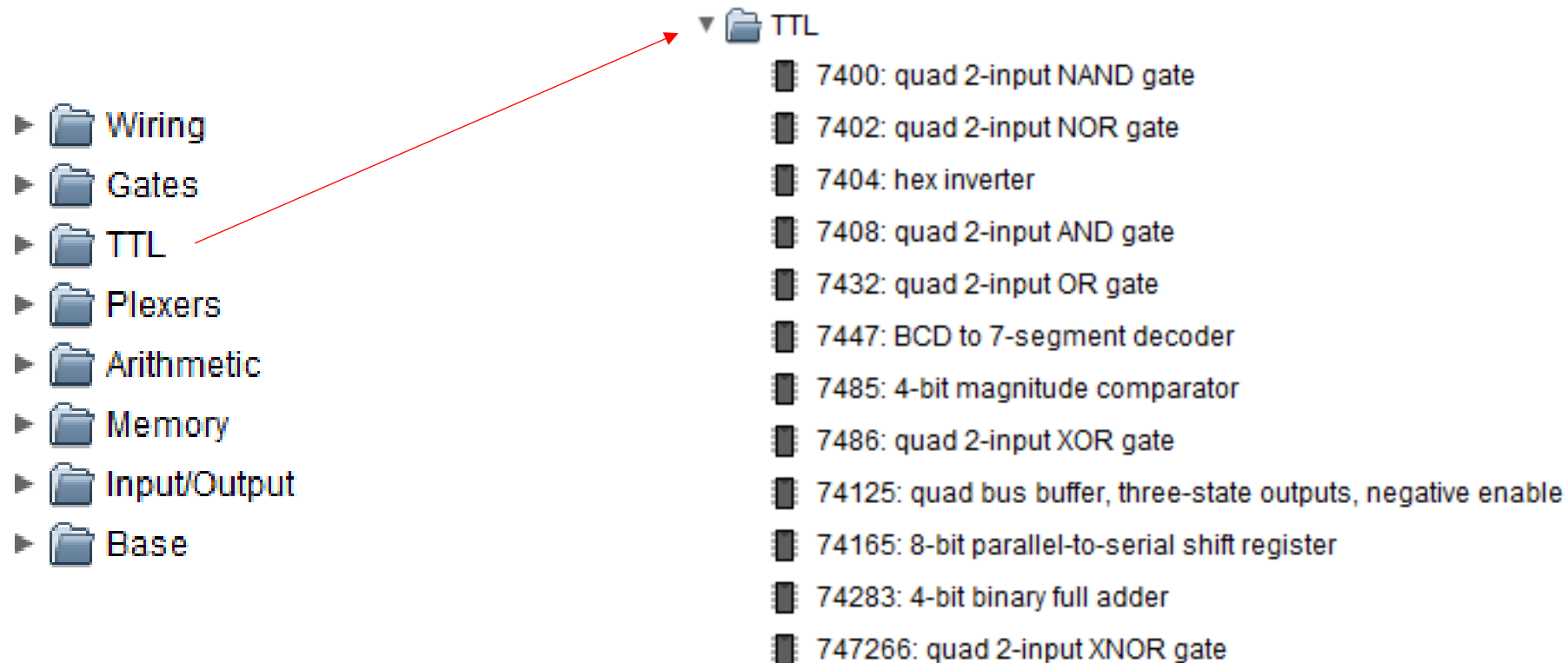


C=1 Y=/X



C=0 Y=高阻

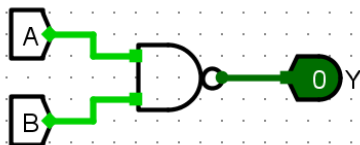
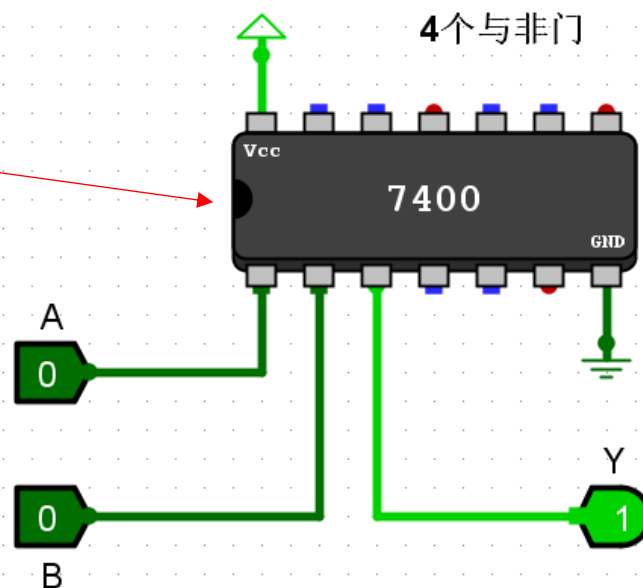
(三) TTL芯片



7400 (4个与非门)

▼ TTL

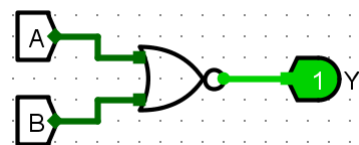
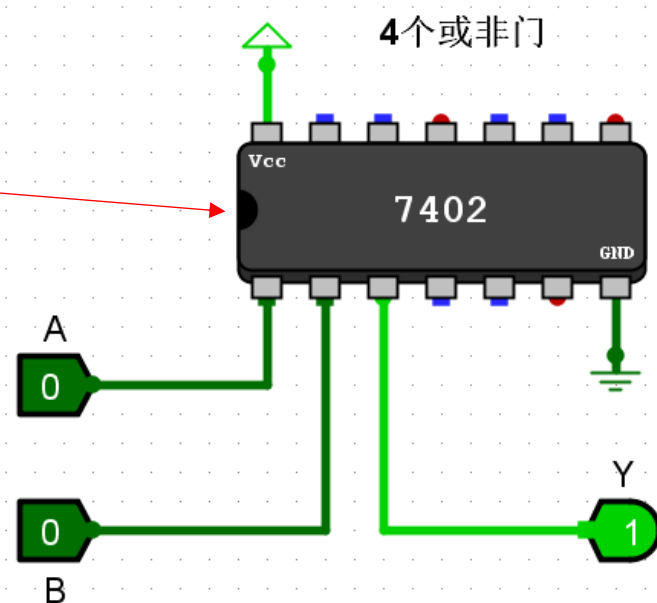
- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate



7402 (4个或非门)

▼ TTL

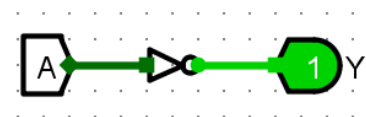
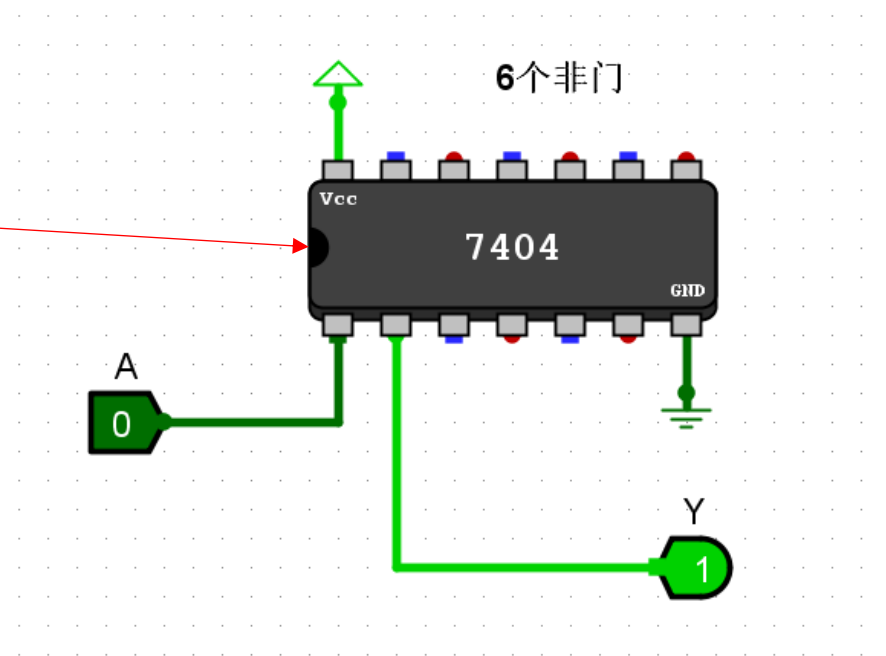
- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate



7404 (6个非门)

▼ TTL

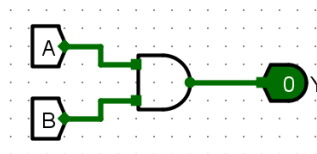
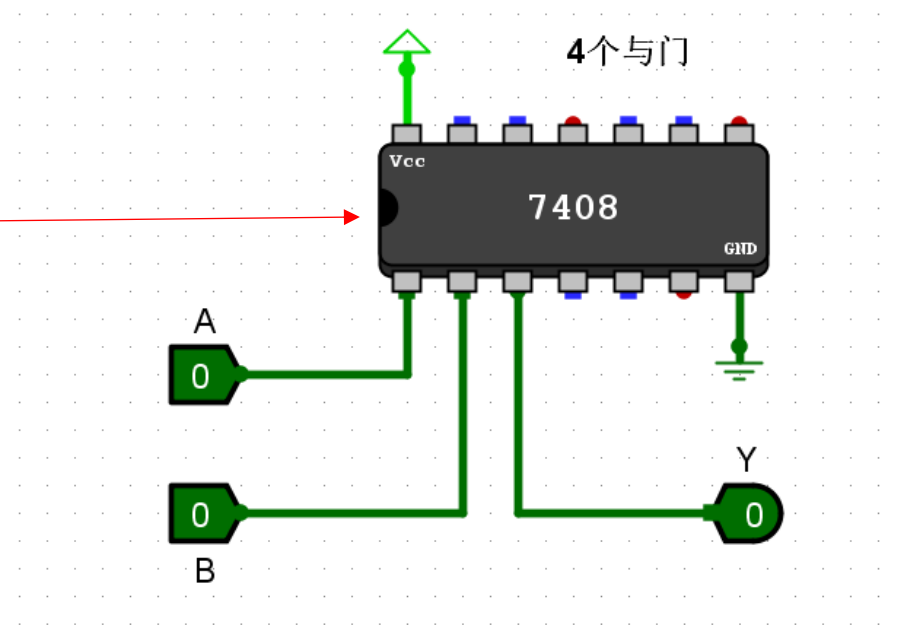
- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate



7408 (4个与门)

▼ TTL

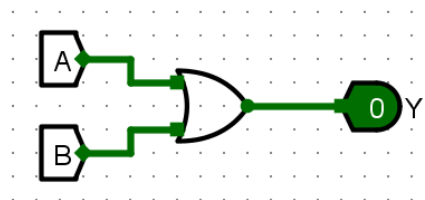
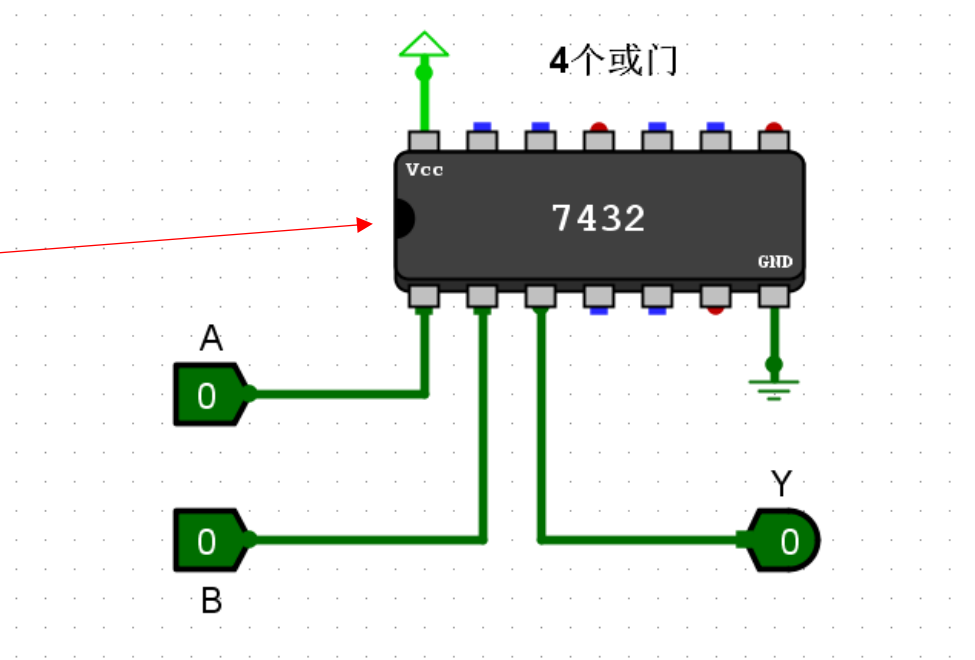
- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate



7432 (4个或门)

▼ TTL

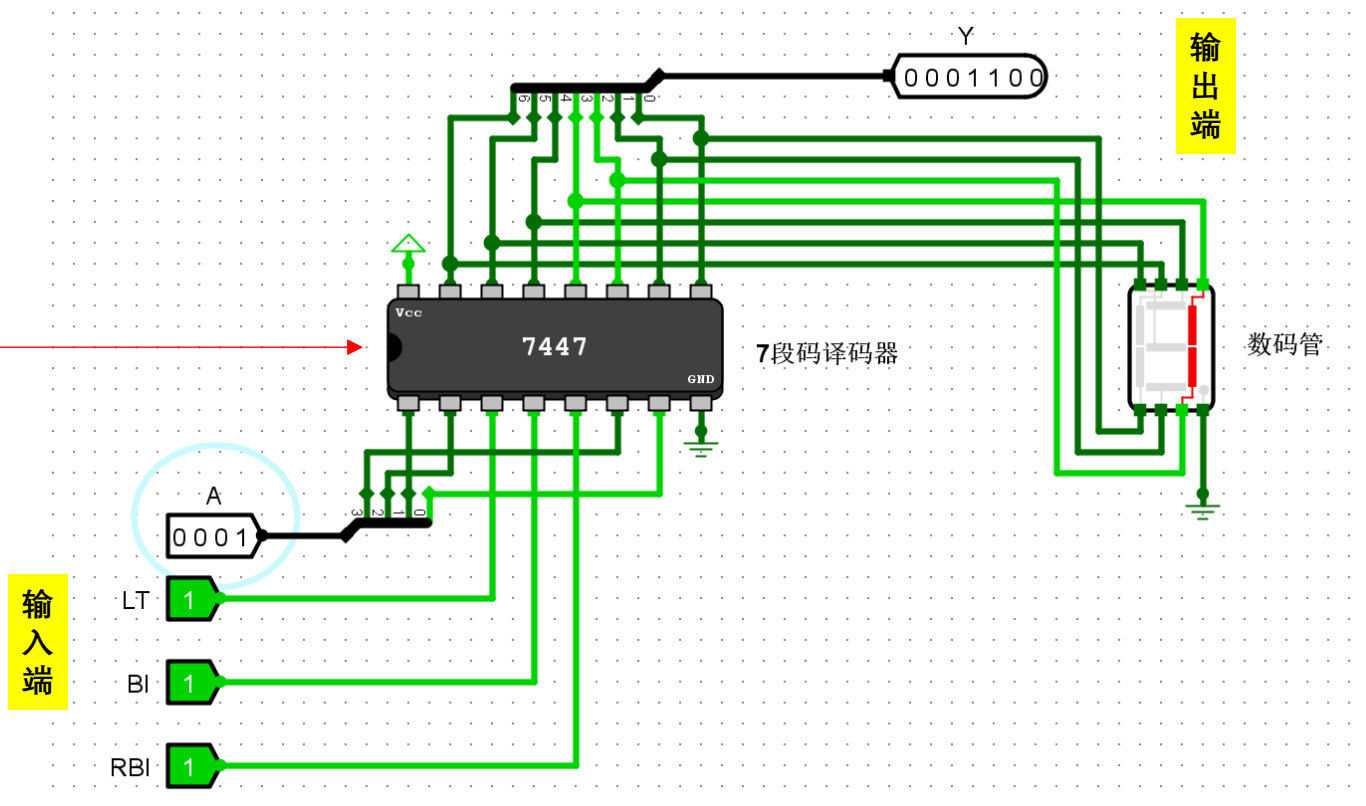
- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate



7447 (7段码译码器)

▼ TTL

- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate

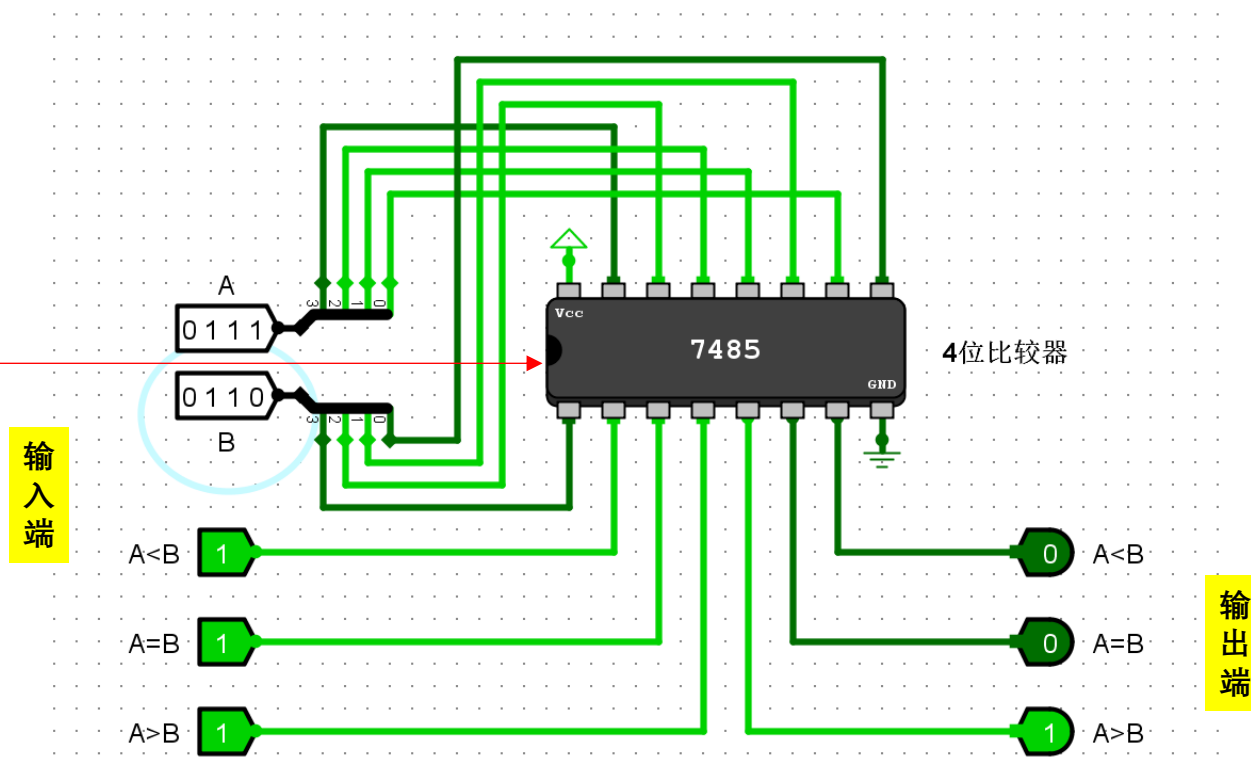


置LT=1, BI=1, RBI=1, A为任意数, 则数码管显示A对应的十六进制数, Y为A对应的七段码

7485 (4位比较器)

▼ TTL

- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate

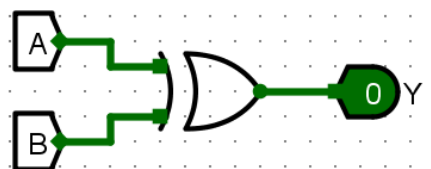
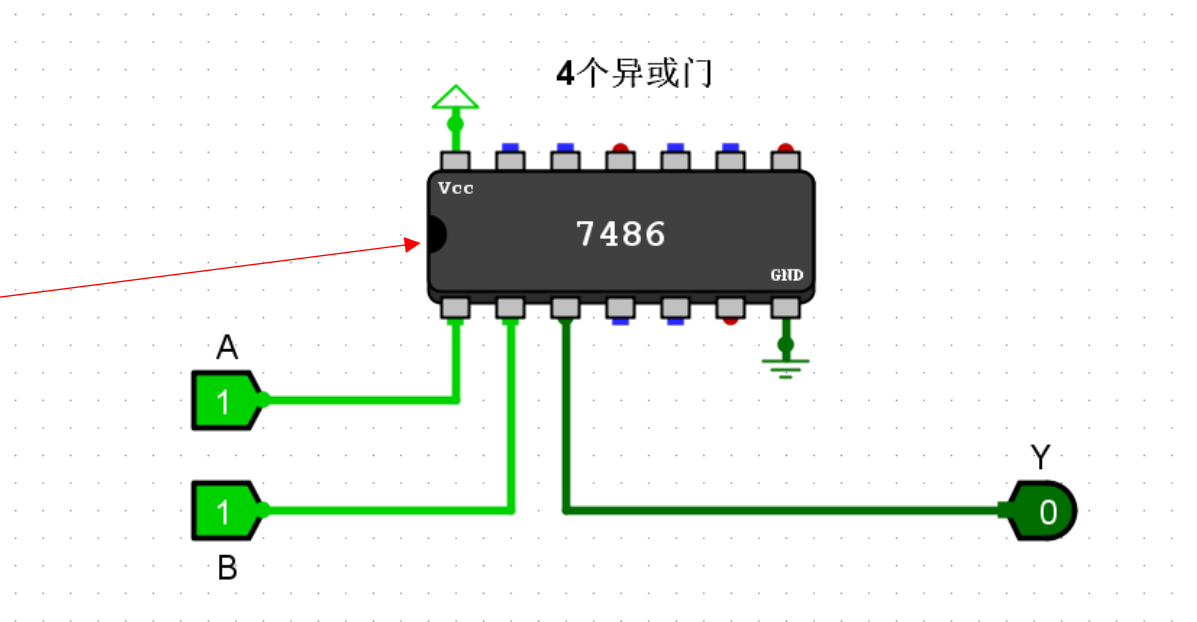


置输入‘A<B’=1、‘A=B’=1、‘A>B’=1，A、B为任意数；如果A<B（无符号数），则输出‘A<B’=1；如果A=B（无符号数），则输出‘A=B’=1；如果A>B（无符号数），则输出‘A>B’=1

7486 (4个异或门)

▼ TTL

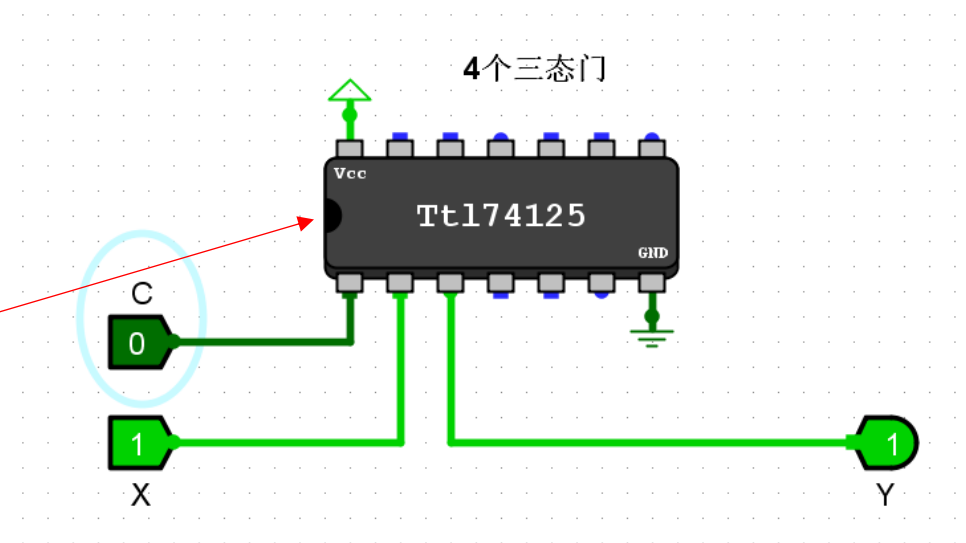
- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate



74125 (4个三态门)

▼ TTL

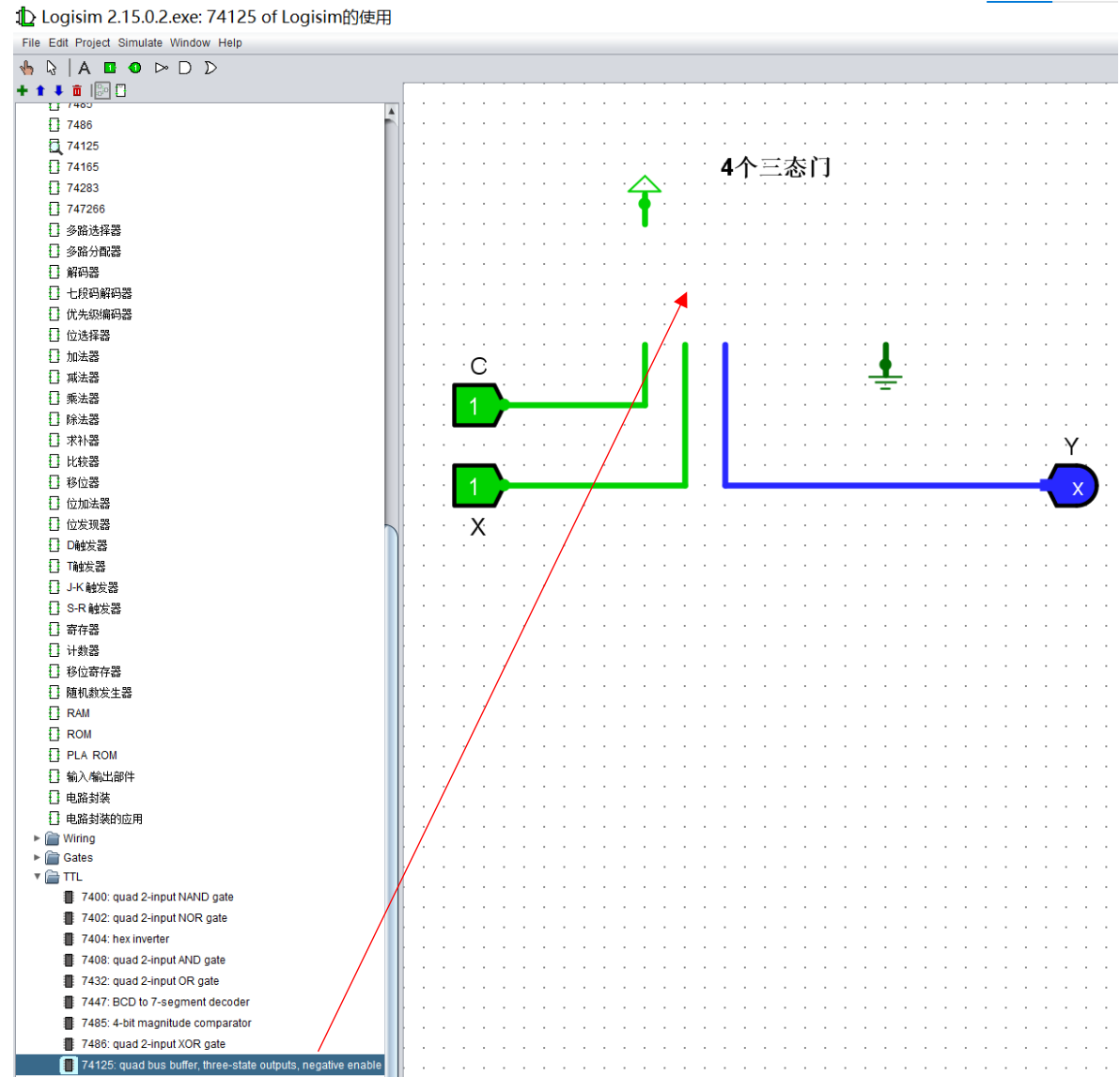
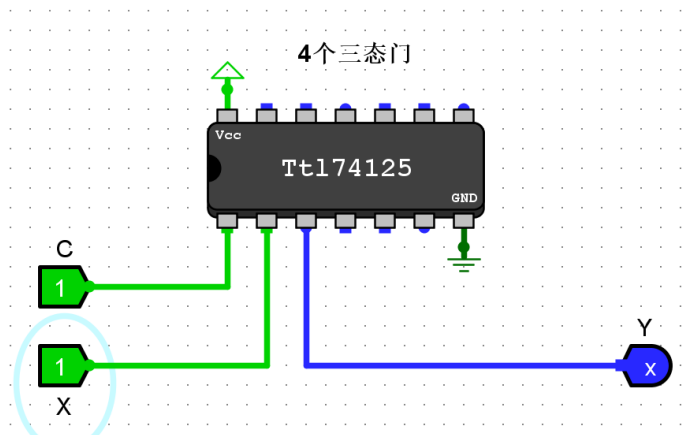
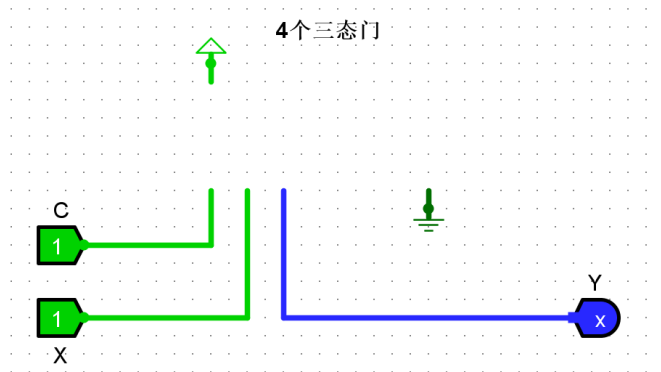
- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate



C=0 Y=X

C=1 Y=高阻

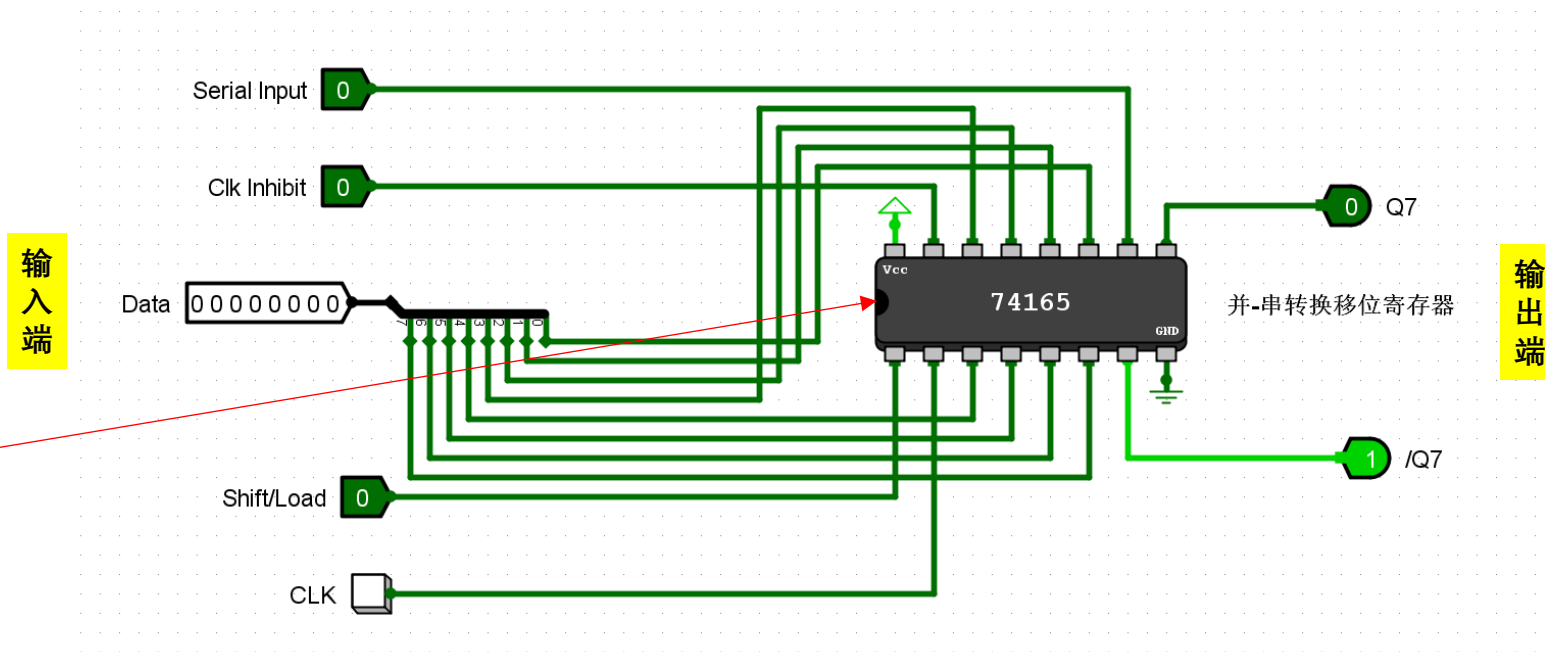
- 如果出现下面的情况，请从TTL器件库中将74125芯片拖到图中。

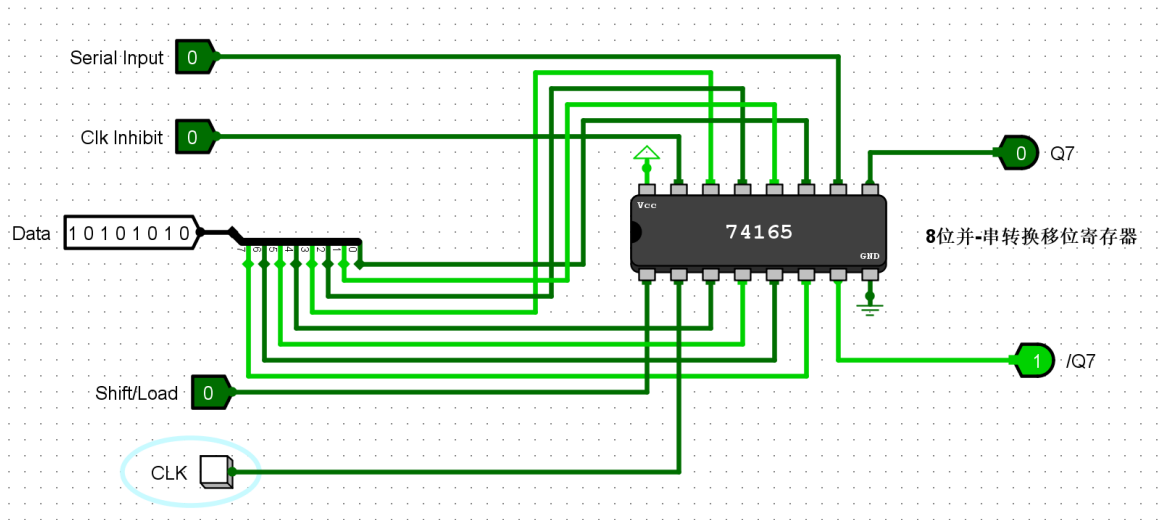


74165 (8位并-串转换移位器)

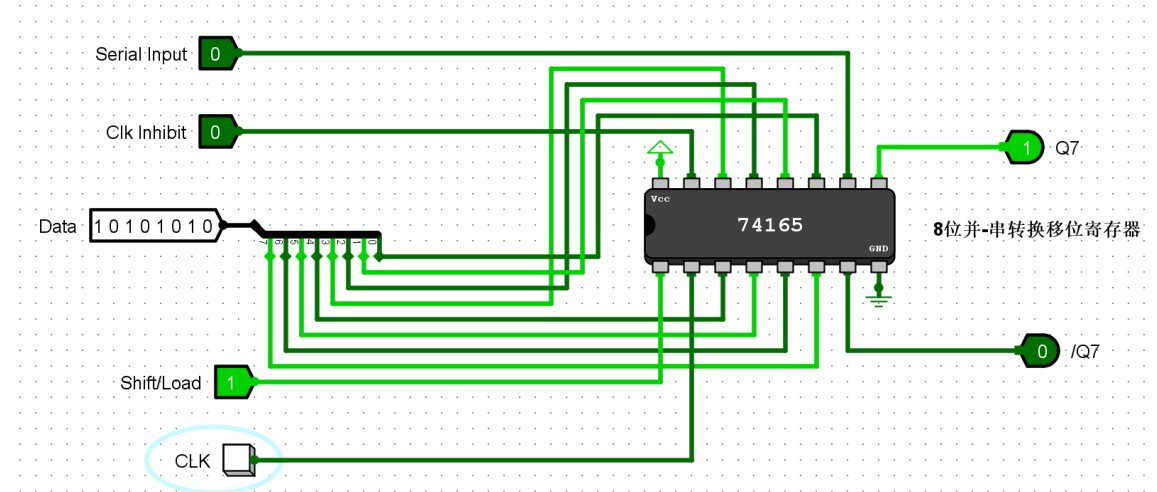
▼ TTL

- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate





首先置Data为任意数，Shift/Load=0，Serial Input=0，Clk Inhibit=0；按CLK，将Data装入74165的寄存器中

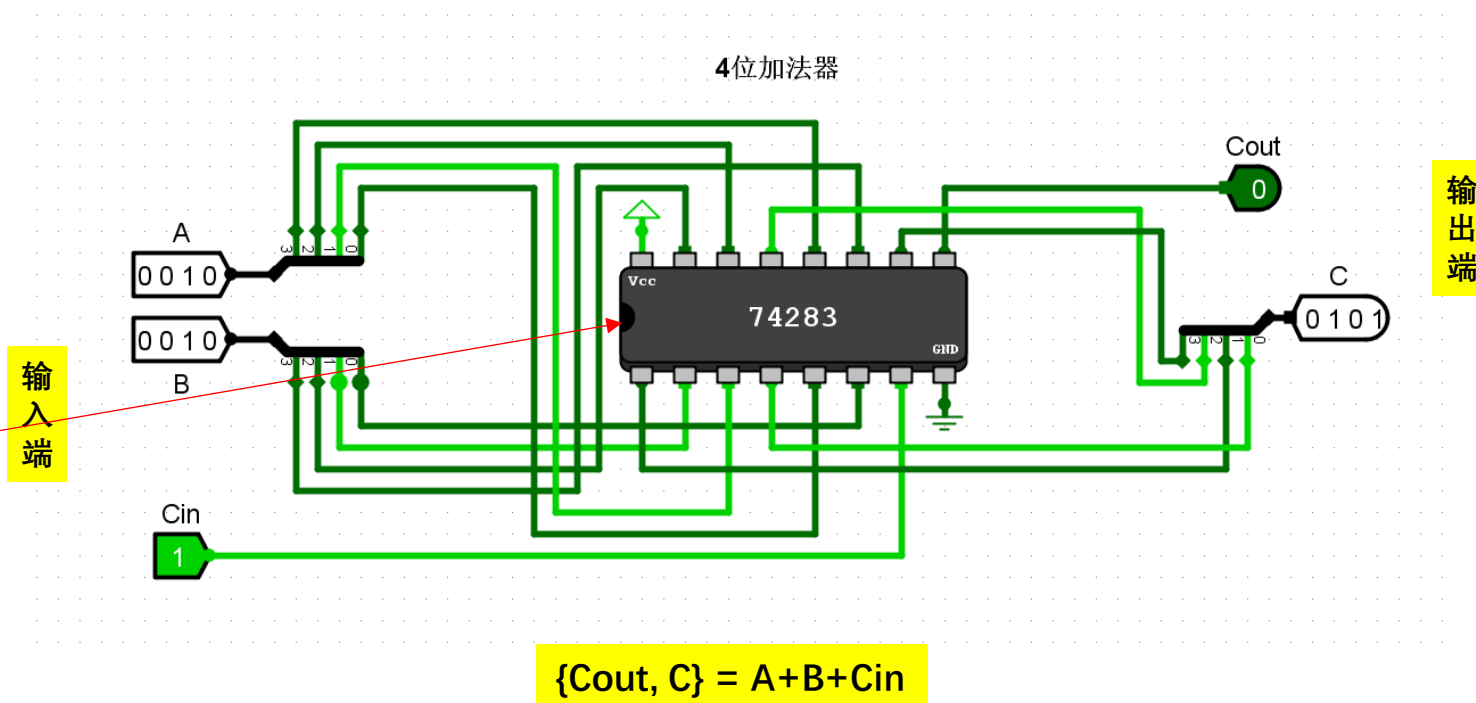


然后置Shift/Load=1，Serial Input=0，Clk Inhibit=0；按CLK多次，则将Data从Q7、/Q7中串行输出

74283 (4位加法器)

▼ TTL

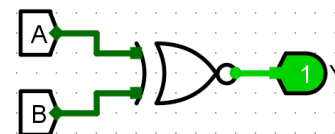
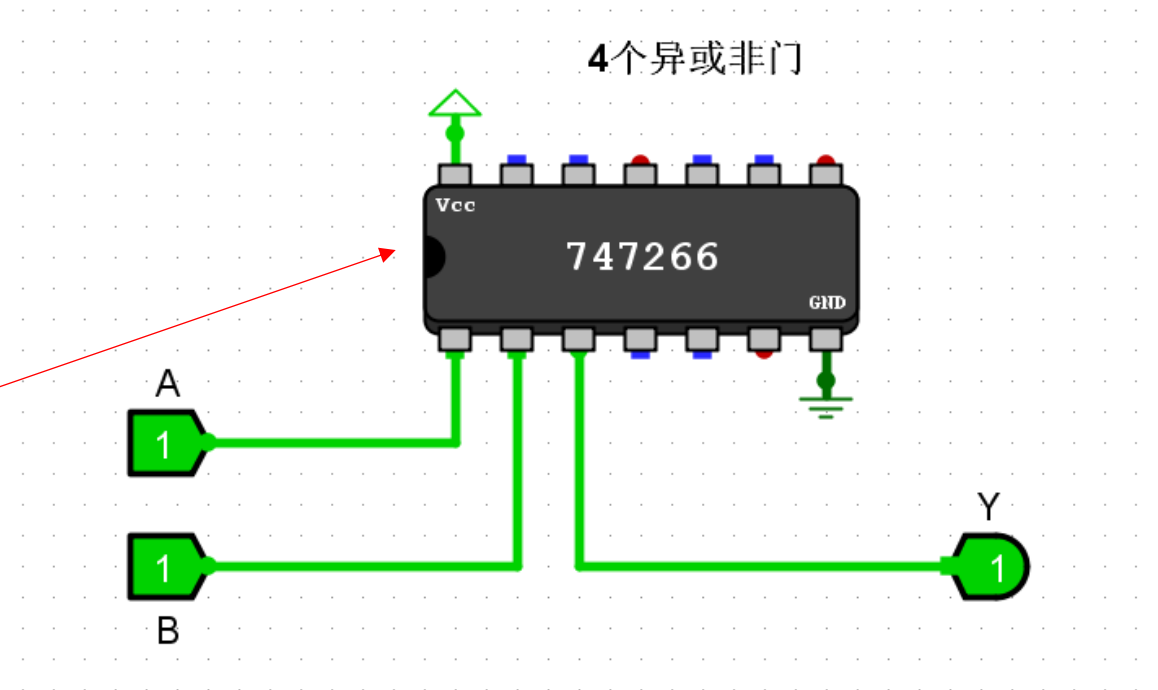
- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 74266: quad 2-input XNOR gate



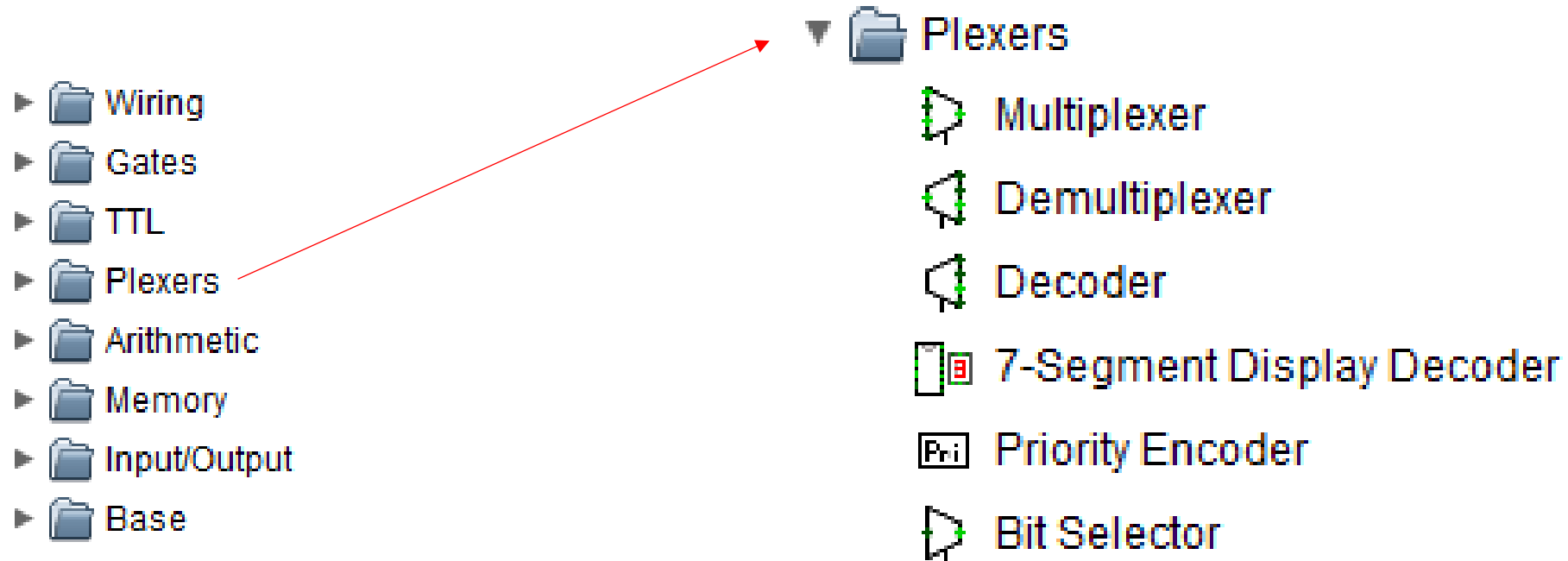
747266 (4个异或非门)

▼ TTL

- 7400: quad 2-input NAND gate
- 7402: quad 2-input NOR gate
- 7404: hex inverter
- 7408: quad 2-input AND gate
- 7432: quad 2-input OR gate
- 7447: BCD to 7-segment decoder
- 7485: 4-bit magnitude comparator
- 7486: quad 2-input XOR gate
- 74125: quad bus buffer, three-state outputs, negative enable
- 74165: 8-bit parallel-to-serial shift register
- 74283: 4-bit binary full adder
- 747266: quad 2-input XNOR gate



(四) 多路复用器 (Plexers)



多路选择器 (Multiplexer)

▼ Plexers

Multiplexer

Demultiplexer

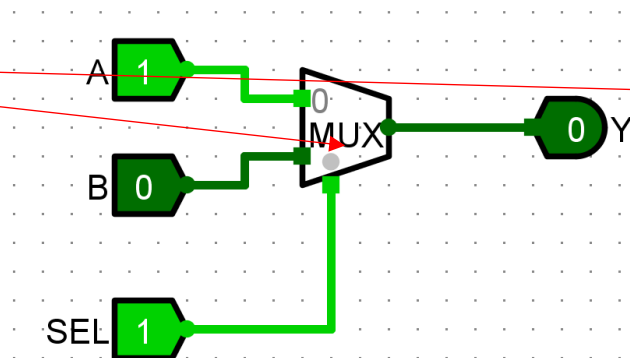
Decoder

7-Segment Display Decoder

Priority Encoder

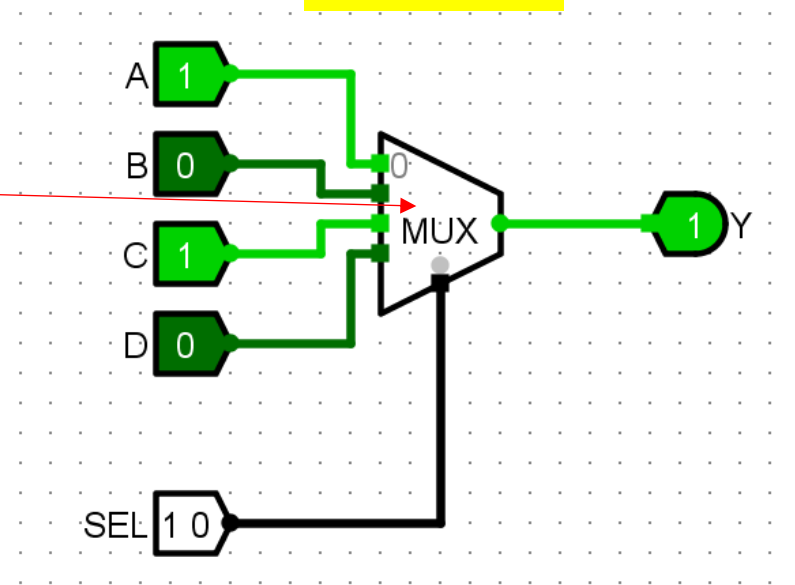
Bit Selector

2路选择器



SEL=0 Y=A
SEL=1 Y=B

4路选择器



SEL=00 Y=A
SEL=01 Y=B
SEL=10 Y=C
SEL=11 Y=D

多路分配器 (Demultiplexer)

▼ Plexers

Multiplexer

Demultiplexer

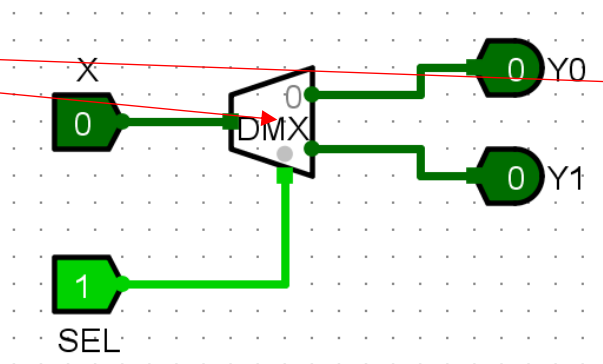
Decoder

7-Segment Display Decoder

Priority Encoder

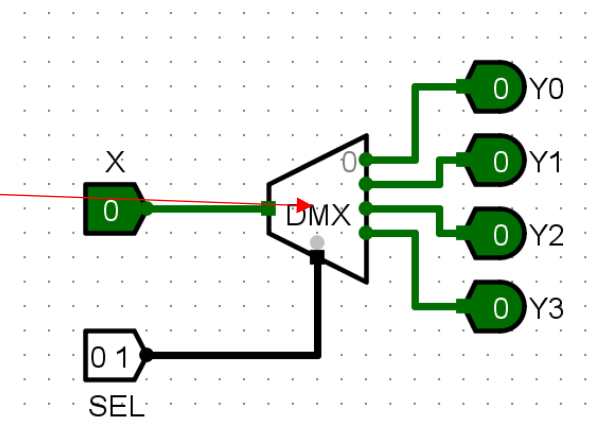
Bit Selector

2路分配器



SEL=0 Y0=X
SEL=1 Y1=X

4路分配器



SEL=00 Y0=X
SEL=01 Y1=X
SEL=10 Y2=X
SEL=11 Y3=X

解码器 (Decoder)

▼ Plexers

Multiplexer

Demultiplexer

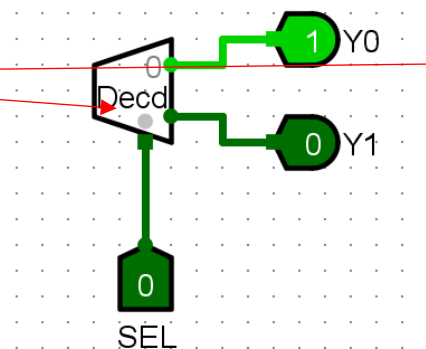
Decoder

7-Segment Display Decoder

Priority Encoder

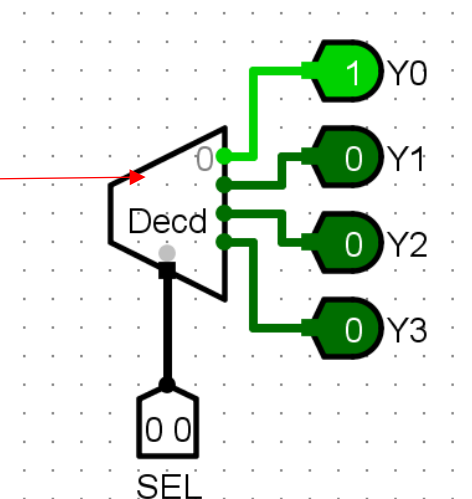
Bit Selector

1位解码器



SEL=0	Y0=1	Y1=0
SEL=1	Y0=0	Y1=1

2位解码器



SEL=00	Y0=1	Y1=0	Y2=0	Y3=0
SEL=01	Y0=0	Y1=1	Y2=0	Y3=0
SEL=10	Y0=0	Y1=0	Y2=1	Y3=0
SEL=11	Y0=0	Y1=0	Y2=0	Y3=1

七段码解码器 (7-Segment Display Decoder)

▼ Plexers

Multiplexer

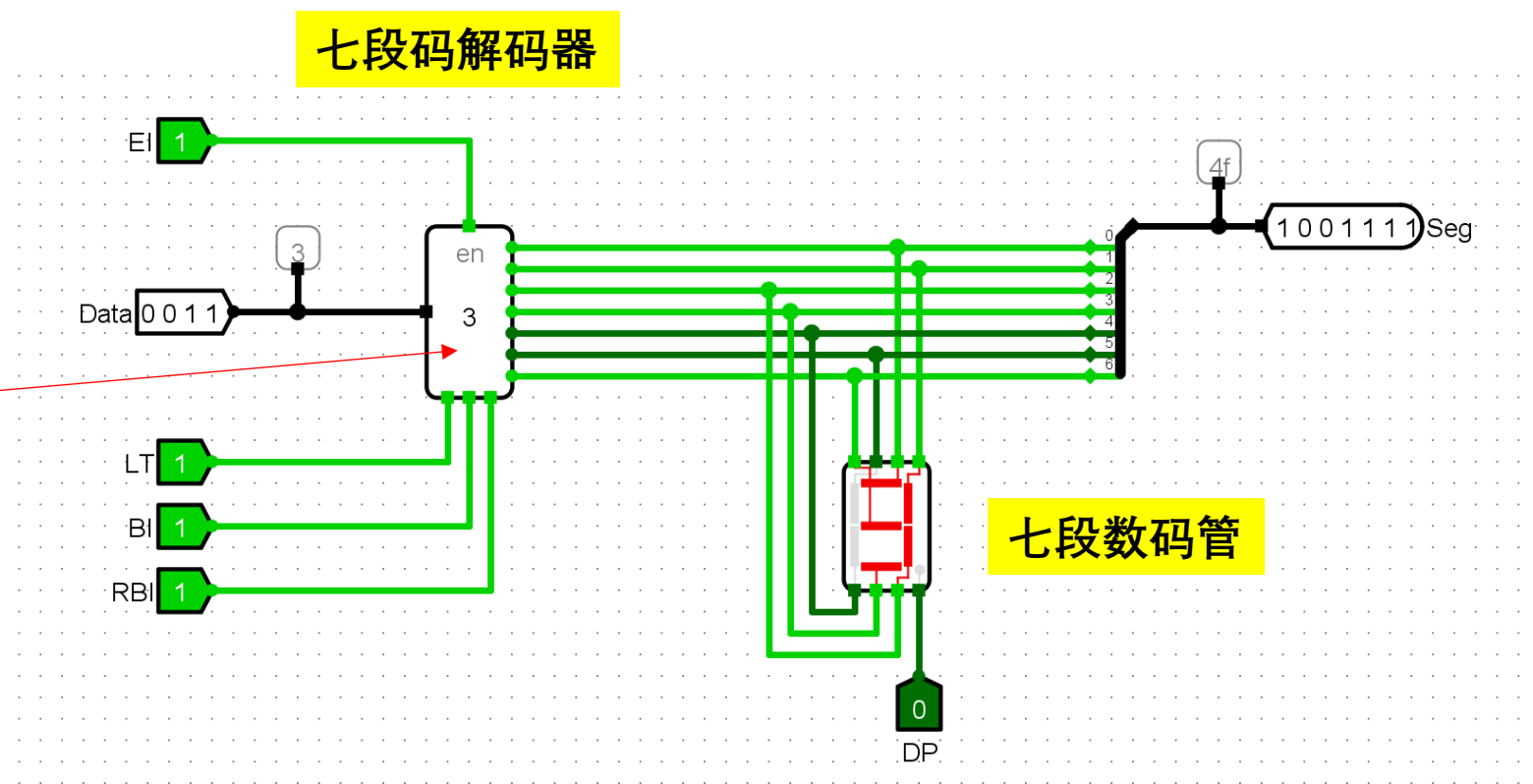
Demultiplexer

Decoder

7-Segment Display Decoder

Priority Encoder

Bit Selector



优先级编码器 (Priority Encoder)

▼ Plexers

Multiplexer

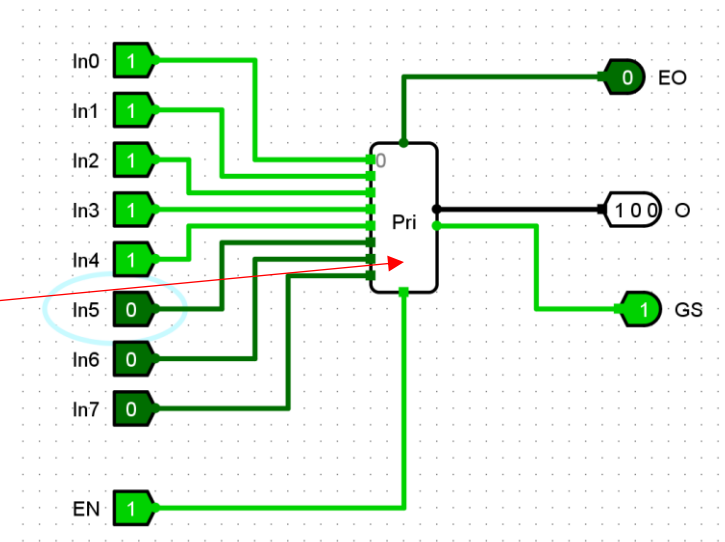
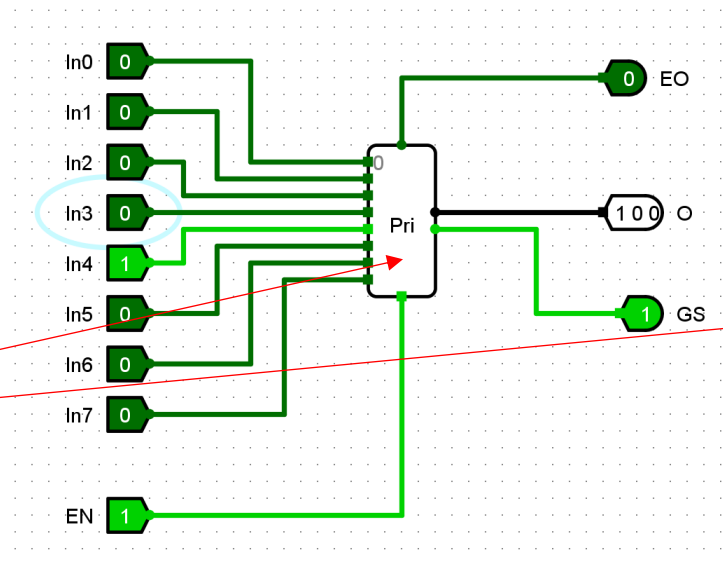
Demultiplexer

Decoder

7-Segment Display Decoder

Priority Encoder

Bit Selector



输出O为输入中最高位1的位置

位选择器 (Bit Selector)

▼ Plexers

Multiplexer

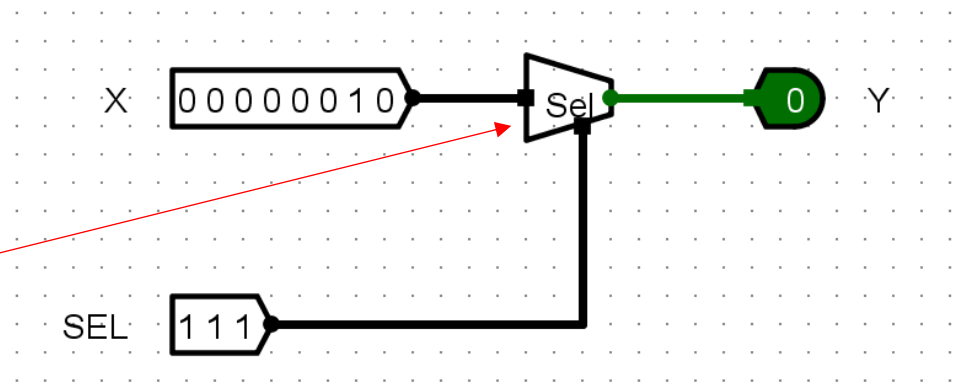
Demultiplexer

Decoder

7-Segment Display Decoder

Priority Encoder

Bit Selector



$X = X_7 X_6 X_5 X_4 X_3 X_2 X_1 X_0$

SEL=000 $Y = X_0$

SEL=001 $Y = X_1$

SEL=010 $Y = X_2$

SEL=011 $Y = X_3$

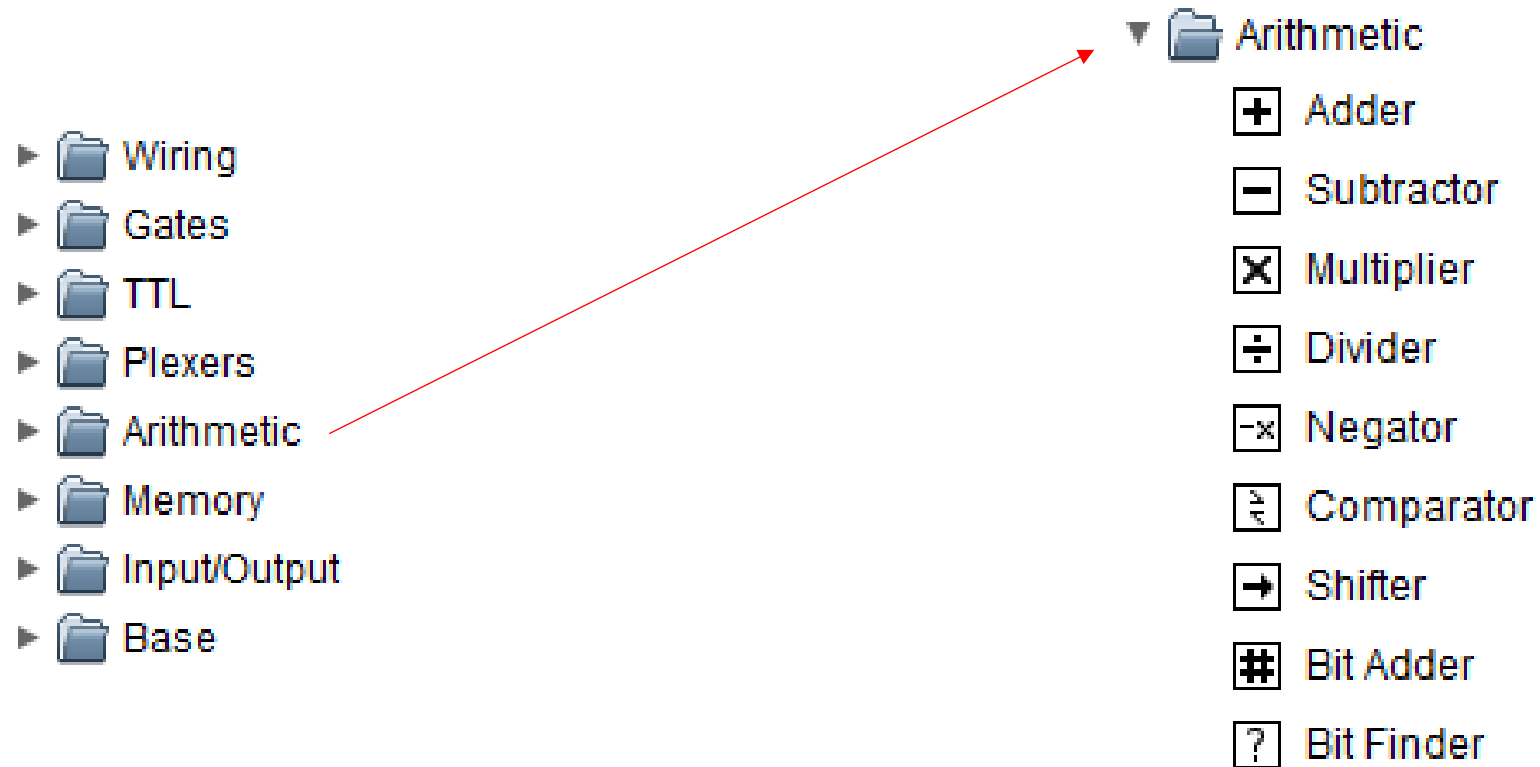
SEL=100 $Y = X_4$

SEL=101 $Y = X_5$

SEL=110 $Y = X_6$

SEL=111 $Y = X_7$

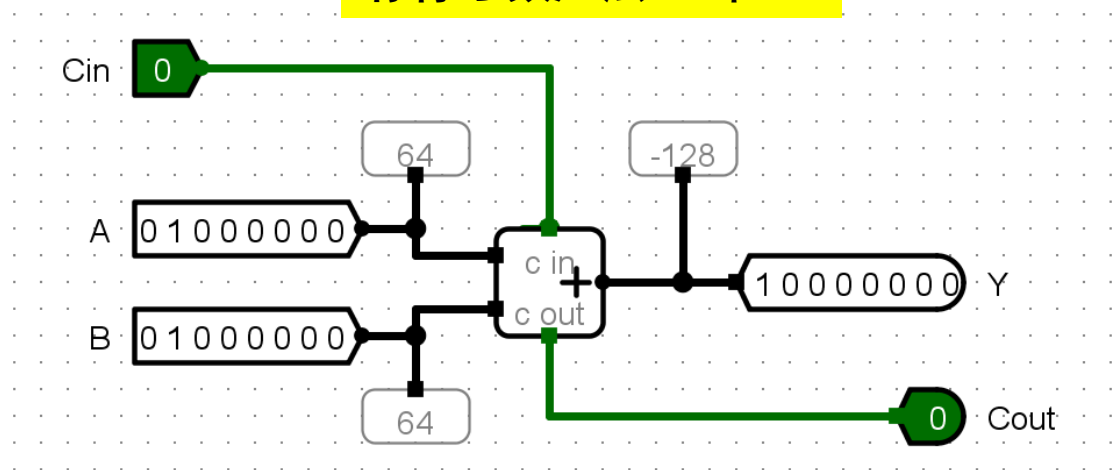
(五) 算术运算部件 (Arithmetic)



- Arithmetic
 - Adder**
 - Subtractor
 - Multiplier
 - Divider
 - Negator
 - Comparator
 - Shifter
 - Bit Adder
 - Bit Finder

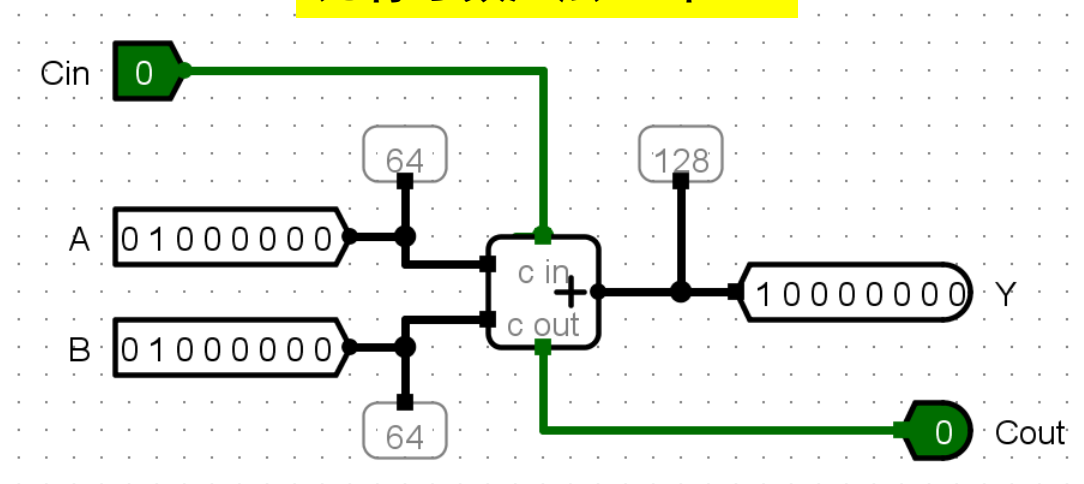
加法器 (Adder)

有符号数加法 (8位)



$$\{ Cout, Y \} = A + B + Cin$$

无符号数加法 (8位)

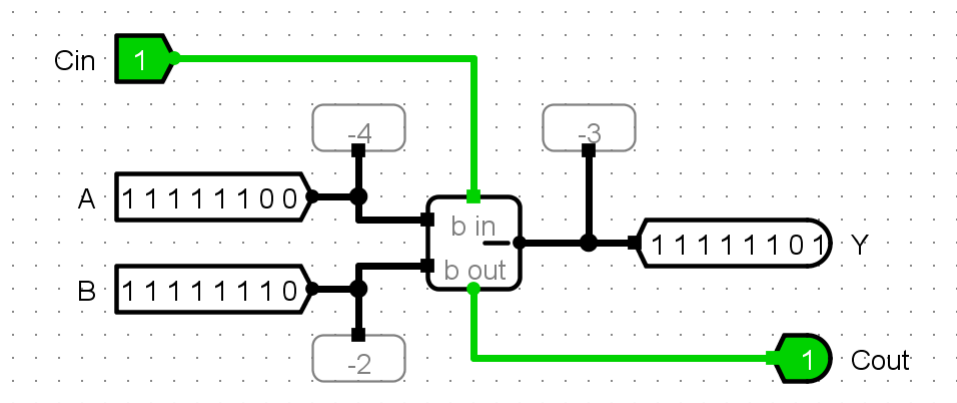


$$\{ Cout, Y \} = A + B + Cin$$

- Arithmetic
 - Adder
 - Subtractor 
 - Multiplier
 - Divider
 - Negator
 - Comparator
 - Shifter
 - Bit Adder
 - Bit Finder

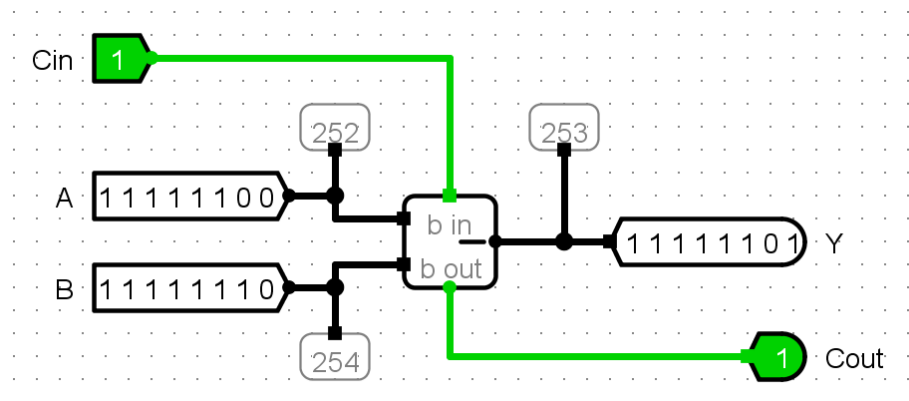
减法器 (Subtractor)

有符号数减法 (8位)



$$\{ Cout, Y \} = A - B - Cin$$

无符号数减法 (8位)

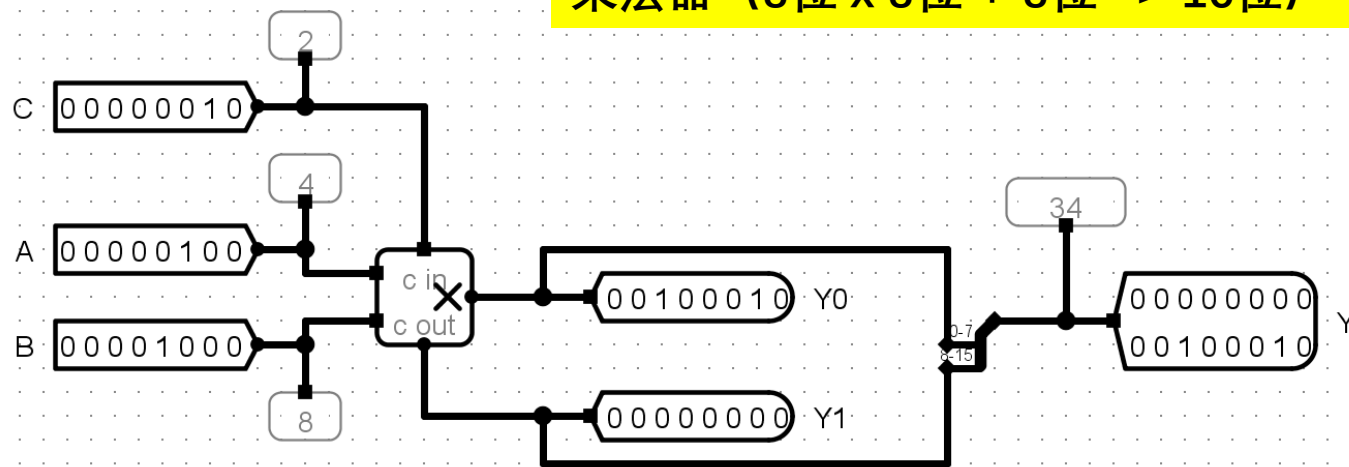


$$\{ Cout, Y \} = A - B - Cin$$

- Arithmetic
 - Adder
 - Subtractor
 - Multiplier**
 - Divider
 - Negator
 - Comparator
 - Shifter
 - Bit Adder
 - Bit Finder

乘法器 (Multiplier)

乘法器 (8位 x 8位 + 8位 -> 16位)

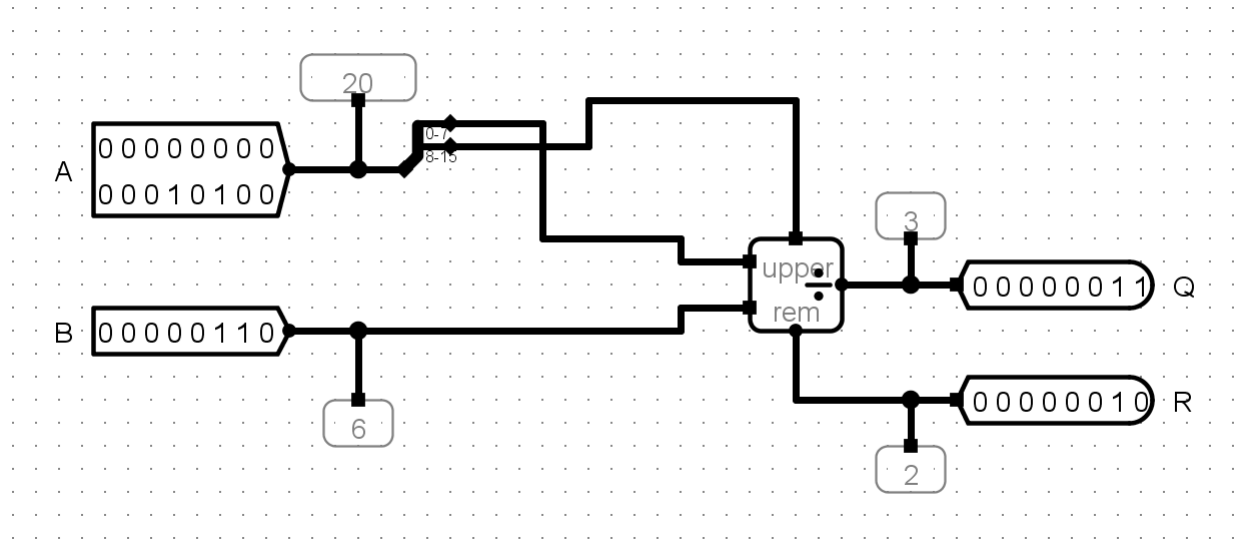


$$Y = \{ Y1, Y0 \} = \{ A \times B + C \}$$

- Arithmetic
 - Adder
 - Subtractor
 - Multiplier
 - Divider
 - Negator
 - Comparator
 - Shifter
 - Bit Adder
 - Bit Finder

除法器 (Divider)

除法器 (16位 ÷ 8位 商Q=8位 余数R=8位)



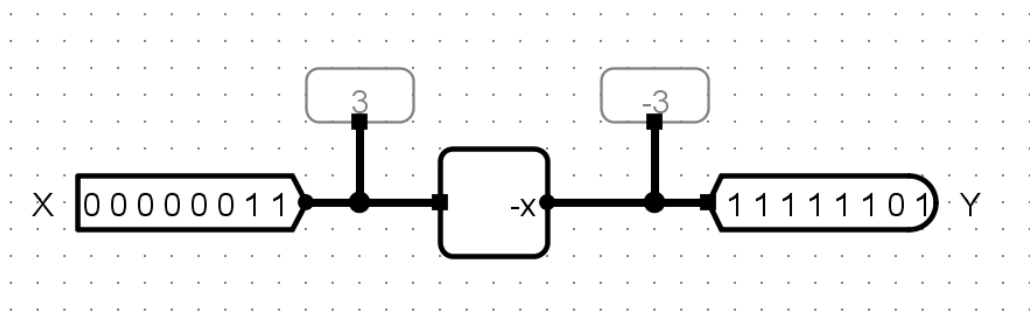
$$\{ Q, R \} = A \div B$$

- Arithmetic
 - Adder
 - Subtractor
 - Multiplier
 - Divider
 - Negator ←
 - Comparator
 - Shifter
 - Bit Adder
 - Bit Finder

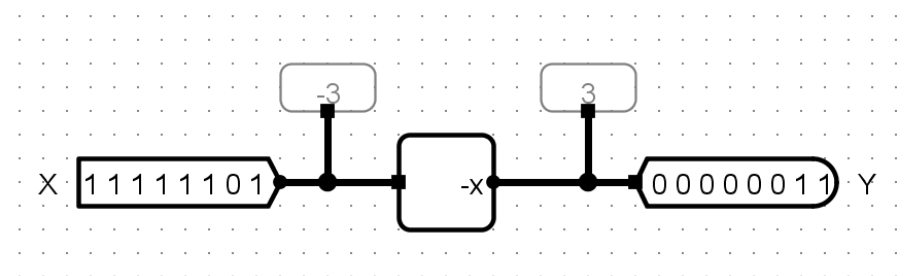
求补器 (Negator)

$Y = X$ 的补数 (不是补码)

$Y = X$ 的模 - X



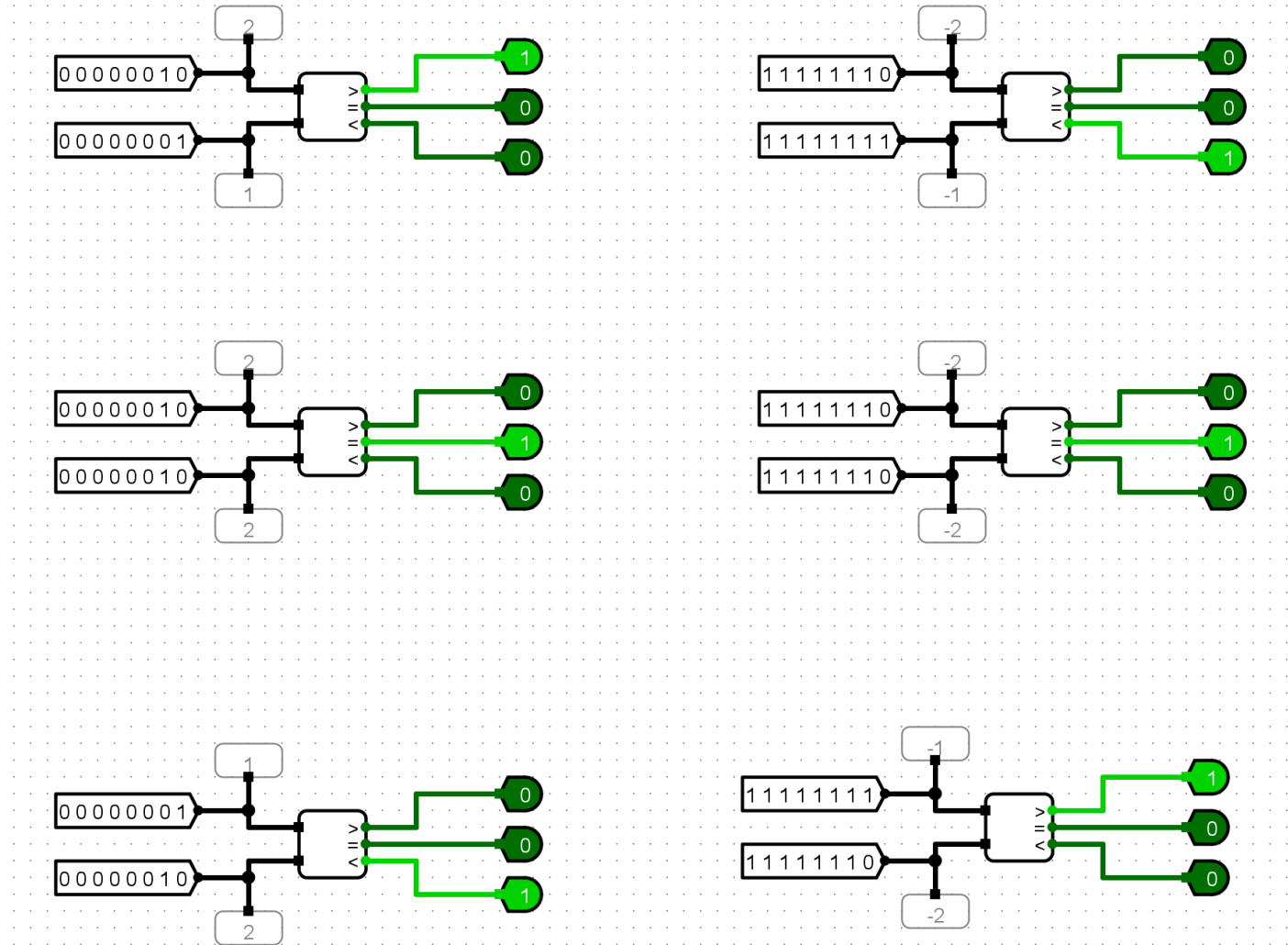
$Y = X$ 的模 - $X = 2^8 - X$



$Y = X$ 的模 - $X = 2^8 - X$

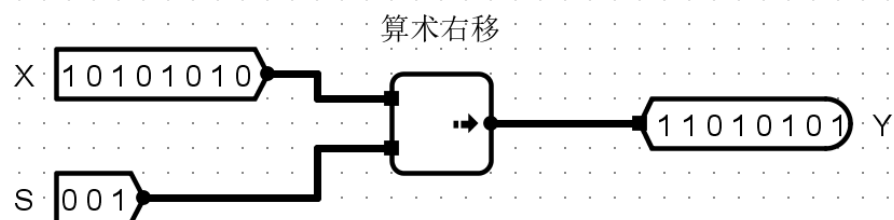
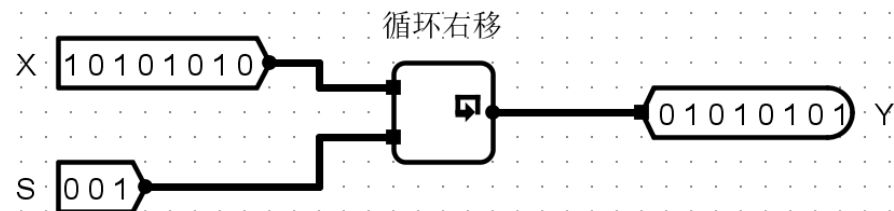
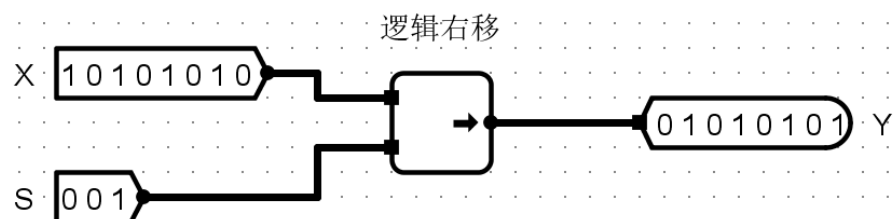
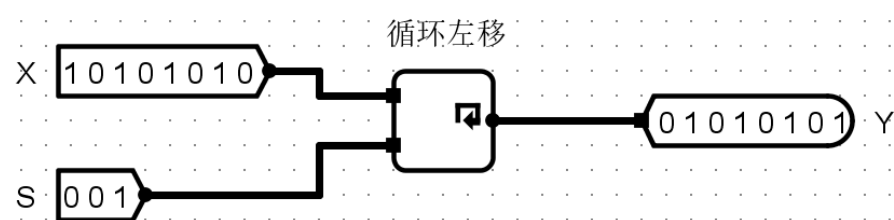
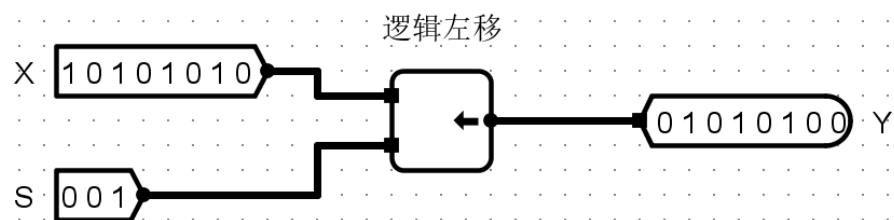
- Arithmetic
- + Adder
 - Subtractor
 - × Multiplier
 - ÷ Divider
 - × Negator
 - ⚖ Comparator ←
 - Shifter
 - ⊞ Bit Adder
 - ⚔ Bit Finder

比较器 (Comparator)



- Arithmetic
 - Adder
 - Subtractor
 - Multiplier
 - Divider
 - Negator
 - Comparator
 - Shifter ←
 - Bit Adder
 - Bit Finder

移位器 (Shifter)

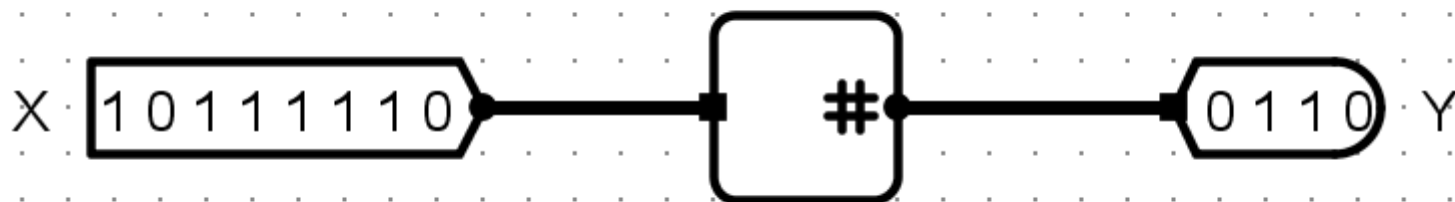


$Y = X \text{ 移位 } S \text{ 次}$

算术左移 = 逻辑左移

- Arithmetic
 - Adder
 - Subtractor
 - Multiplier
 - Divider
 - Negator
 - Comparator
 - Shifter
 - Bit Adder
 - Bit Finder

位加法器 (Bit Adder)



Y = X中1的个数

- Arithmetic
 - Adder
 - Subtractor
 - Multiplier
 - Divider
 - Negator
 - Comparator
 - Shifter
 - Bit Adder
 - Bit Finder ←

位发现器 (Bit Finder)



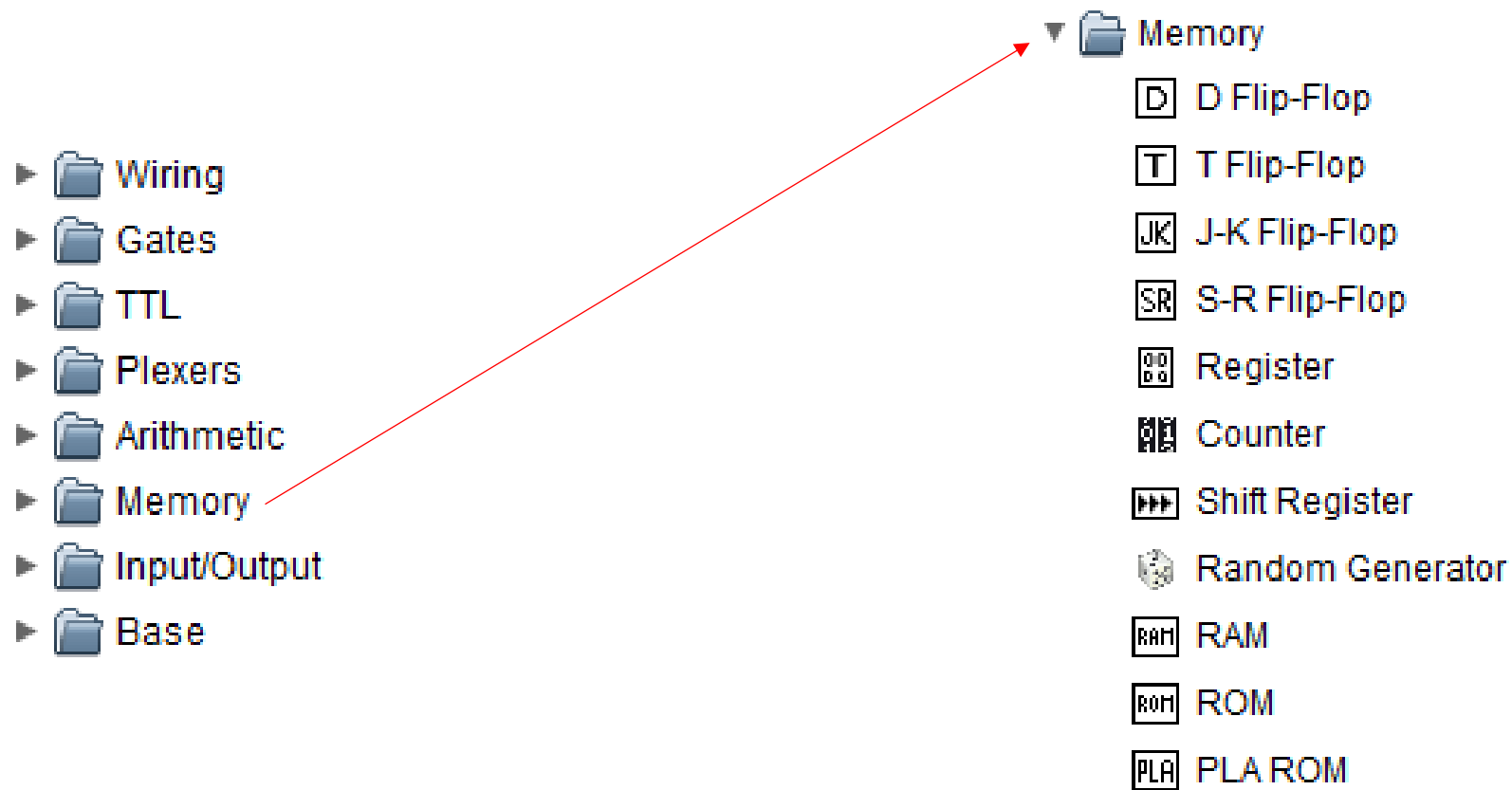
如果 $X \neq 0$ ，则 $Y = X$ 中从右开始第1个“1”的位置

并且置 $S=1$

最右边位置=0，最左边位置=7

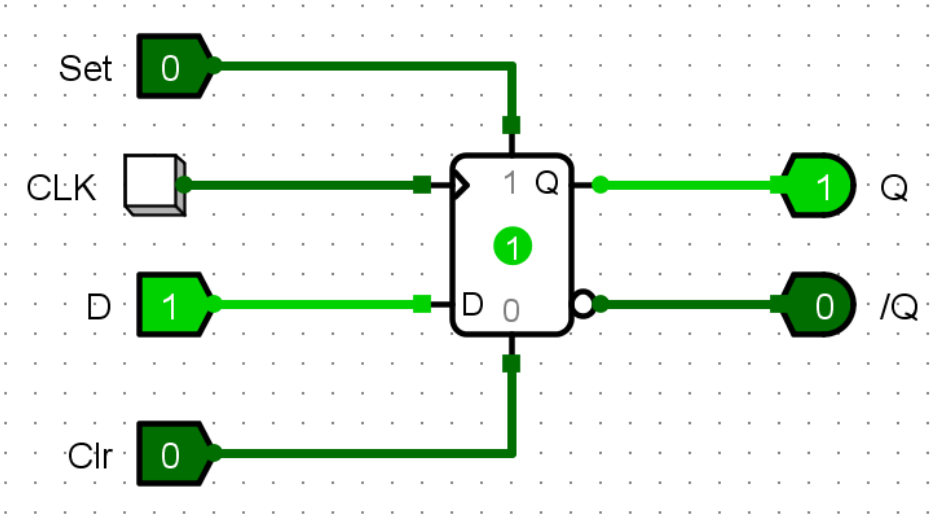
如果 $X = 0$ ，则 $Y = 000$ ，并且置 $S=0$

(六) 存储单元 (Memory)



- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM
 - PLA ROM

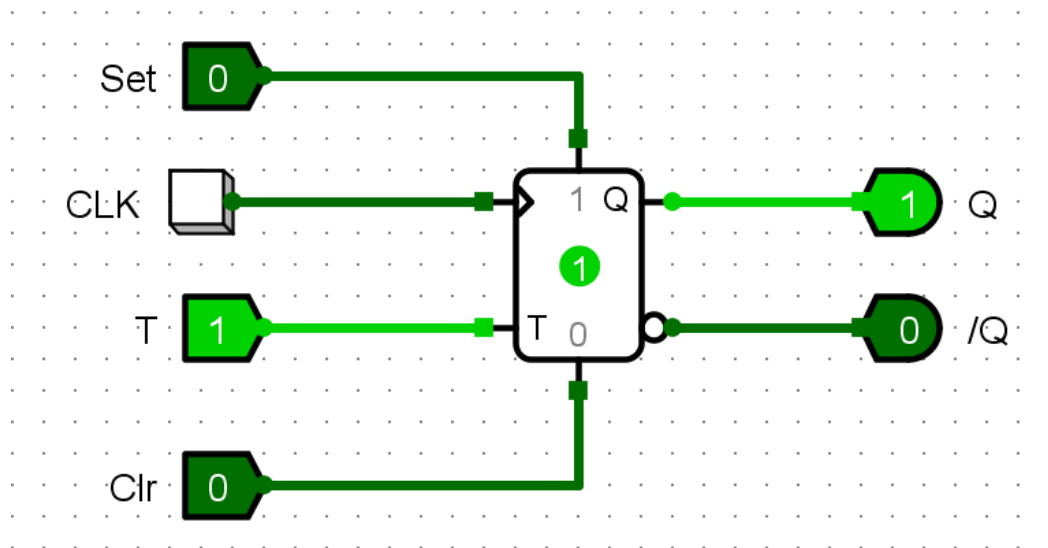
D触发器 (D Flip-Flop)



输入				输出	
CLK	Set	Clr	D	Q	/Q
上升沿	0	0	0	0	1
上升沿	0	0	1	1	0
x	0	1	x	0	1
x	1	0	x	1	0
x	1	1	x	0	1

- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM
 - PLA ROM

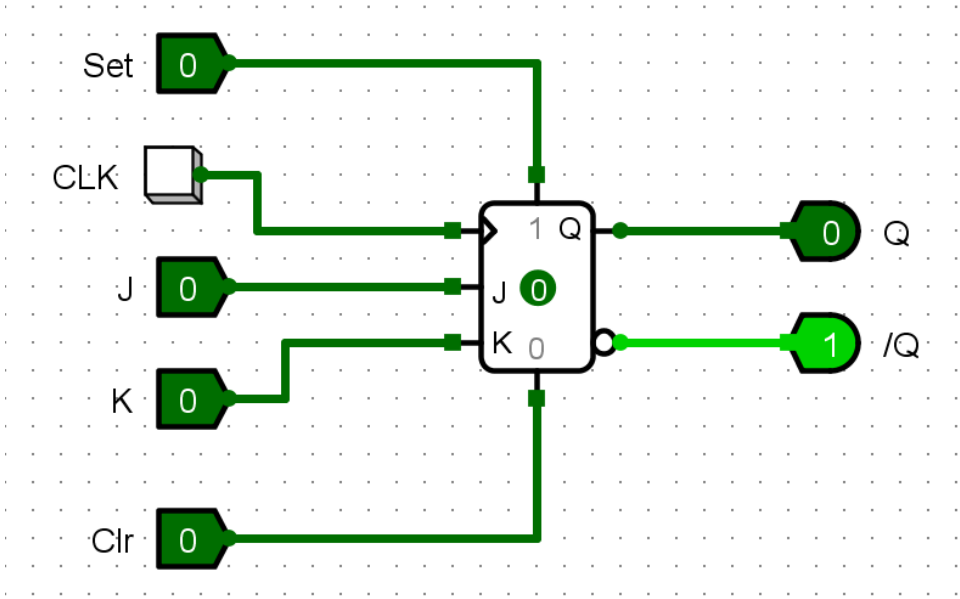
T触发器 (T Flip-Flop)



输入				输出	
CLK	Set	Clr	T	Q	/Q
上升沿	0	0	0	保持不变	保持不变
上升沿	0	0	1	翻转	翻转
x	0	1	x	0	1
x	1	0	x	1	0
x	1	1	x	0	1

- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM
 - PLA ROM

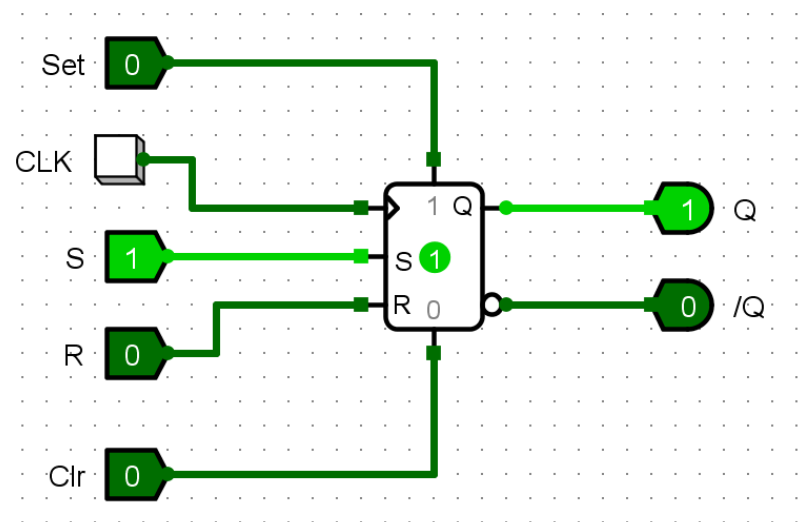
J-K触发器 (J-K Flip-Flop)



输入					输出	
CLK	Set	Clr	J	K	Q	/Q
上升沿	0	0	0	0	保持不变	保持不变
上升沿	0	0	0	1	0	1
上升沿	0	0	1	0	1	0
上升沿	0	0	1	1	翻转	翻转
x	0	1	x	x	0	1
x	1	0	x	x	1	0
x	1	1	x	x	0	1

- ▼ Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop ←
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM
 - PLA ROM

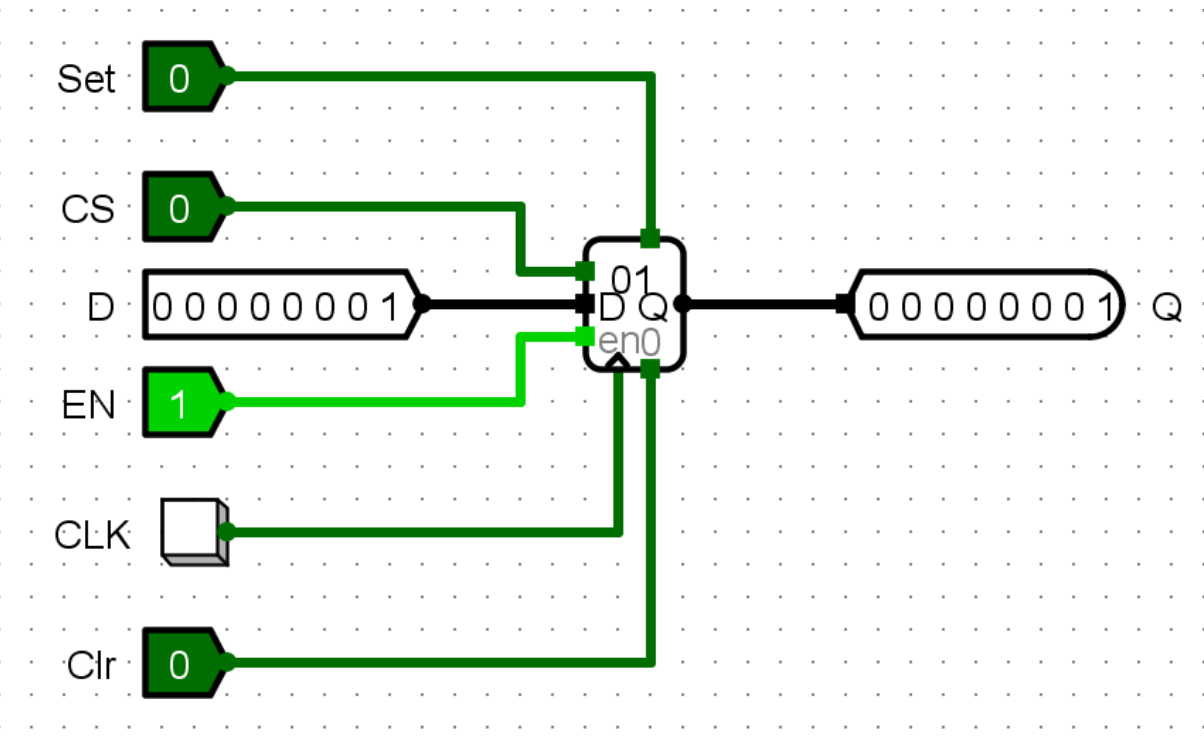
S-R触发器 (S-R Flip-Flop)



输入					输出	
CLK	Set	Clr	S	R	Q	/Q
上升沿	0	0	0	0	保持不变	保持不变
上升沿	0	0	0	1	0	1
上升沿	0	0	1	0	1	0
上升沿	0	0	1	1	保持不变	保持不变
x	0	1	x	x	0	1
x	1	0	x	x	1	0
x	1	1	x	x	0	1

- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM
 - PLA ROM

寄存器 (Register)

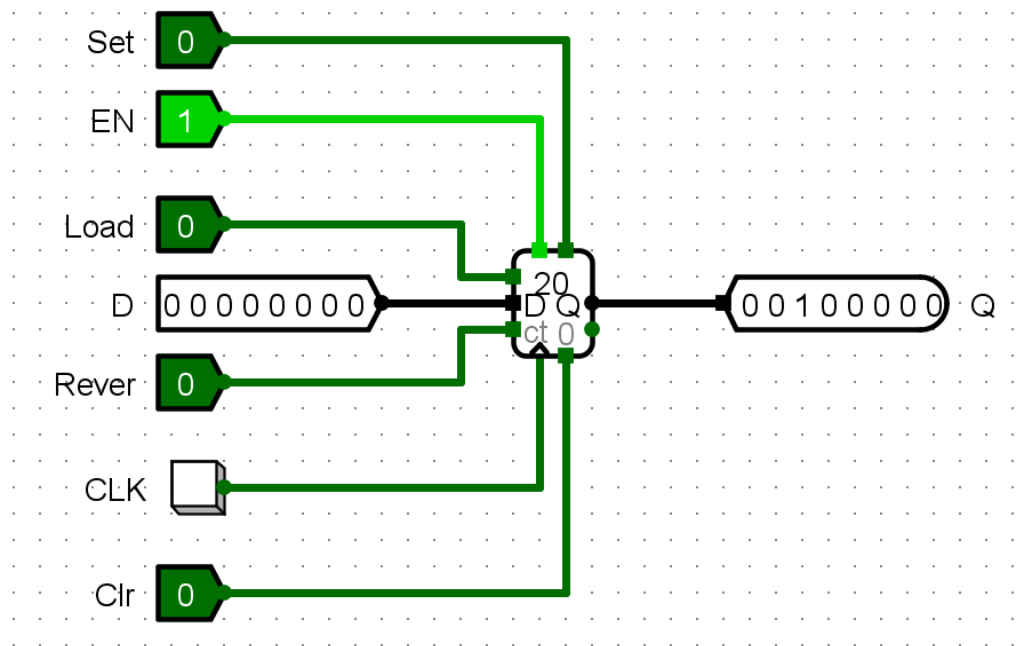


8位寄存器

输入						输出
CLK	CS	EN	Set	Clr	D	Q
x	1	x	x	x	任意	高阻
上升沿	0	1	0	0	任意	D
x	0	0	0	0	任意	保持不变
x	0	x	1	0	任意	全 1
x	0	x	0	1	任意	全 0
x	0	x	1	1	任意	全 0

- ▼ Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter ←
 - Shift Register
 - Random Generator
 - RAM
 - ROM
 - PLA ROM

计数器 (Counter)

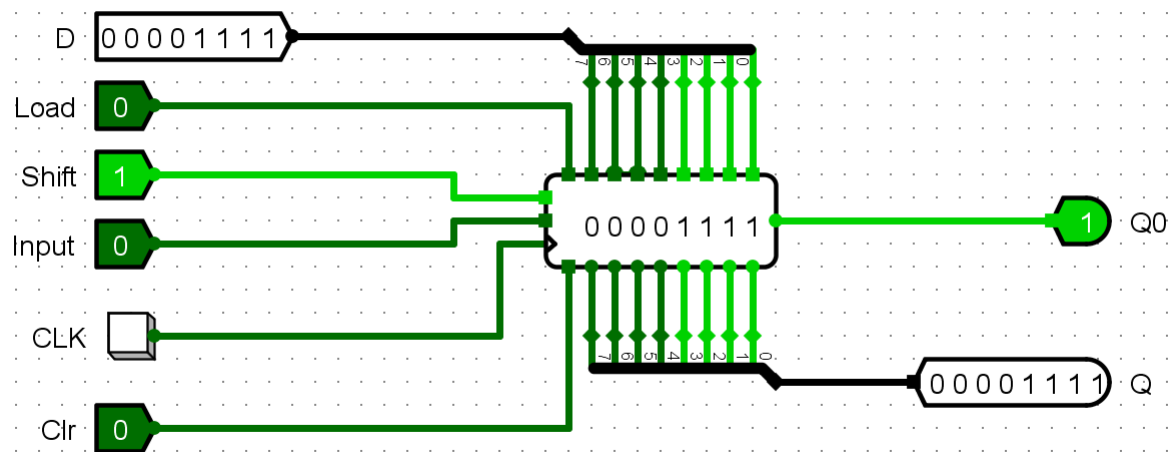


8位计数器

输入							输出
CLK	EN	Load	Rever	Set	Clr	D	Q
上升沿	1	0	0	0	0	任意	计数 (加 1)
上升沿	1	0	1	0	0	任意	计数 (减 1)
上升沿	1	1	0	0	0	任意	D (装入)
x	x	x	x	1	0	任意	全 1
x	x	x	x	0	1	任意	全 0
x	x	x	x	1	1	任意	全 0

- ▼ Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register ←
 - Random Generator
 - RAM
 - ROM
 - PLA ROM

移位寄存器 (Shift Register)

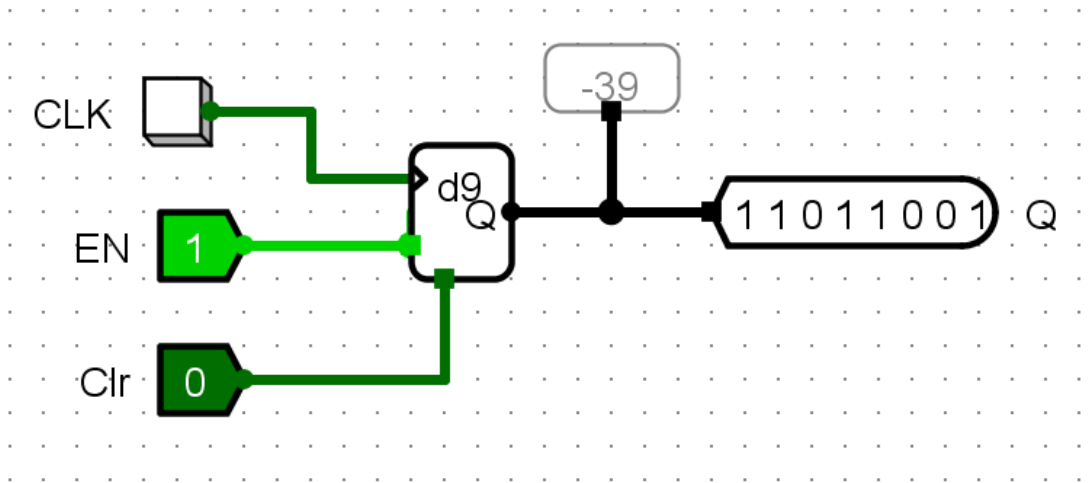


输入						输出	
CLK	Load	Shift	Input	Clr	D	Q	Q0
上升沿	1	0	0	0	任意	D (装入)	Q 的最低位
上升沿	0	1	0	0	任意	移位 (右移)	
上升沿	1	1	0	0	任意	移位 (右移)、并将 1 从左边移入	
x	x	x	x	1	任意	全 0	

8位移位寄存器

- ▼ Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator ←
 - RAM
 - ROM
 - PLA ROM

随机数发生器 (Random Generator)



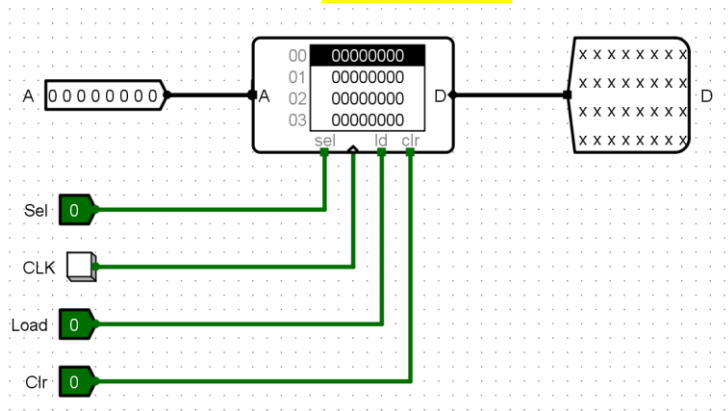
输入			输出
CLK	EN	Clr	Q
上升沿	1	0	产生一个随机数
上升沿	0	0	保持不变

8位随机数发生器

- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM ←
 - ROM
 - PLA ROM

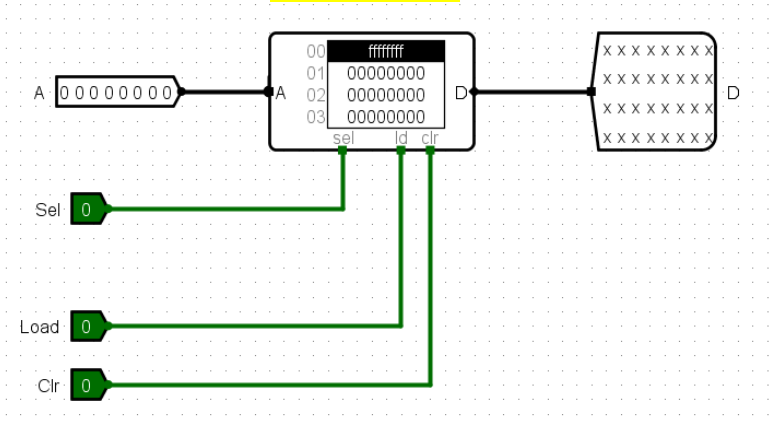
RAM（随机存取存储器）

同步模式



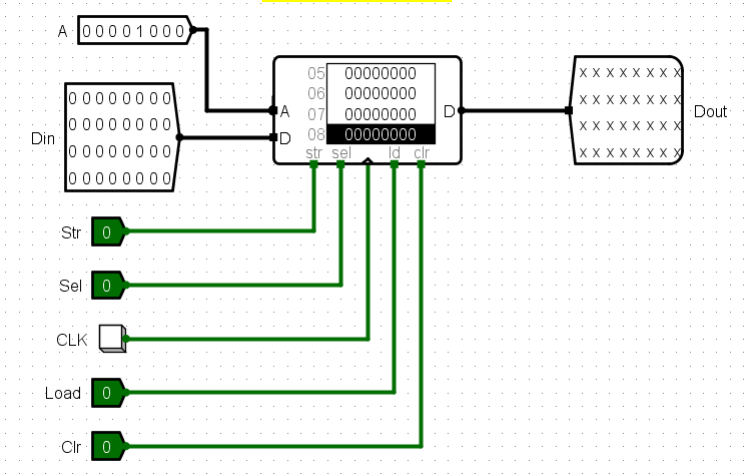
RAM	
Address Bit Width	8
Data Bit Width	32
Label	
Label Font	Dialog Plain 12
Label Color	#000000
Data Interface	One synchronous load/store port
Sel Active On:	Low Level

异步模式



RAM	
Address Bit Width	8
Data Bit Width	32
Label	
Label Font	Dialog Plain 12
Label Color	#000000
Data Interface	One asynchronous load/store port
Sel Active On:	Low Level

分离模式



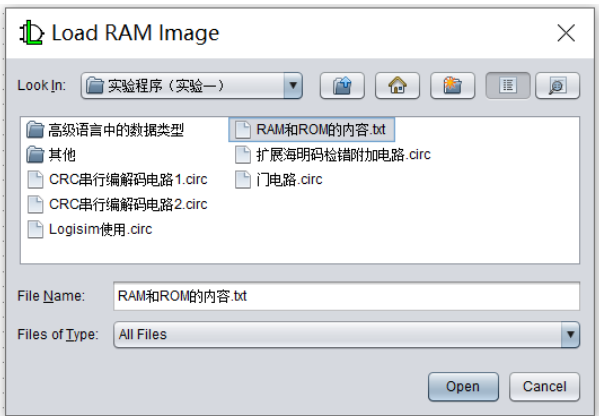
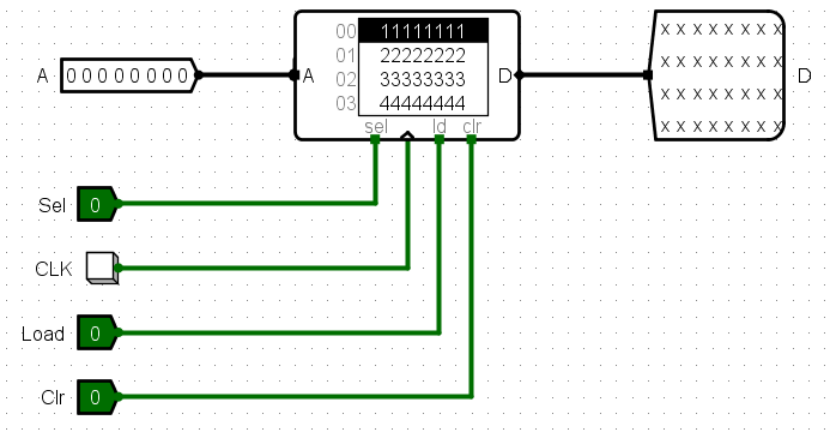
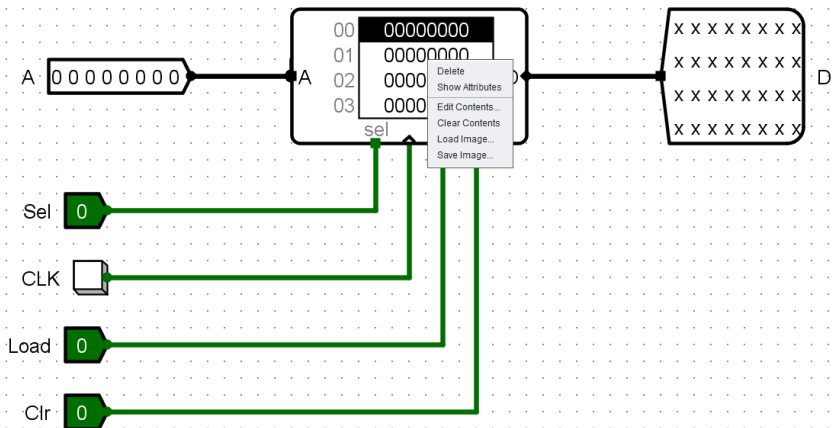
RAM	
Address Bit Width	8
Data Bit Width	32
Label	
Label Font	Dialog Plain 12
Label Color	#000000
Data Interface	Separate load and store ports
Sel Active On:	Low Level

- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM
 - PLA ROM

同步模式RAM存储器的操作

装入数据

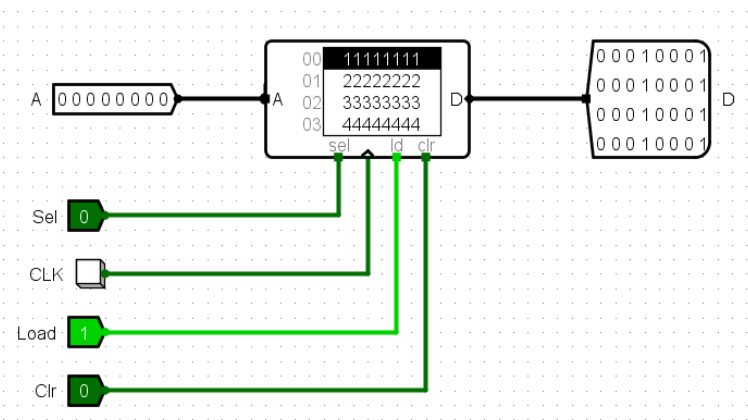
Sel=0、Load=0、
Clr=0; 按鼠标右键，选
择“Load Image”，将文
本文件中的内容装入存
储器中



RAM和ROM的内容.txt - 记事本
文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)
v2.0 raw
11111111
22222222
33333333
44444444
55555555
66666666
77777777
88888888
99999999
aaaaaaaa
bbbbbbbb
cccccccc
dddddddd
eeeeeeee
ffffff

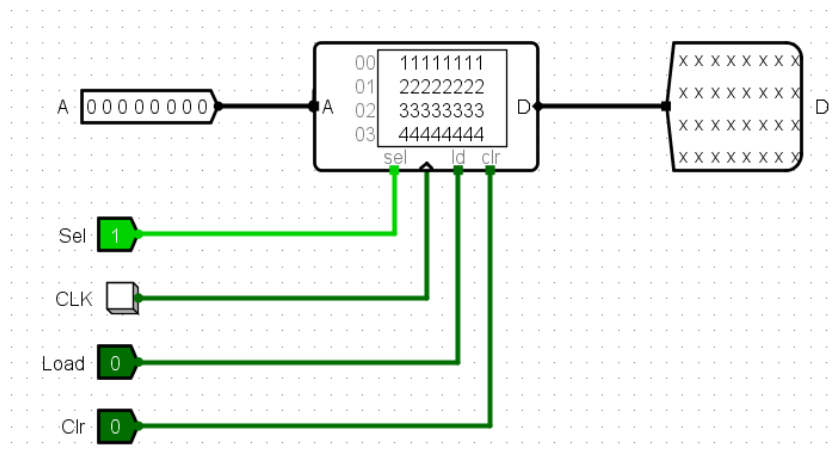
- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM ←
 - ROM
 - PLA ROM

同步模式RAM存储器的操作



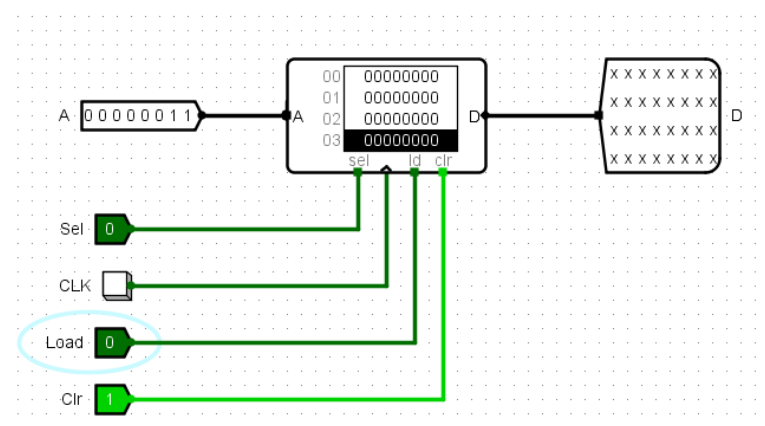
读出数据

Sel=0、Load=1、
Clr=0；给定地址A，则
对应的内容输出到D



输出高阻

Sel=1、Load=0、
Clr=0时，输出为高阻

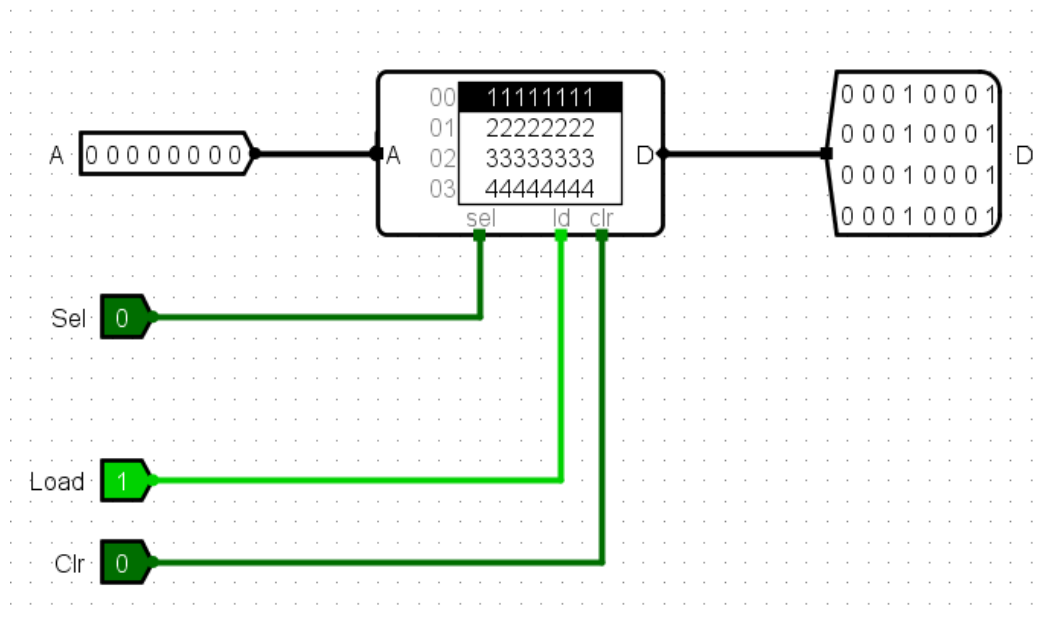


内容清零

Sel=0、Load=0、
Clr=1时，存储器中的内
容清零

- ▼ Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM ←
 - ROM
 - PLA ROM

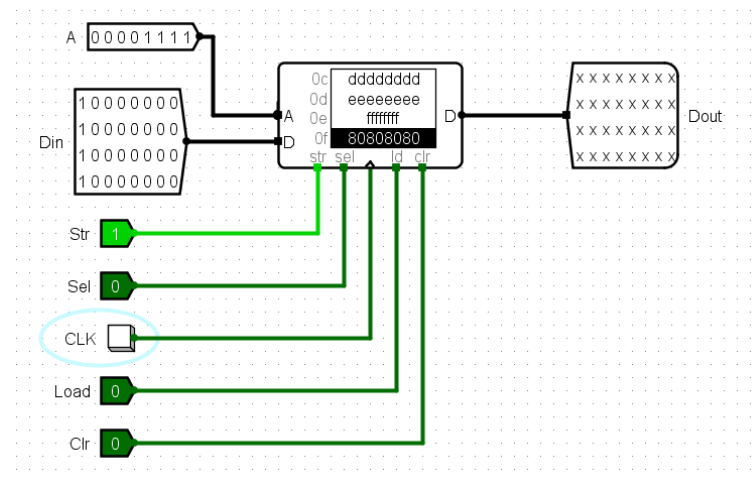
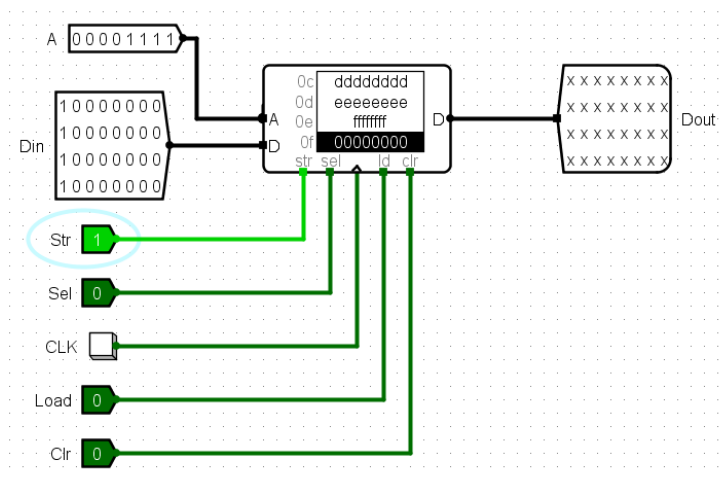
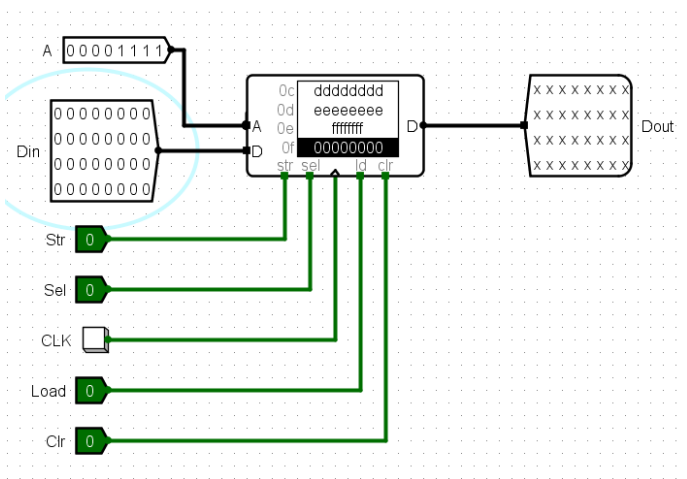
异步模式RAM存储器的操作



除了没有时钟信号CLK外，其它与同步模式相同

- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM ←
 - ROM
 - PLA ROM

分离模式RAM存储器的操作



装入数据

Str=0、Sel=0、
Load=0、Clr=0；按鼠标右键，选择“Load Image”，将文本文件中的内容装入存储器中

写入数据（1）

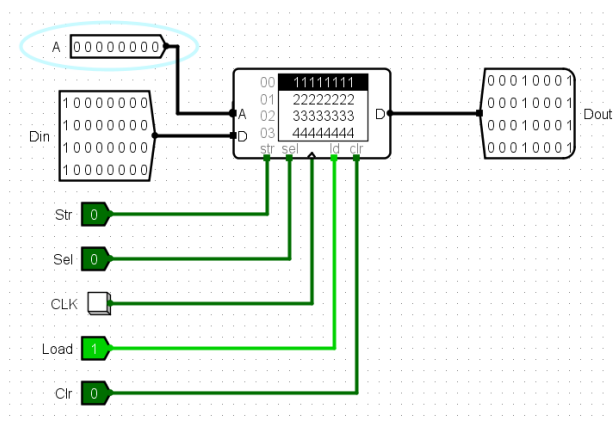
Str=1、Sel=0、
Load=0、Clr=0；
A=00001111、
Din=80808080H

写入数据（2）

按CLK键，则将Din写入到00001111存储单元

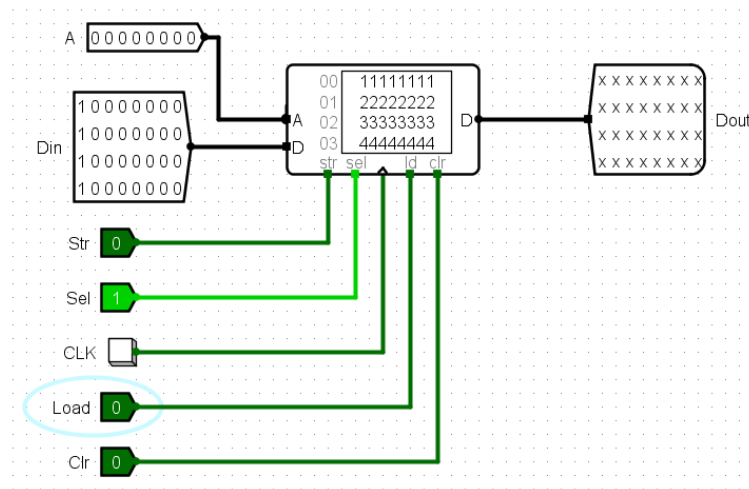
- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM ←
 - ROM
 - PLA ROM

分离模式RAM存储器的操作



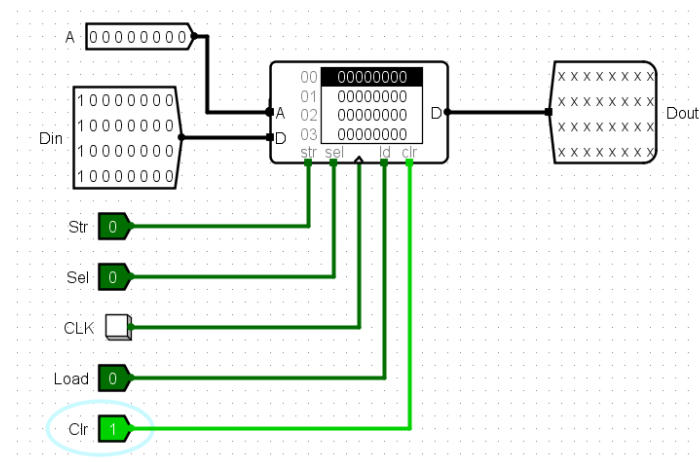
读出数据

Str=0、Sel=0、
Load=1、Clr=0；给定
地址A，则对应的内容
输出到D中



输出高阻

Str=0、Sel=1、
Load=0、Clr=0；
A=00001111、
Din=80808080H

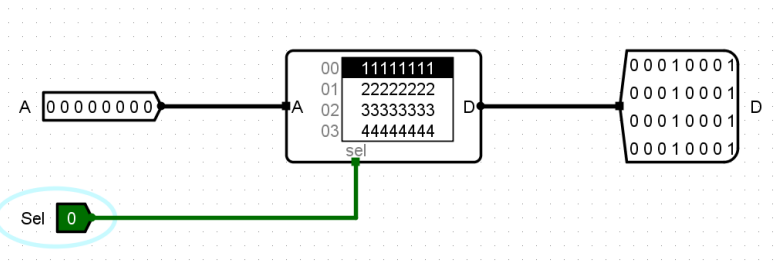


内容清零

Str=0、Sel=0、
Load=0、Clr=1；
A=00001111、
Din=80808080H

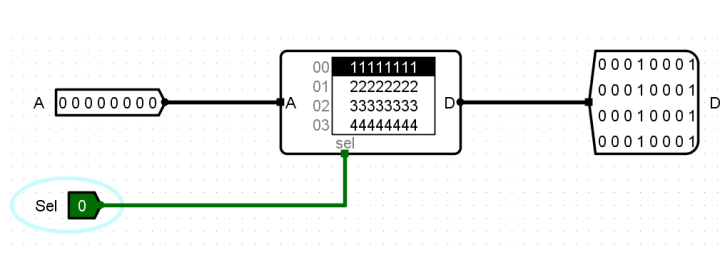
- Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM ←
 - PLA ROM

ROM (只读存储器)



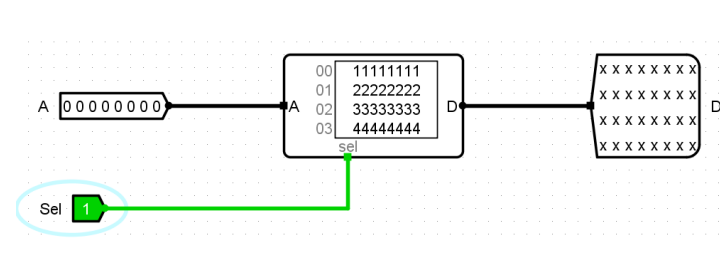
装入数据

Sel=0; 按鼠标右键，选择“Load Image”，将文本文件中的内容装入存储器中



读出数据

Sel=0; 给定地址A，则对应的内容输出到D中



输出高阻

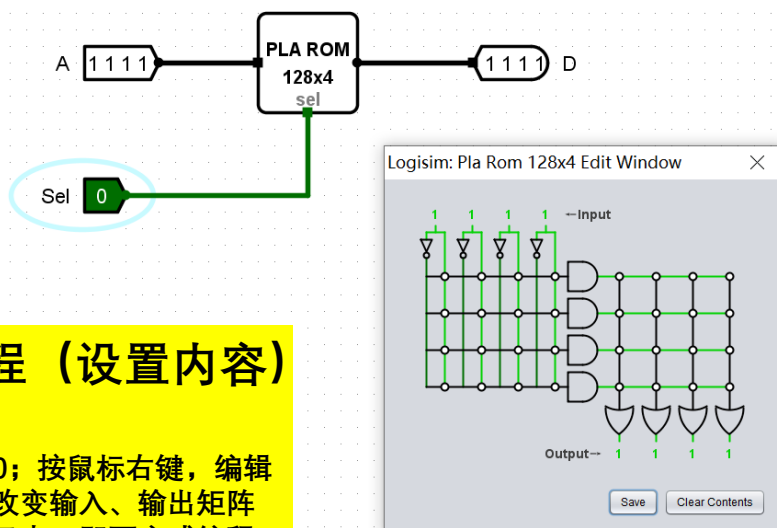
Sel=1时，输出为高阻

- ▼ Memory
 - D Flip-Flop
 - T Flip-Flop
 - J-K Flip-Flop
 - S-R Flip-Flop
 - Register
 - Counter
 - Shift Register
 - Random Generator
 - RAM
 - ROM
 - PLA ROM ←

PLA ROM

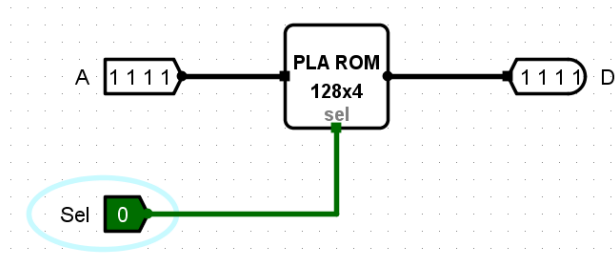
(可编程逻辑阵列只读存储器)

Programmable Logic Arrays: 可编程逻辑阵列



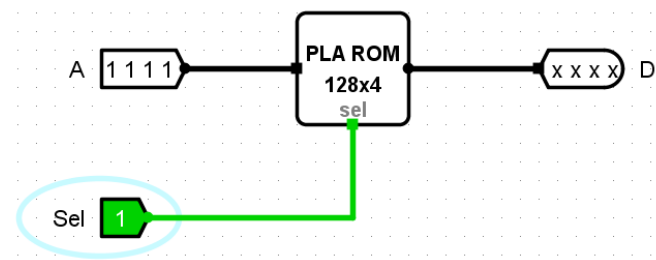
编程 (设置内容)

Sel=0; 按鼠标右键, 编辑框, 改变输入、输出矩阵的交叉点, 即可完成编程 (设置16个单元的内容)



读出数据

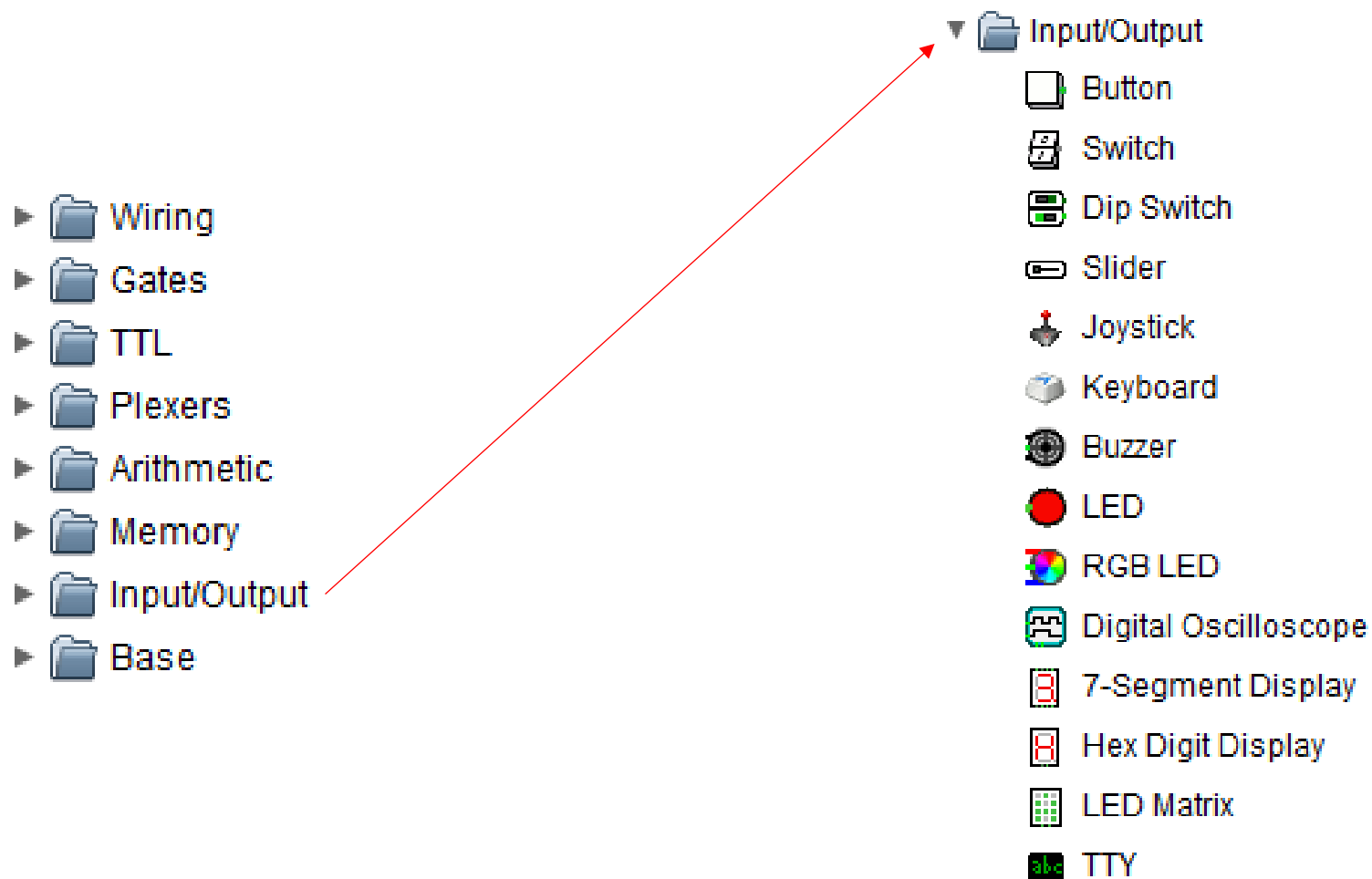
Sel=0; 给定地址A, 则对应的内容输出到D中



输出高阻

Sel=1时, 输出为高阻

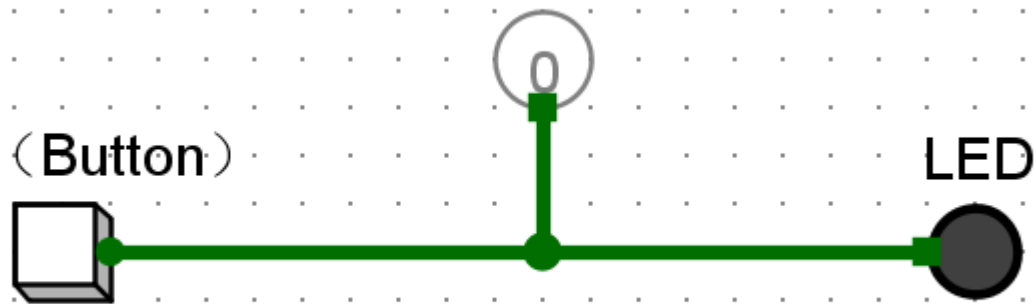
(七) 输入/输出部件 (Input/Output)



- ▼ Input/Output
 - Button 
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

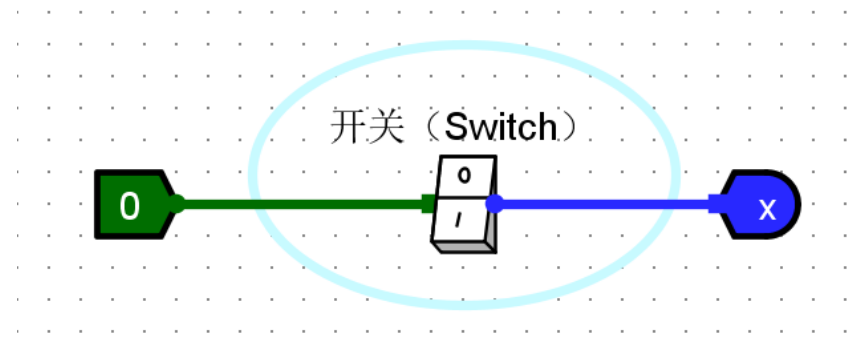
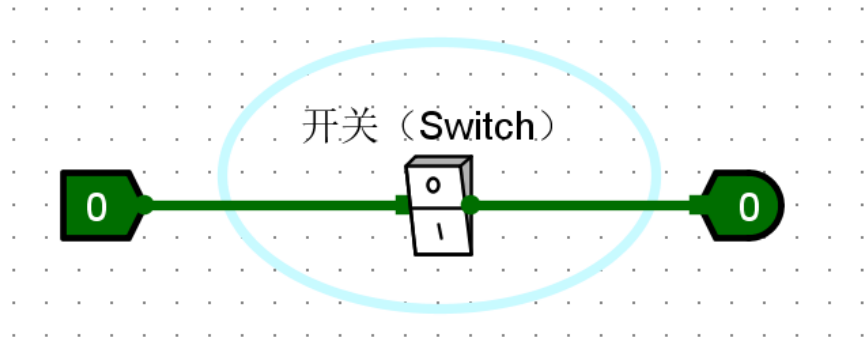
按钮 (Button)

按钮 (Button)



- ▼ Input/Output
 - Button
 - Switch 
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

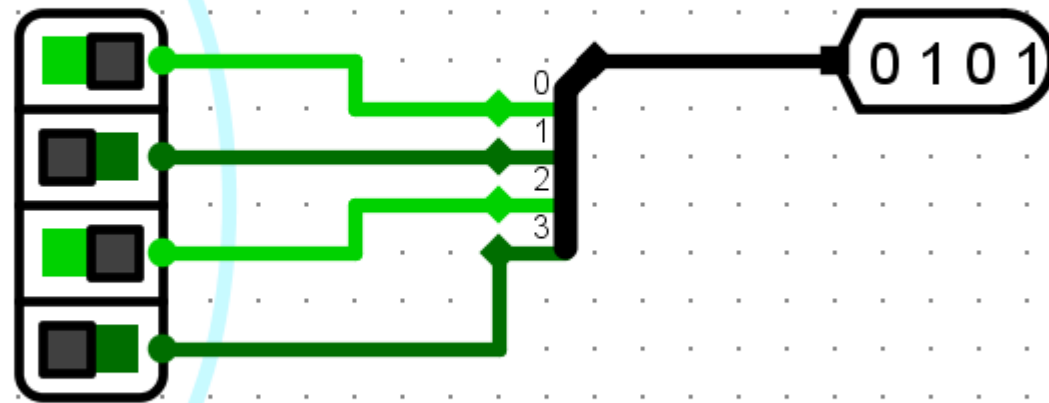
开关 (Switch)



- Input/Output
 - Button
 - Switch
 - Dip Switch 
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

DIP开关 (DIP Switch)

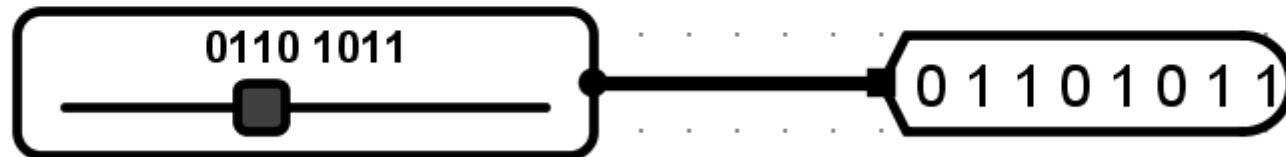
DIP开关 (DIP Switch)



- ▼ Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider ←
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

滑杆 (Slider)

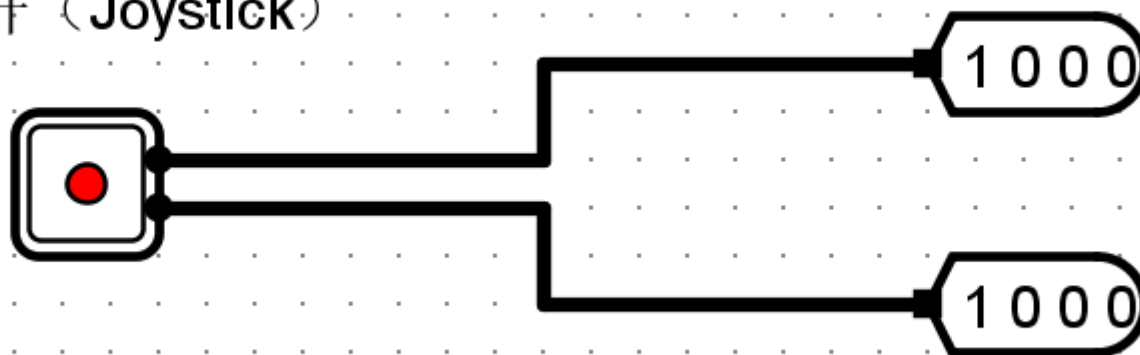
滑杆 (Slider)



- ▼ Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick ←
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

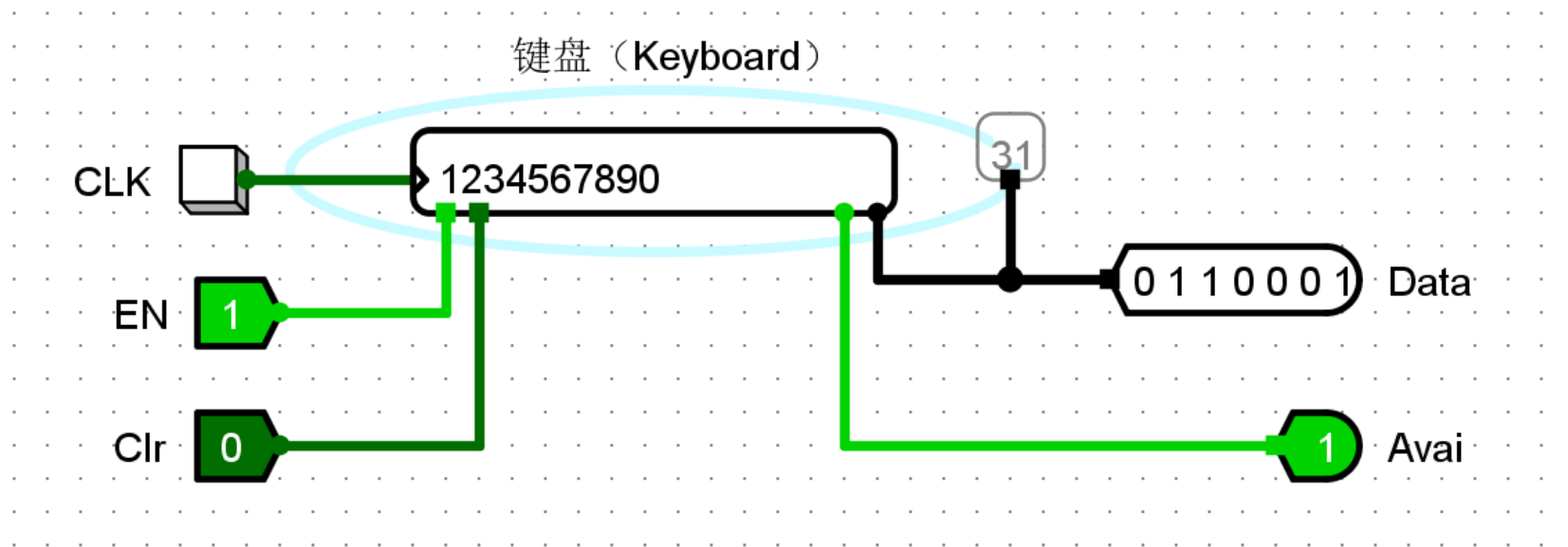
游戏杆 (Joystick)

游戏杆 (Joystick)



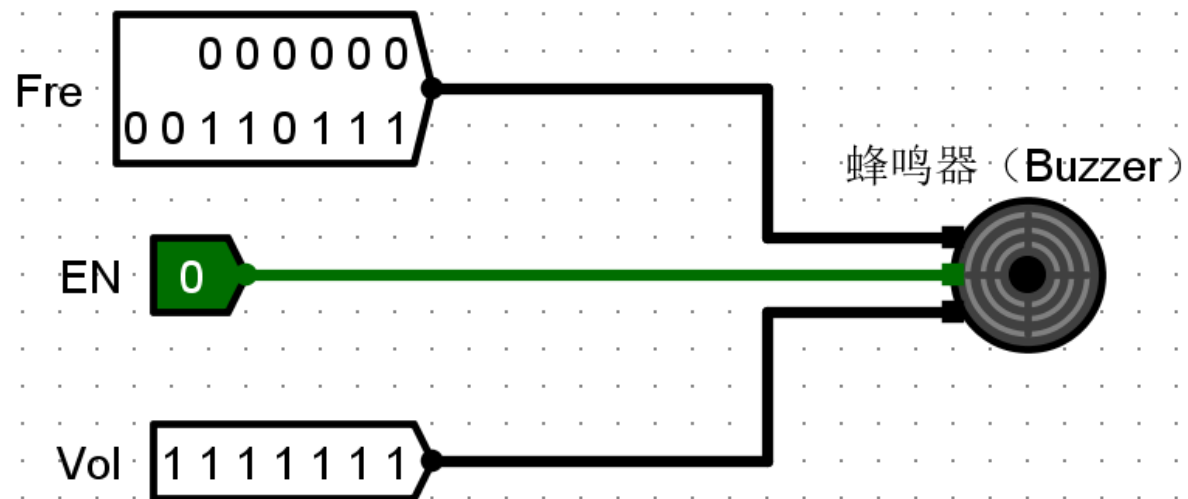
- Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

键盘 (Keyboard)



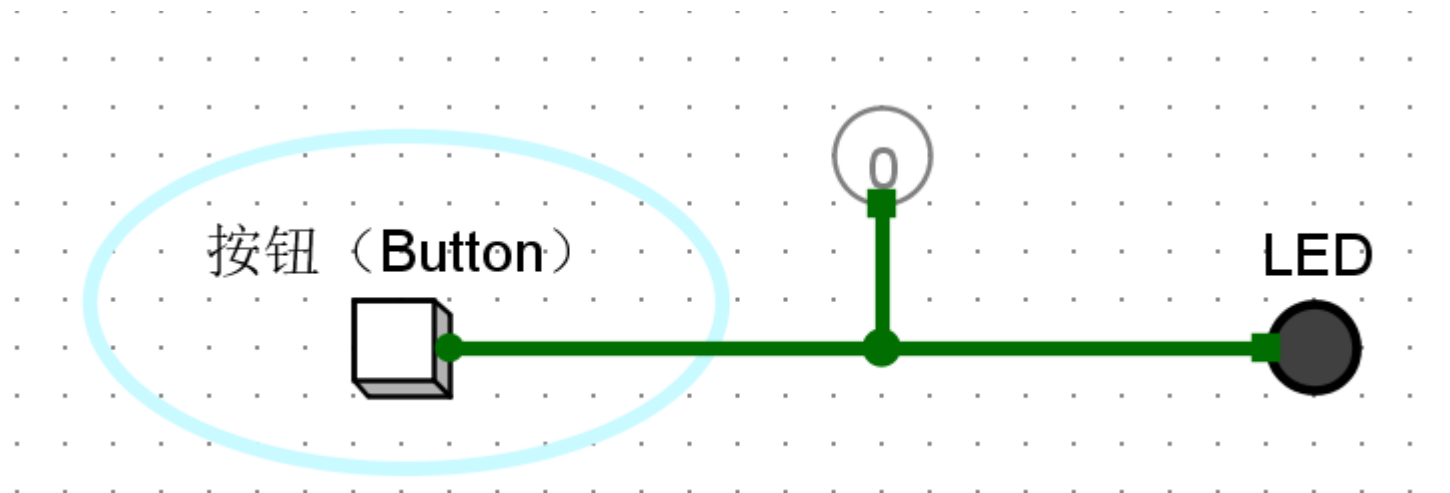
- Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

蜂鸣器 (Buzzer)



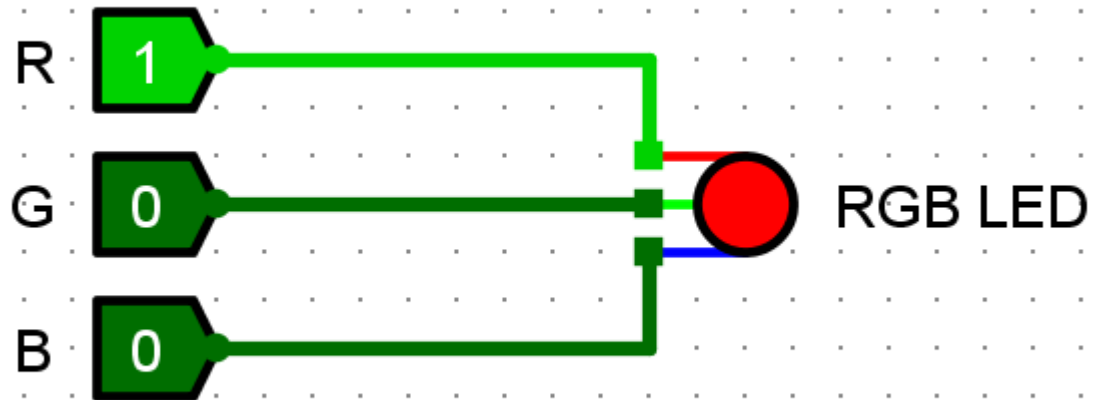
- ▼ Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED ←
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

LED灯



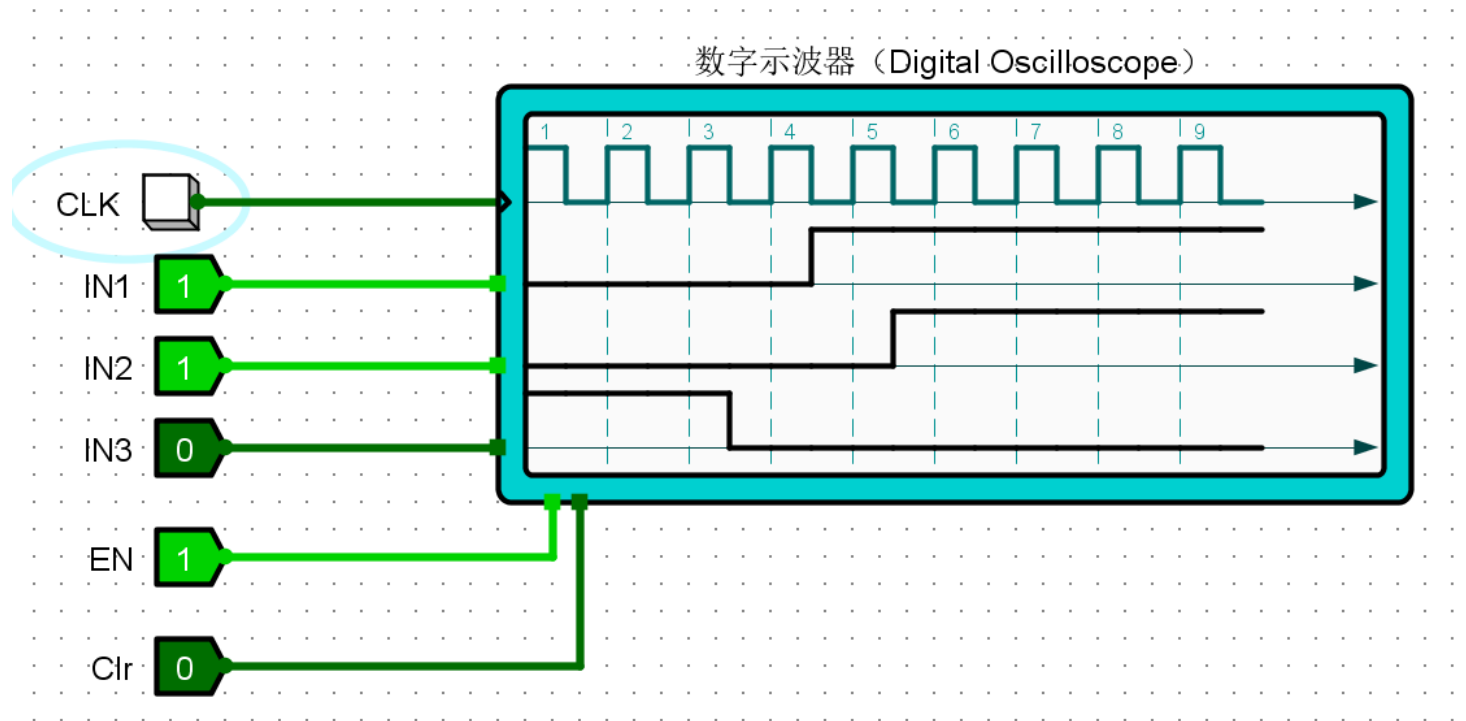
- ▼ Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED ←
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

三色灯 (RGB LED)



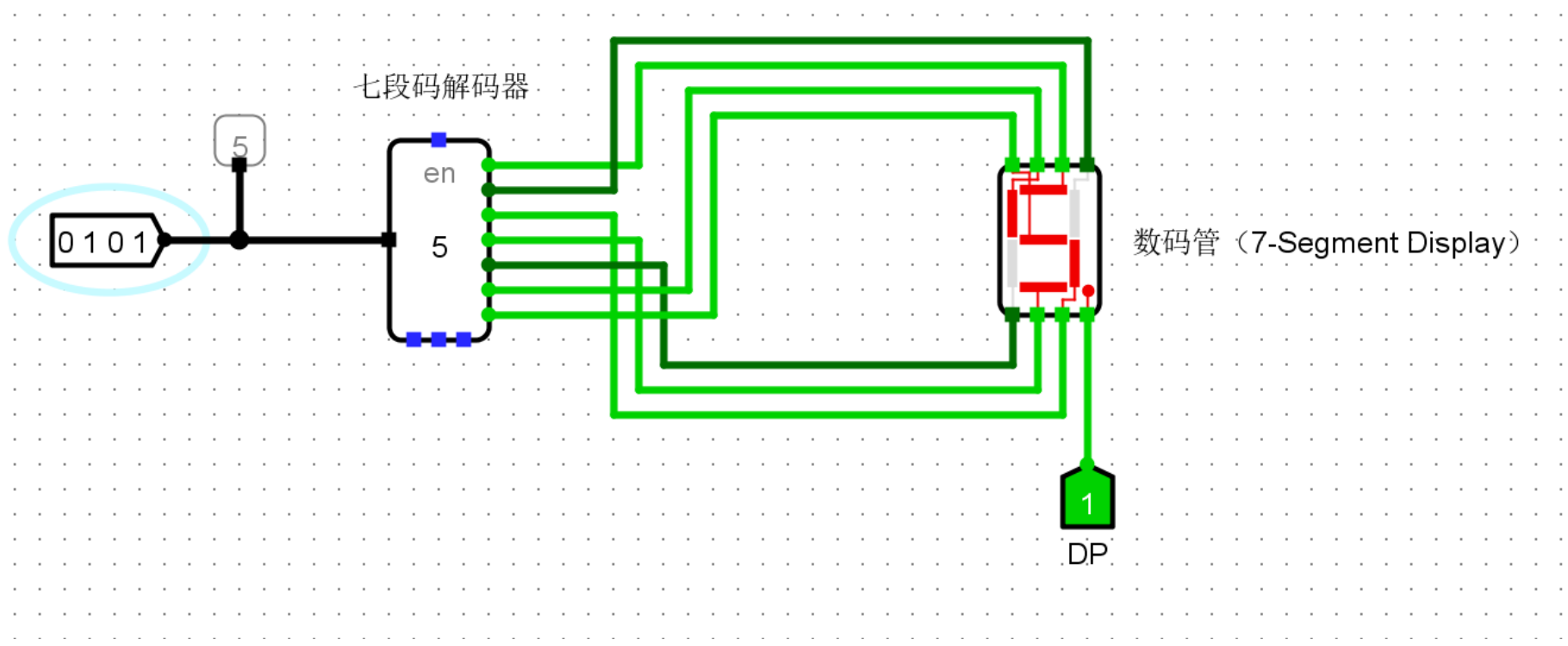
- Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope ←
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

数字示波器 (Digital Oscilloscope)



- Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display ←
 - Hex Digit Display
 - LED Matrix
 - TTY

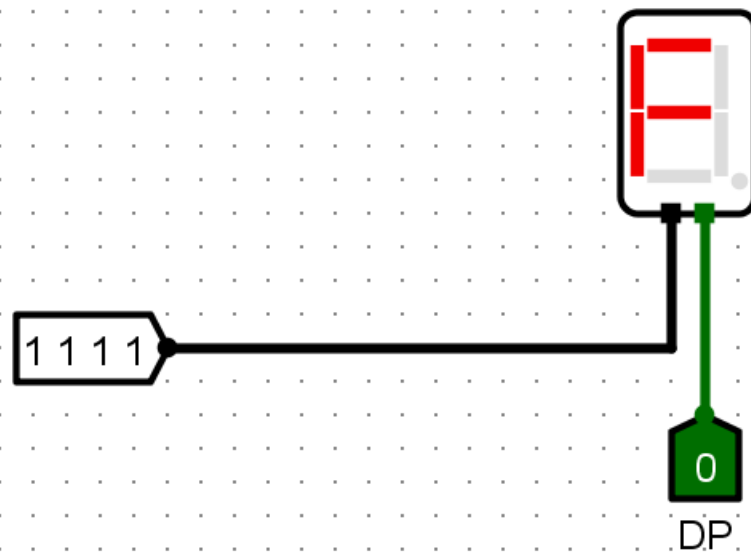
数码管 (7-Segment Display)



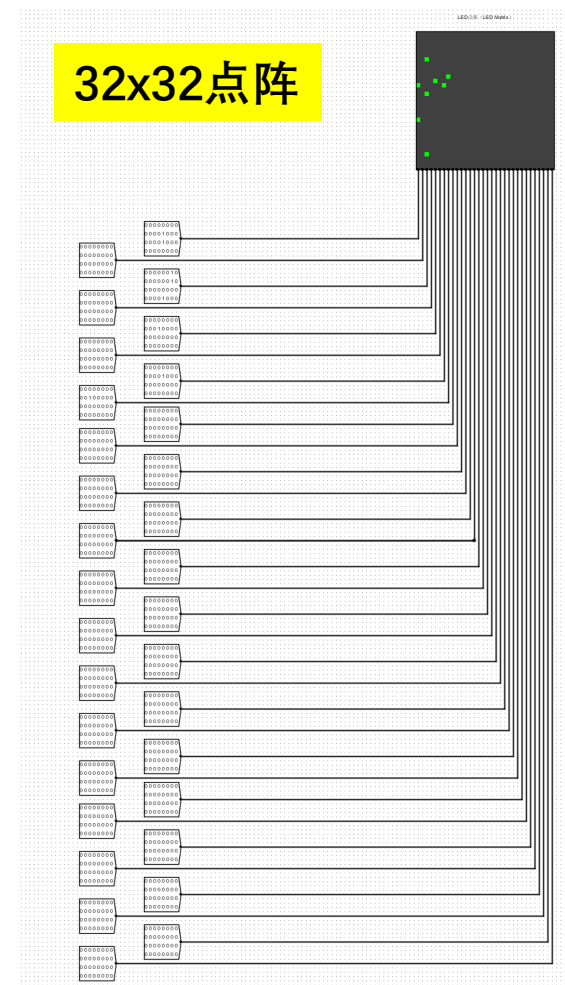
十六进制数码管 (Hex Digital Display)

- Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display ←
 - LED Matrix
 - TTY

十六进制数码管 (Hex Digital Display)

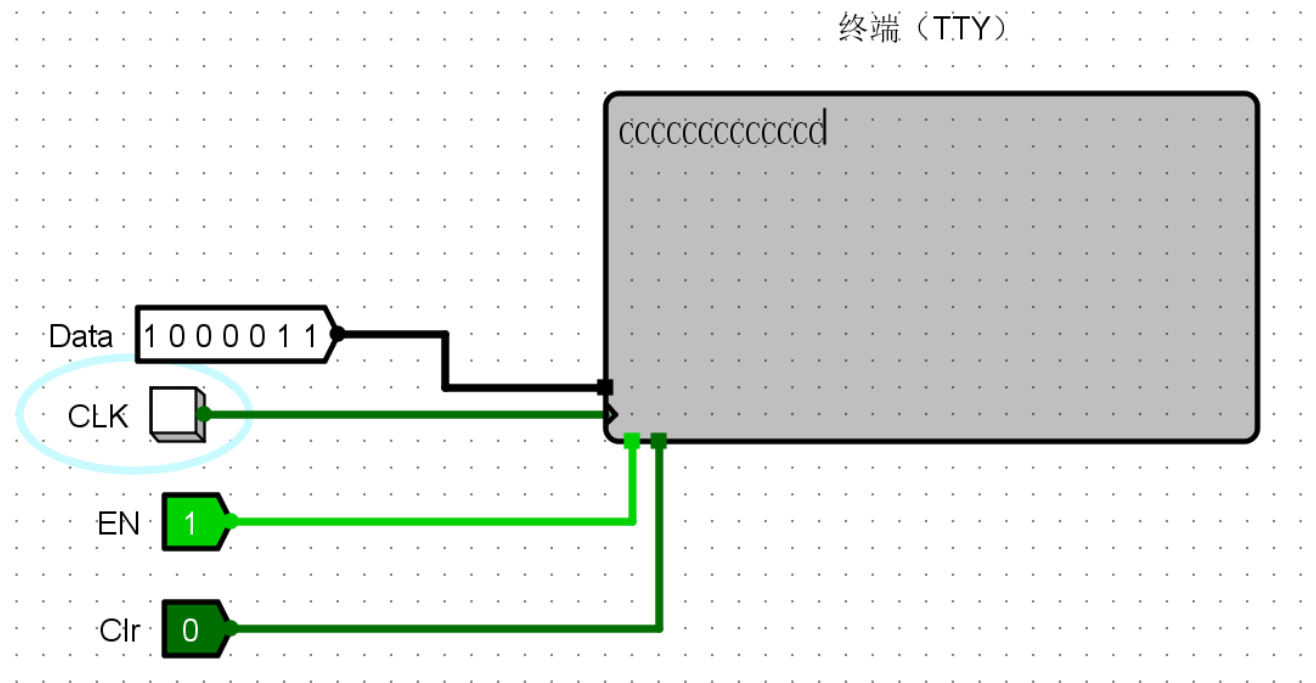


- Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY

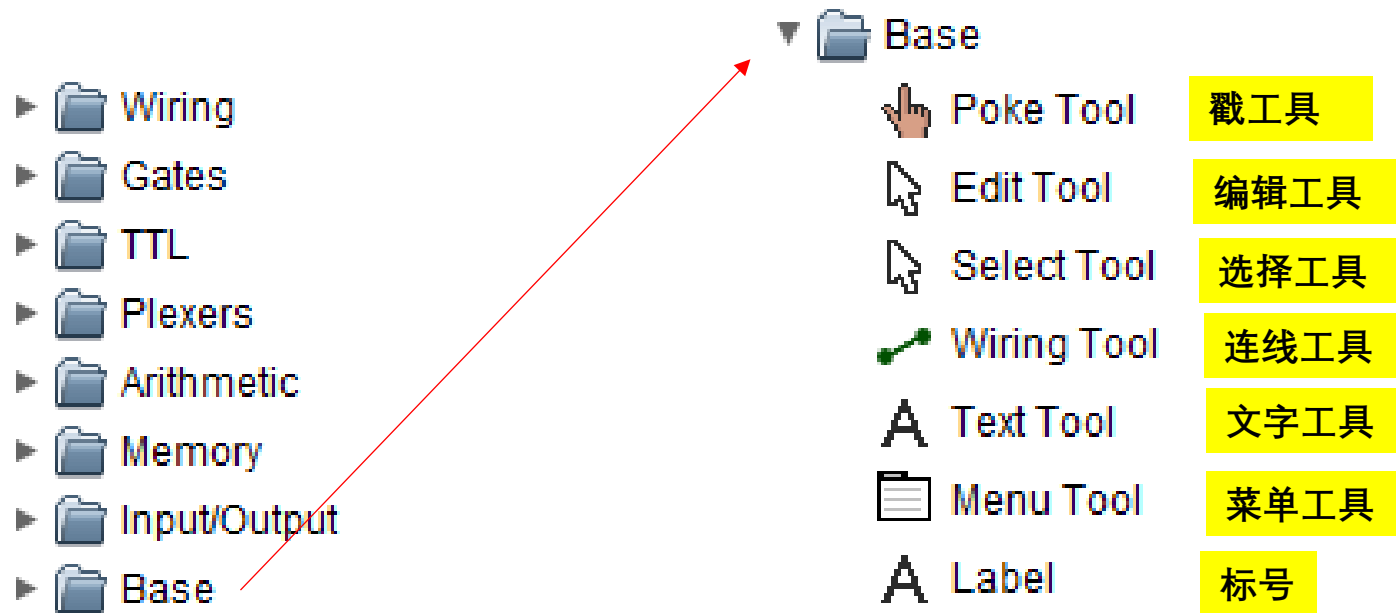


- ▼ Input/Output
 - Button
 - Switch
 - Dip Switch
 - Slider
 - Joystick
 - Keyboard
 - Buzzer
 - LED
 - RGB LED
 - Digital Oscilloscope
 - 7-Segment Display
 - Hex Digit Display
 - LED Matrix
 - TTY ←

终端 (TTY)



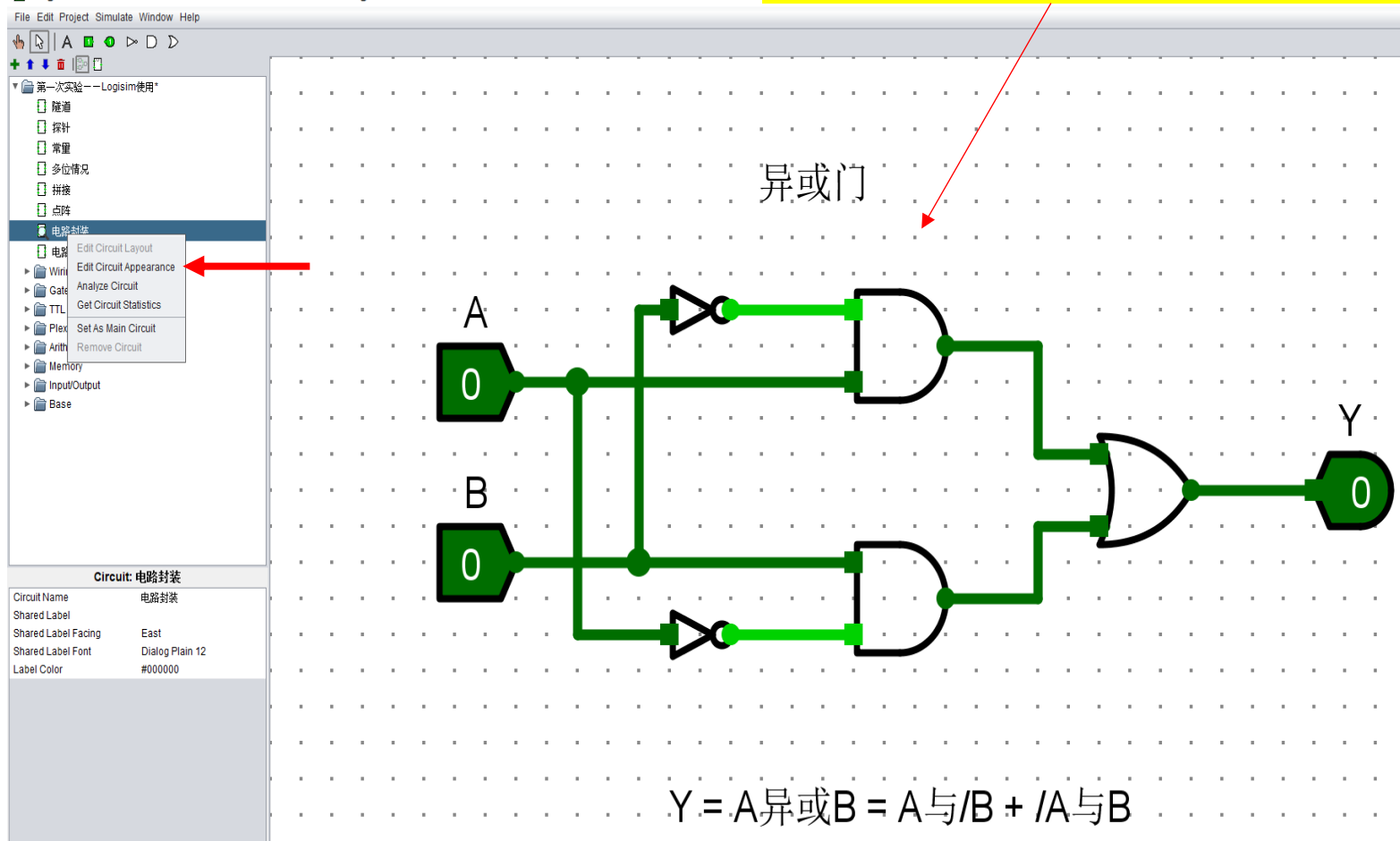
(八) 基本操作 (Base)



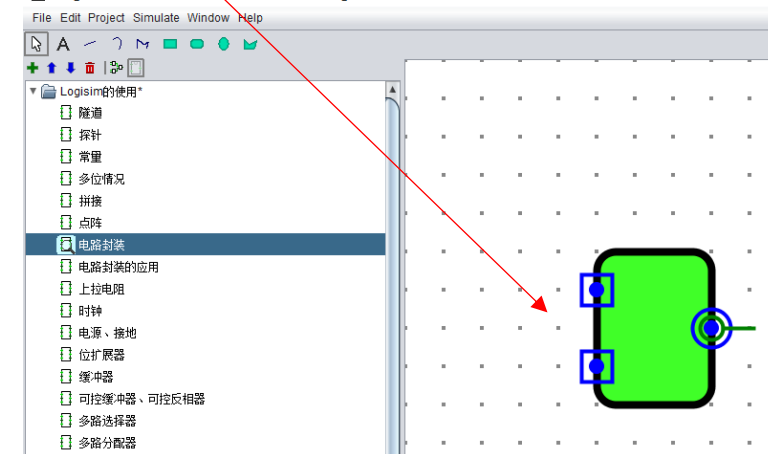
(九) 电路封装

将左边的异或门电路封装成右边的电路

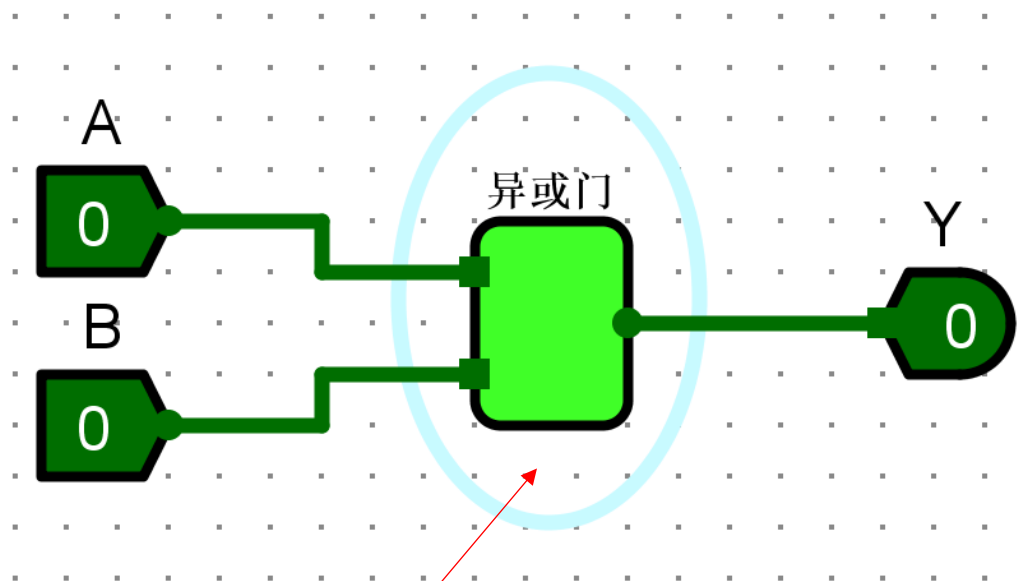
Logisim 2.16.1.0.exe: 电路封装 of 第一次实验——Logisim使用



Logisim 2.15.0.2.exe: 电路封装 of Logisim的使用



电路封装的应用



这里的异或门电路就是上面封装好的电路